

Lecture notes in Statistical modeling (draft)

Biljana Tojtovska Ribarski,
Data Mining, Statistical modeling courses at FCSE, Skopje

May 11, 2026

Contents

1	Fundamental concepts in Probability	4
1.1	Random variables	7
2	Discrete distributions	8
2.1	Bernouli distribution	8
2.2	Binomial distribution	9
2.3	Poisson distribution	12
2.4	Geometric distribution	14
2.5	Negative binomial distribution	15
2.6	Hypergeometric distribution	17
3	Continuous Distributions	17
3.1	Uniform Distribution (Continuous)	17
3.2	Exponential Distribution	19
3.3	Normal Distribution	22
3.4	Students' t Distribution	23
3.5	Beta Distribution	24
3.6	Gamma Distribution	26
3.7	Logistic distribution	30
3.8	Weibull distribution	33
3.9	Laplace distribution	34
3.10	Lognormal distribution	35
3.11	Pareto distribution	38
3.12	Cauchy distribution (aka Lorentz Distribution)	41
3.13	How to analyze heavy tailed distributions	43
3.14	Bimodal distributions are their analyzes	44
3.15	Mixture models and applications	48
3.16	Circular statistics	48
3.17	Convolutions and compound distributions	48
4	Families of distributions	48
4.1	Exponential family	48
4.2	Location-Scale Family	49
4.3	Stable distributions	49
4.4	Extreme value distributions	50

5 Solved exercises	56
6 Projects	60
7 Discrete Multivariate Distributions	60
8 Continuous Multivariate Distributions	69
9 Likelihood Function	76
10 Definition of the Likelihood Function	77
11 Projects	84
12 Bayesian statistics	85
12.1 Motivation	85
13 Introduction	85
14 Data from Bernoulli distribution	89
14.1 Choice of parameters of prior distribution	95
14.2 Prior predictive check, sensitivity analysis, data simulation	100
14.2.1 Maximum a posteriori estimation (MAP)	105
14.2.2 Bayesian confidence region aka credible intervals.	107
14.2.3 Posterior analysis	108
14.2.4 ROPE	108
14.2.5 Posterior predictive check	108
14.3 Conjugate Priors	108
15 Data from Binomial distribution	108
16 Data from Geometrical distribution	113
17 Data from Exponential distribution	117
18 Data from Poisson distribution	118
19 Normal - normal model	124
20 Inference and decision making	125
21 Important formulas	128
22 Improvement of the material	129
23 Bayesian decision theory, loss function, minimizing expected loss	129
24 Markov Chain Monte Carlo (MCMC) simulations with applications	129
25 Application in Machine Learning	129
26 Notes on R and additional reading	129

27 Bayesian Linear regression	130
27.1 Convolution of Random Variables	130
27.2 Additive Noise Model	130
27.3 Likelihood for a Sample	131
27.4 Bayesian Linear Regression	131
27.4.1 One-Dimensional Regression ($p = 1$)	131
27.4.2 Multivariate Regression ($p > 1$)	131
27.5 Connection to Ridge and Lasso	131
27.6 Summary	132

1 Fundamental concepts in Probability

In this section, we present the core concepts of probability theory in a rigorous manner using formal definitions, examples, and illustrative figures.

Definition 1.1 (Probability space). A *probability space* is a triple $(\Omega, \mathcal{F}, P)$, where:

- Ω is the *sample space*, the set of all possible outcomes.
- $\mathcal{F}$ is a σ -algebra of subsets of Ω , whose elements are called *events*.
- $P : \mathcal{F} \rightarrow [0, 1]$ is a *probability measure* satisfying:
 1. $P(\Omega) = 1$,
 2. $P(A) \geq 0$ for all $A \in \mathcal{F}$,
 3. (Countable additivity) For any countable collection of disjoint events $\{A_i\}_{i=1}^{\infty} \subset \mathcal{F}$,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

Example 1.2. Consider a fair coin toss. Here, $\Omega = \{\text{Heads}, \text{Tails}\}$, $\mathcal{F}$ can be taken as the power set of Ω , and the probability measure is defined by $P(\{\text{Heads}\}) = P(\{\text{Tails}\}) = 0.5$.

□

Basic properties of probability

1. **Non-negativity:** For any event A ,

$$P(A) \geq 0.$$

2. **Normalization:** The probability of the entire sample space is 1:

$$P(\Omega) = 1.$$

3. **Monotonicity:** If $A \subseteq B$, then

$$P(A) \leq P(B).$$

4. **Complement Rule:** For any event A , the probability of its complement is given by

$$P(A^c) = 1 - P(A).$$

Additive properties of probability

1. **Finite Additivity (Inclusion-Exclusion):** For any two events A and B ,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

2. **Countable Additivity:** If $\{A_i\}_{i=1}^{\infty}$ is a countable collection of pairwise disjoint events, then

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

Definition 1.3 (Conditional probability). Given two events A and B with $P(B) > 0$, the *conditional probability* of A given B is defined as

$$P(A | B) = \frac{P(A \cap B)}{P(B)}.$$

This definition quantifies the probability of event A occurring under the assumption that event B has occurred.

Example 1.4. Consider drawing a card from a standard deck of 52 cards. Let A be the event “the card is a queen” and B be the event “the card is a heart”. Since there is only one queen of hearts,

$$P(A \cap B) = \frac{1}{52}, \quad P(B) = \frac{13}{52},$$

and thus

$$P(A | B) = \frac{\frac{1}{52}}{\frac{13}{52}} = \frac{1}{13}.$$

□

Definition 1.5 (Independence of events). Two events A and B are said to be *independent* if the occurrence of one does not affect the probability of the other. Formally, A and B are independent if

$$P(A \cap B) = P(A)P(B).$$

An equivalent condition is that if $P(B) > 0$, then

$$P(A | B) = P(A).$$

Example 1.6. Consider two independent tosses of a fair coin. Let A be the event “the first toss results in Heads” and B the event “the second toss results in Heads”. Since

$$P(A) = \frac{1}{2}, \quad P(B) = \frac{1}{2}, \quad \text{and} \quad P(A \cap B) = \frac{1}{4},$$

we have $P(A \cap B) = P(A)P(B)$, confirming that the two events are independent.

□

Properties of probability involving independence

1. **Independence of complements:** If A and B are independent, then:

$$P(A \cap B^c) = P(A)P(B^c), \quad P(A^c \cap B) = P(A^c)P(B),$$

and

$$P(A^c \cap B^c) = P(A^c)P(B^c).$$

Example: For independent events A and B , we have

$$P(A \cap B^c) = P(A) - P(A \cap B) = P(A) - P(A)P(B) = P(A)(1 - P(B)) = P(A)P(B^c).$$

Properties involving conditional probability

1. **Multiplication rule:** For any two events A and B with $P(A) > 0$,

$$P(A \cap B) = P(A) P(B | A).$$

Equivalently, if $P(B) > 0$,

$$P(A \cap B) = P(B) P(A | B).$$

2. **Chain rule:** For events $A_1, A_2, \dots, A_n$ (with the necessary probabilities positive), the joint probability is given by

$$P\left(\bigcap_{i=1}^n A_i\right) = P(A_1) P(A_2 | A_1) P(A_3 | A_1 \cap A_2) \cdots P(A_n | A_1 \cap \cdots \cap A_{n-1}).$$

Bayes' Formula and the Law of total Probability

Definition 1.7 (Law of total Probability). Let $\{H_1, H_2, \dots, H_n\}$ be a partition of the sample space Ω i.e. $H_i \cap H_j = \emptyset$ for $i \neq j$ and $\bigcup_i H_i = \Omega$, such that $P(H_i) > 0$ for each i . Then, for any event A ,

$$P(A) = \sum_{i=1}^n P(A | H_i) P(H_i).$$

Definition 1.8 (Bayes' Formula). For events A and B with $P(A) > 0$,

$$P(B | A) = \frac{P(A | B)P(B)}{P(A)}.$$

More generally, if $\{B_i\}$ is a partition of Ω then for any event A with $P(A) > 0$,

$$P(H_j | A) = \frac{P(A | H_j)P(H_j)}{\sum_i P(A | H_i)P(H_i)}.$$

Example 1.9. Suppose a diagnostic test for a disease is 99% accurate. Let D denote the event that a patient has the disease and T the event that the test is positive. Assume the disease prevalence is $P(D) = 0.01$. Then, if

$$P(T | D) = 0.99 \quad \text{and} \quad P(T | D^c) = 0.01,$$

the law of total probability gives

$$P(T) = P(T | D)P(D) + P(T | D^c)P(D^c) = 0.99 \times 0.01 + 0.01 \times 0.99 = 0.0099 + 0.0099 = 0.0198.$$

□

Example 1.10. Continuing with the diagnostic test scenario, we wish to compute the probability that a patient has the disease given a positive test result. Denote D as the event “patient has the disease” and T as the event “test is positive”. Using Bayes' formula,

$$P(D | T) = \frac{P(T | D)P(D)}{P(T)} = \frac{0.99 \times 0.01}{0.0198} \approx 0.5.$$

Thus, despite the high accuracy of the test, the probability that a patient has the disease given a positive result is only about 50%, primarily due to the low prevalence of the disease. □

1.1 Random variables

Definition 1.11 (Random variable). A *random variable* is a measurable function

$$X : \Omega \rightarrow \mathbb{R},$$

which assigns a real number to each outcome in the sample space.

Definition 1.12 (Discrete random variable and probability mass Function). A random variable X is *discrete* if the set of all possible values of X , R_X is at most countable. Its distribution is characterized by a *probability mass function (PMF)* p_X defined by

$$p_X(x) = P(X = x), \quad \text{for } x \in R_X$$

with the property that

$$\sum_{x \in R_X} p_X(x) = 1.$$

Example 1.13. Let X be the number of heads in two tosses of a fair coin. Then, X takes values in $\{0, 1, 2\}$ with

$$P(X = 0) = \frac{1}{4}, \quad P(X = 1) = \frac{1}{2}, \quad P(X = 2) = \frac{1}{4}.$$

□

Definition 1.14 (Continuous random variable and density function). A random variable X is *continuous* if its distribution is absolutely continuous with respect to the Lebesgue measure. In this case, there exists a *density function* p_X such that for any Borel set $B \subseteq \mathbb{R}$,

$$P(X \in B) = \int_B p_X(x) dx,$$

and

$$\int_{-\infty}^{\infty} p_X(x) dx = 1.$$

Example 1.15. If X is uniformly distributed on the interval $[0, 1]$, then

$$p_X(x) = \begin{cases} 1, & \text{if } x \in [0, 1], \\ 0, & \text{otherwise.} \end{cases}$$

□

Remark. The probability mass function in the discrete case and the probability density function in the continuous case are not the same nor do they have the same properties. Still we will often use the same notation $p_X(x)$ or simply $p(x)$ and the meaning can be understood from the context. Other notation for the density in this lecture notes is $f(x)$.

Definition 1.16 (Cumulative distribution function). For any random variable X , the *cumulative distribution function (CDF)* F_X is defined as

$$F_X(x) = P(X \leq x).$$

The CDF is non-decreasing, right-continuous, and satisfies

$$\lim_{x \rightarrow -\infty} F_X(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow \infty} F_X(x) = 1.$$

For a continuous random variable, the relationship between the density p_X and the CDF is given by

$$F_X(x) = \int_{-\infty}^x p_X(t) dt,$$

and for a discrete random variable,

$$F_X(x) = \sum_{t \in R_X, t \leq x} p_X(t).$$

2 Discrete distributions

2.1 Bernouli distribution

Bernoulli Distribution	
What it models	Binary outcomes (success/failure, 1/0).
Parameters	p
Range of Values	$\{0, 1\}$
Expectation and Variance	$E[X] = p, \quad Var[X] = p(1 - p)$
R Commands	Random Generation: <code>rbinom(n, size = 1, prob = p)</code> PMF: <code>dbinom(x, size = 1, prob = p)</code> CDF: <code>pbinom(x, size = 1, prob = p)</code> Quantiles: <code>qbinom(prob, size = 1, prob = p)</code>

Table 1: Bernoulli Distribution

2.2 Binomial distribution

Binomial Distribution	
What it models	Number of successes in n independent Bernoulli trials.
Parameters	n (number of trials), p (success probability in each trial)
Range of Values	$\{0, 1, \dots, n\}$
Probability Mass Function (PMF)	$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n.$
Expectation and Variance	$E[X] = np, \quad \text{Var}(X) = np(1 - p)$
Important Characteristics	• Skewness depends on p: Symmetric if $p = 0.5$, right-skewed if $p < 0.5$, and left-skewed if $p > 0.5$. • The distribution becomes approximately normal for large n (Central Limit Theorem). • Discrete distribution with bounded support $[0, n]$.
R Commands	Random Generation: <code>rbinom(n, size = n, prob = p)</code> PMF: <code>dbinom(x, size = n, prob = p)</code> CDF: <code>pbinom(x, size = n, prob = p)</code> Quantiles: <code>qbinom(prob, size = n, prob = p)</code>

Table 2: Binomial Distribution

Example 2.1. In a factory each batch of products consists of 20 items. Historically, about 30% of items are defective. Thus, the number of defective items per batch can be modeled by a Binomial distribution $X \sim \text{Binomial}(20, 0.3)$. The mean of this distribution is $EX = np = 20 \cdot 0.3 = 6$, and its variance is $DX = np(1 - p) = 20 \cdot 0.3 \cdot 0.7 = 4.2$.

To illustrate the model, we simulate data for 15 batches (a small sample) and 100 batches (a larger sample). We then compare the distribution of simulated defective counts with the theoretical PMF.

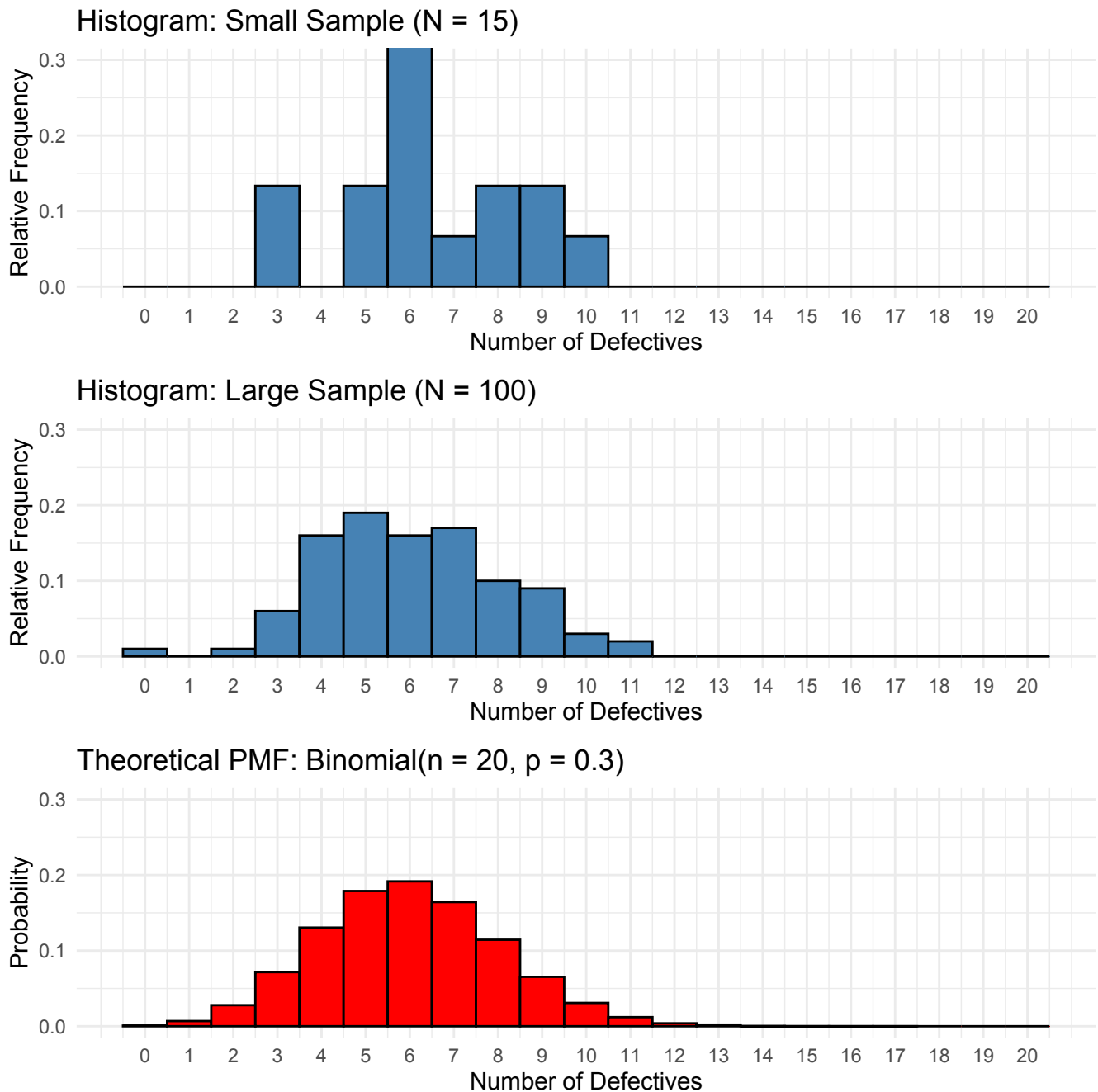


Figure 1: Example for binomial distribution

□

Example 2.2. (Exampe from finance)

In a credit portfolio, suppose a bank lends to 50 borrowers and each borrower has a 10% probability of default. Assuming defaults occur independently, the total number of defaults in the portfolio follows a Binomial distribution, $X \sim \text{Binomial}(n=50, p=0.1)$.

The mean number of defaults is $EX = 50 \cdot 0.1 = 5$ and the variance is $DX = 50 \cdot 0.1 \cdot 0.9 = 4.5$. This model helps assess the risk of loss due to defaults.

□

Example 2.3. (Example from medicine)

In a clinical trial 30 patients are treated with a new treatment. For each of the patients there is

70% chance of success. Then, the number of patients who respond positively to the treatment follows X Binomial (n=30, p=0.7). $\square$

Example 2.4. Modeling successful packet transmissions

Consider a scenario in a computer network where a server sends 20 packets during a given interval, and each packet is successfully transmitted with probability 95%. The number of successful transmissions can be modeled by X Binomial (n=20, p=0.95). $\square$

Example 2.5. Risk of cystic fibrosis in children

Cystic fibrosis (CF) is an autosomal recessive disorder. For a child to be affected by CF, they must inherit two copies of the mutated gene (one from each parent). If both parents are carriers of the mutated allele, then according to Mendelian genetics each child has the following probabilities:

- 25% chance of inheriting two mutated alleles and being affected by CF,
- 50% chance of being a carrier (inheriting one mutated allele) but remaining unaffected,
- 25% chance of inheriting two normal alleles and not being a carrier.

Since only the 25% outcome results in a child affected by CF, we can model the number of affected children in a family with n children as a Binomial random variable, i.e. $X \sim \text{Binomial}(n, 0.25)$.

Let us consider a family with 4 children, where both parents are carriers of the CF gene. Then the number of children with CF, X , is modeled as $X \sim \text{Binomial}(4, 0.25)$.

This model allows us to answer questions such as: - What is the probability that exactly 1 of the 4 children is affected by CF?

$$P(X = 1) = \binom{4}{1} (0.25)^1 (0.75)^3.$$

- What is the expected number of affected children?

$$E[X] = 4 \times 0.25 = 1.$$

- What is the variance in the number of affected children?

$$\text{Var}(X) = 4 \times 0.25 \times 0.75 = 0.75.$$

To better understand the problem it is always good to simulate some data, to get a better sense of the problem $\square$

2.3 Poisson distribution

Poisson Distribution	
What it models	Number of events occurring in a fixed interval of time or space when events occur at a constant mean rate and independently of the previous events.
Parameters	$\lambda > 0$ (mean number of occurrences in the interval)
Range of Values	$\{0, 1, 2, \dots\}$ (non-negative integers)
Probability Mass Function (PMF)	$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k = 0, 1, 2, \dots$
Expectation and Variance	$E[X] = \lambda, \quad \text{Var}(X) = \lambda$
Important Characteristics	• Skewed for small λ; becomes more symmetric as λ increases. • The variance equals the mean, which is a defining property of the Poisson distribution. • A limiting case of the Binomial distribution when $n \rightarrow \infty$ and $p \rightarrow 0$ such that $np = \lambda$. • Discrete distribution with a probability mass concentrated around λ.
R Commands	Random Generation: <code>rpois(n, lambda = λ)</code> PMF: <code>dpois(x, lambda = λ)</code> CDF: <code>ppois(x, lambda = λ)</code> Quantiles: <code>qpois(prob, lambda = λ)</code>

Table 3: Poisson Distribution

Example 2.6. Modeling the number of trades per minute with a Poisson distribution

In high-frequency trading, the number of trades executed in a given minute can be modeled by a Poisson distribution. Suppose on average 100 trades occur per minute. This means the number of trades X in one minute follows a Poisson distribution with parameter $\lambda = 100$. The mean and variance of X are both 100.

We present the results from simulations of the model with $\lambda = 2$ and $\lambda = 100$. See which values of the random variable will be more probable.

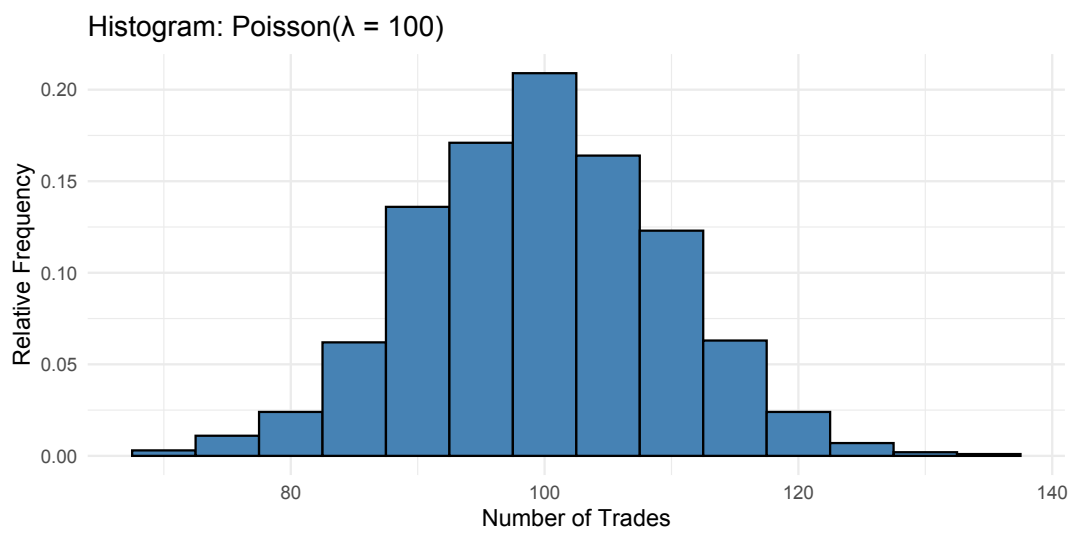
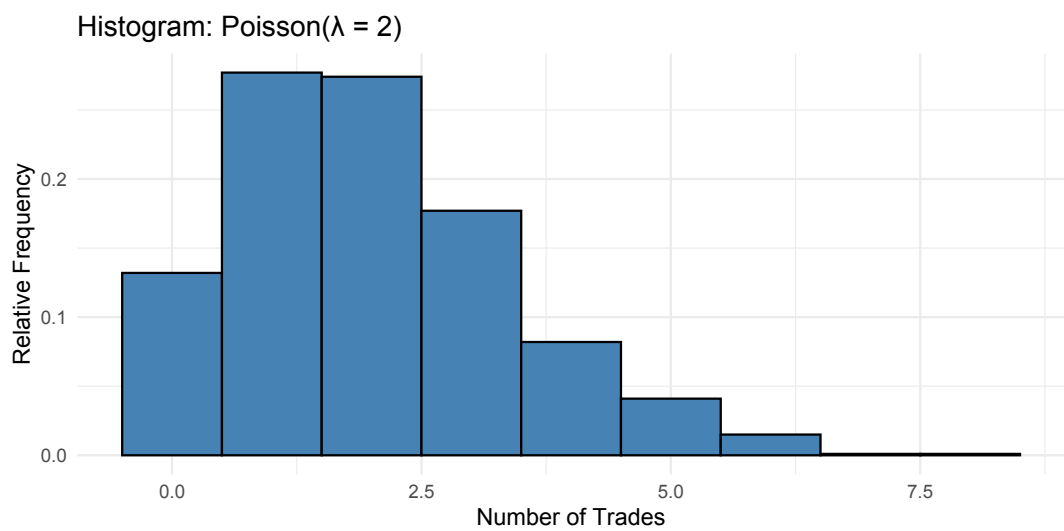


Figure 2: Data simulated from Poisson distribution

□

2.4 Geometric distribution

Geometric Distribution	
What it models	Number of trials until the first success (or equivalently, the number of failures before the first success). Remark: R uses support $\{0, 1, 2, \dots\}$.
Parameters	p (probability of success per trial, $0 < p \leq 1$)
Range of Values	$\{1, 2, 3, \dots\}$ or $\{0, 1, 2, \dots\}$ (depending on convention)
Probability Mass Function (PMF)	$P(X = k) = (1 - p)^{k-1}p, \quad k = 1, 2, 3, \dots$ $P(X = k) = (1 - p)^k p, \quad k = 0, 1, 2, \dots \quad (\text{R convention})$
Expectation and Variance	$E[X] = 1/p$ (if support is $\{1, 2, \dots\}$) $E[X] = (1 - p)/p$ (if support is $\{0, 1, 2, \dots\}$) $\text{Var}(X) = (1 - p)/p^2$ (both cases)
Important Characteristics	• Always positively skewed, with a long right tail. • The mean depends on p: The lower p, the larger the expected number of trials. • Memoryless property: If X is the number of trials until the first success, then after any number of failures, the distribution of the remaining number of trials is unchanged. (See later remark on memoryless property.) • Special case of the Negative Binomial Distribution with $r = 1$.
R Commands	Random Generation: <code>rgeom(n, prob = p)</code> PMF: <code>dgeom(x, prob = p)</code> CDF: <code>pgeom(x, prob = p)</code> Quantiles: <code>qgeom(prob, prob = p)</code>

Table 4: Geometric Distribution

Example 2.7. Modeling packet transmission attempts with a Geometric distribution

In computer networks, a packet may need several attempts to be transmitted successfully. Suppose the probability of successful transmission in a single attempt is $p=0.2$. The number of attempts X until the first success can be modeled by a Geometric distribution and when defined like this the range of all possible values are $1, 2, \dots$. We can also count the number of failures before the first success, and this is the way geometric distribution is implemented in R. In this case all possible values of X are $0, 1, 2, \dots$.

In our example, the mean number of failures is $\frac{1-p}{p} = \frac{4(1-p)}{p} = 4$, so on average 5 attempts (if adding the successful attempt) are needed.

Simulate a larger data set with different values for p ($p=0.2$, $p=0.5$, $p=0.8$), analyze the histogram and the characteristic values.

□

2.5 Negative binomial distribution

Negative Binomial Distribution	
What it models	Number of failures until achieving r successes in independent Bernoulli trials.
Parameters	r (number of successes), p (probability of success per trial, $0 < p \leq 1$)
Range of Values	$\{0, 1, 2, \dots\}$ (number of failures before the r^{th} success)
Probability Mass Function (PMF)	$P(X = k) = \binom{k + r - 1}{k} p^r (1 - p)^k, \quad k = 0, 1, 2, \dots$
Expectation and Variance	$E[X] = \frac{r(1 - p)}{p}, \quad \text{Var}(X) = \frac{r(1 - p)}{p^2}$ Note: These formulas assume X counts the number of failures before the r^{th} success.
Important Characteristics	• Always positively skewed, with a long right tail. • Reduces to the geometric distribution when $r = 1$. • Overdispersed relative to the Poisson distribution ($\text{Var}(X) > E[X]$), making it useful for modeling count data with extra variance. • Models real-world scenarios where the number of trials required to observe r successes varies, such as modeling customer purchases before a user churns or biological mutation counts.
R Commands	Random Generation: <code>rnbinom(n, size = r, prob = p)</code> PMF: <code>dnbinom(x, size = r, prob = p)</code> CDF: <code>pnbinom(x, size = r, prob = p)</code> Quantiles: <code>qnbinom(prob, size = r, prob = p)</code>

Table 5: Negative Binomial Distribution

In the classic formulation, the Negative Binomial random variable X represents the number of **failures** before the r -th success is achieved. In this case, the total number of trials is $X + r$. Alternatively, some sources define the Negative Binomial distribution as the total number of trials needed to achieve r successes. Here, the variable would be $Y = X + r$, where X is defined as above.

In many statistical software packages (for example, R's `rnbinom()` function), the distribution is parameterized in terms of the **number of failures** before the r -th success. In that setting, the mean and variance are given by: $EX = \frac{r(1-p)}{p}$, $DX = \frac{r(1-p)}{p^2}$, where p is the probability of success in each trial.

Example 2.8. Modeling sales calls with the Negative binomial distribution

Salesperson makes cold calls to potential clients - the salesperson contacts different potential customers, often chosen from a list of leads, who haven't been previously engaged or expressed interest in the product

or service. Each call results in a successful sale with probability $p = 0.2$, independent of other calls. The salesperson continues making calls until achieving $r = 3$ successful sales.

The number of unsuccessful calls (failures) made before the third successful sale is modeled by a Negative Binomial distribution:

$$X \sim \text{NegBin}(r = 3, p = 0.2),$$

where X counts the number of failures before the 3rd success.

The expected number of failures is given by

$$E[X] = \frac{r(1-p)}{p} = \frac{3(1-0.2)}{0.2} = \frac{3 \times 0.8}{0.2} = 12.$$

On average, the salesperson is expected to make 12 unsuccessful calls before achieving 3 sales.

The variance in the number of failures is

$$\text{Var}(X) = \frac{r(1-p)}{p^2} = \frac{3(0.8)}{(0.2)^2} = \frac{2.4}{0.04} = 60.$$

This indicates a high degree of variability in the number of calls required.

The mean of 12 unsuccessful calls (failures) suggests that, in addition to the 3 successful calls, about 15 calls are needed on average to reach the target. The variance of 60 reflects substantial variability, meaning that while the average might be 15 calls, in practice the number could be considerably higher or lower.

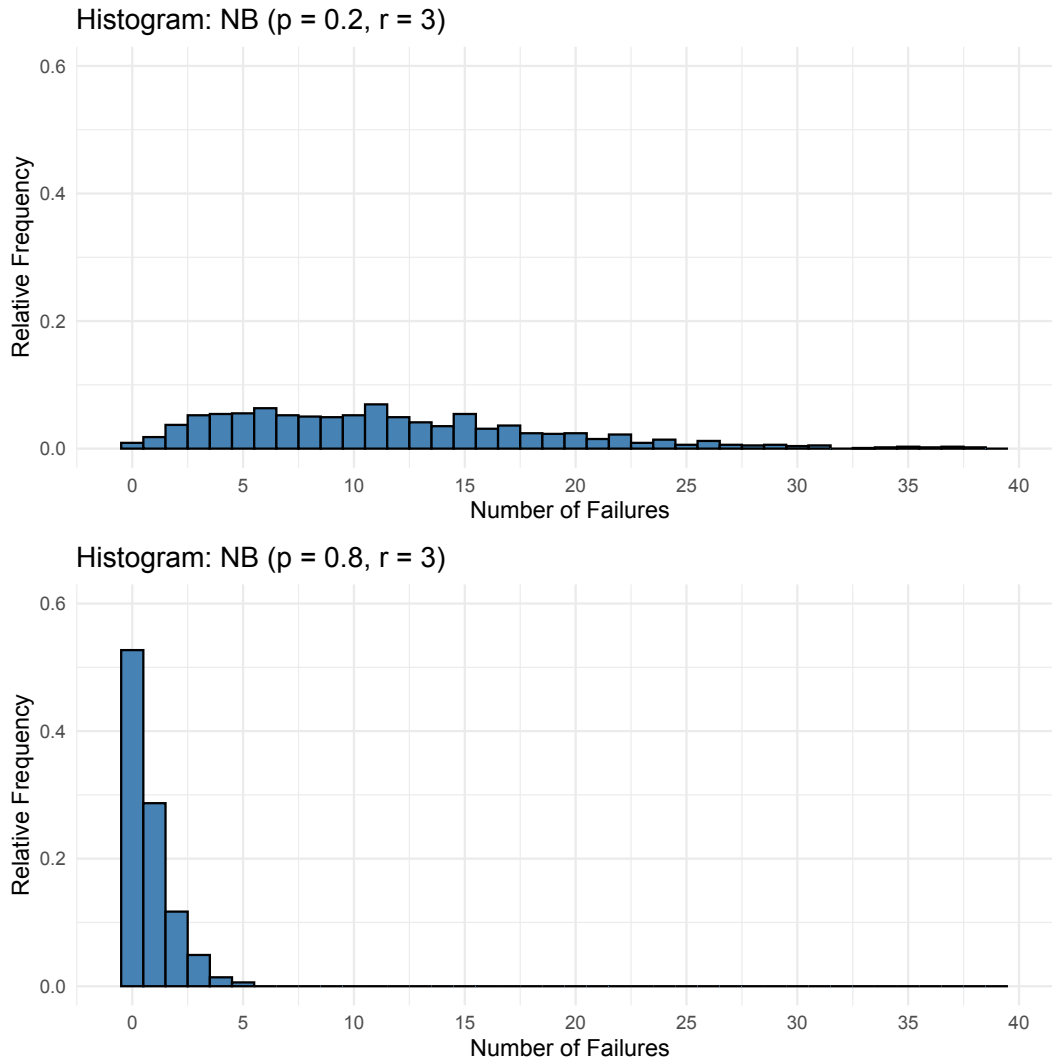


Figure 3: Data simulated from Negative binomial distribution

2.6 Hypergeometric distribution

Hypergeometric Distribution	
What it models	Number of successes in n draws without replacement from a population of size N containing K successes.
Parameters	N (population size), K (number of successes in population), n (sample size)
Range of Values	$\{\max(0, n - (N - K)), \dots, \min(n, K)\}$
Probability Mass Function (PMF)	$P(X = k) = \frac{\binom{K}{k} \binom{N-K}{n-k}}{\binom{N}{n}}, \quad \max(0, n - (N - K)) \leq k \leq \min(n, K).$
Expectation and Variance	$E[X] = n \frac{K}{N}, \quad \text{Var}(X) = n \frac{K}{N} \frac{N - K}{N} \frac{N - n}{N - 1}$
Important Characteristics	• Unlike the Binomial distribution, sampling is done without replacement, meaning trials are not independent. • When the sample size n is much smaller than N, the Hypergeometric distribution approximates a Binomial distribution with parameters n and K/N. • Used in finite population settings, such as quality control (defective vs. non-defective items) and card games.
R Commands	Random Generation: <code>rhyper(nn, m = K, n = N-K, k = n)</code> PMF: <code>dhyper(x, m = K, n = N-K, k = n)</code> CDF: <code>phyper(x, m = K, n = N-K, k = n)</code> Quantiles: <code>qhyper(prob, m = K, n = N-K, k = n)</code>

Table 6: Hypergeometric Distribution

3 Continuous Distributions

3.1 Uniform Distribution (Continuous)

The continuous Uniform distribution assigns equal probability to all values in an interval $[a, b]$.

Uniform Distribution	
What it models	Continuous variables that are equally likely to take any value within a specified interval $[a, b]$.
Parameters	a (minimum), b (maximum).
Range of Values	$[a, b]$.
Probability Density Function (PDF)	$f(x) = \frac{1}{b-a}, \quad a \leq x \leq b.$
Expectation and Variance	$E[X] = \frac{a+b}{2}, \quad \text{Var}(X) = \frac{(b-a)^2}{12}.$
Important Characteristics	• The distribution is symmetric and has constant probability density over $[a, b]$. • No skewness since every value in the interval is equally likely. • The kurtosis is $-\frac{6}{5}$, which is lower than the normal distribution, indicating a flat shape. • The cumulative distribution function (CDF) is a linear function. • Often used as a reference distribution for generating random numbers and for modeling cases where no prior information favors certain values over others.
R Commands	Random Generation: <code>runif(n, min = a, max = b)</code> Density: <code>dunif(x, min = a, max = b)</code> CDF: <code>punif(x, min = a, max = b)</code> Quantiles: <code>qunif(prob, min = a, max = b)</code>

Table 7: Uniform Distribution

Example 3.1. Tolerances in manufacturing

In many manufacturing processes, machines are calibrated to produce parts within a specific tolerance range. For instance, suppose a machine is designed to produce metal rods that are exactly 10.00 cm long, but due to unavoidable small variations in the production process, the actual length of each rod is equally likely to fall anywhere between 9.95 cm and 10.05 cm.

In this case, it is reasonable to assume that the measurement error (or the variation in rod lengths) is uniformly distributed over the interval $[9.95, 10.05]$. That is, any length in this interval is equally likely. The Uniform distribution is ideal for this scenario because there is no reason to believe that one error value is more likely than another within the specified tolerance range.

Parameters - $a = 9.95$ and $b = 10.05$.

Range of Values - $[9.95, 10.05]$.

Expectation

$$EX = \frac{a+b}{2} = \frac{9.95 + 10.05}{2} = 10.00 \text{ cm.}$$

Variance

$$\text{Var}(X) = \frac{(b - a)^2}{12} = \frac{(10.05 - 9.95)^2}{12} = \frac{(0.10)^2}{12} \approx 0.00083 \text{ cm}^2.$$

□

3.2 Exponential Distribution

The Exponential distribution models the time between events in a Poisson process.

Exponential Distribution	
What it models	Waiting times between independent events in a Poisson process.
Parameters	λ (rate parameter).
Range of Values	$[0, \infty)$.
Probability Density Function (PDF)	$f(x) = \lambda e^{-\lambda x}, \quad x \geq 0.$
Expectation and Variance	$E[X] = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}.$
Important Characteristics	- The exponential distribution is memoryless: $P(X > s + t \mid X > s) = P(X > t).$ - The skewness is always 2, making it highly right-skewed. - The kurtosis is 6, which is larger than the normal distribution, indicating heavy tails. - It is a special case of the Gamma distribution with shape parameter $k = 1$. - Used extensively in reliability analysis, survival studies, and queueing theory.
R Commands	Random Generation: <code>rexp(n, rate = λ)</code> Density: <code>dexp(x, rate = λ)</code> CDF: <code>pexp(x, rate = λ)</code> Quantiles: <code>qexp(prob, rate = λ)</code>

Table 8: Exponential Distribution

Skewness quantifies the asymmetry of a distribution around its mean. A positive value indicates a longer or heavier right tail, whereas a negative value indicates a longer or heavier left tail. Mathematically, skewness is defined as the third standardized central moment.

Kurtosis describes the tail heaviness of a distribution and its propensity to generate extreme values compared with a normal distribution. Larger kurtosis values indicate heavier tails and a greater likelihood

of outliers. Mathematically, kurtosis is defined as the fourth standardized central moment.

In many practical scenarios, waiting times between random events can be modeled by the Exponential distribution, particularly when the process is memoryless.

Example 3.2. Waiting time for an elevator

Suppose that in an office building, the time between elevator arrivals is modeled as an Exponential random variable. Assume that, on average, an elevator arrives every 2 minutes. This implies that the mean waiting time is 2 minutes. Let X denote the waiting time (in minutes) for an elevator. Then,

$$X \sim \text{Exponential}(\lambda),$$

where the rate parameter λ is defined as

$$\lambda = \frac{1}{\text{Mean waiting time}} = \frac{1}{2} \text{ per minute.}$$

The probability density function (pdf) is given by:

$$p(x) = \lambda e^{-\lambda x} = \frac{1}{2} e^{-x/2}, \quad x \geq 0.$$

We can ask the following question - What is the probability that a person waits less than 3 minutes for the elevator?

The cumulative distribution function (CDF) for the Exponential distribution is:

$$F(x) = 1 - e^{-\lambda x}.$$

Thus, the probability of waiting less than 3 minutes is:

$$P(X < 3) = F(3) = 1 - e^{-\frac{1}{2} \cdot 3} = 1 - e^{-1.5} \approx 1 - 0.2231 = 0.7769.$$

Thus, there is approximately a 77.7% chance that an individual will wait less than 3 minutes for an elevator under the assumption that the waiting times follow an Exponential distribution with a mean of 2 minutes.

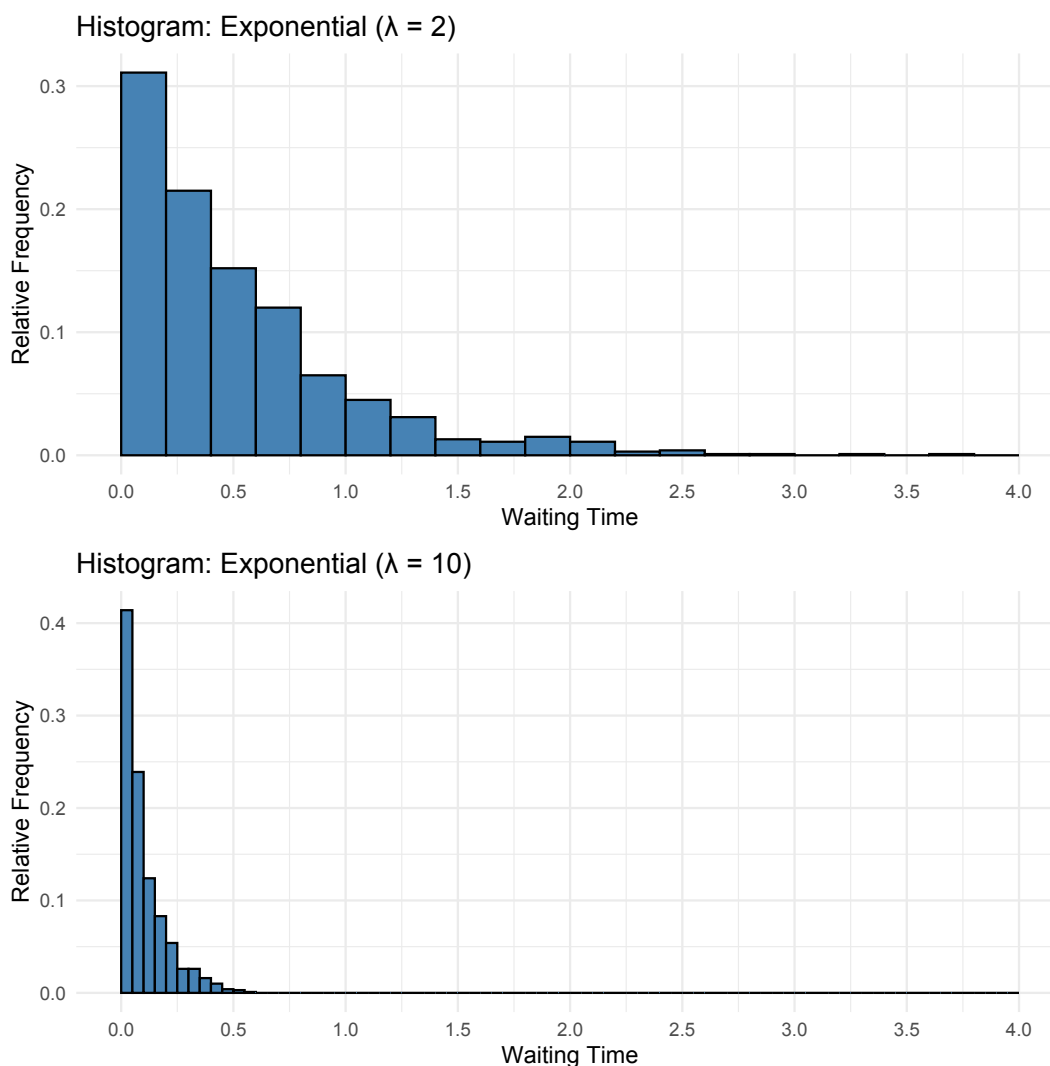


Figure 4: Data simulated from Exponential distribution

□

Remark. Memoryless property

A probability distribution is said to be *memoryless* if the probability of an event occurring in the future is independent of the past. In other words, the distribution "forgets" how much time has already elapsed. More specific, given that the event has not yet occurred by time s , the probability of waiting at least t additional time units is the same as it was at the beginning. Mathematically, if X is a random variable with the memoryless property, then for any $s, t \geq 0$:

$$P(X > s + t \mid X > s) = P(X > t).$$

This equation means that, regardless of how long you have already waited (time s), the probability that you will need to wait at least an additional t units of time is the same as the probability of waiting at least t units of time from the start.

The Exponential distribution is the only continuous distribution with this property (and the Geometric distribution is its discrete counterpart).

Example 3.3. Memoryless property Suppose X represents the waiting time (in minutes) for an elevator, and X follows an Exponential distribution with rate λ . Then, no matter how long you have already waited,

the probability that you must wait at least an additional 5 minutes is always:

$$P(X > s + 5 \mid X > s) = P(X > 5) = e^{-5\lambda}.$$

This illustrates the memoryless property of the Exponential distribution. $\square$

Remark. Exponential distribution can be generalised with Weibull and Gamma distribution. See also details on Erlang distribution

3.3 Normal Distribution

The Normal (or Gaussian) distribution is a foundational continuous distribution in statistics.

Normal Distribution	
What it models	Continuous data that tend to cluster around a central value with symmetric variations. Frequently used in natural and social sciences due to the Central Limit Theorem.
Parameters	μ (mean, location parameter), σ (standard deviation, scale parameter).
Range of Values	$(-\infty, \infty)$.
Probability Density Function (PDF)	$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$
Expectation and Variance	$E[X] = \mu, \quad \text{Var}(X) = \sigma^2.$
Important Characteristics	• The distribution is symmetric around the mean μ. • Skewness is 0, meaning it has no skew. • Kurtosis is 3, (equivalently, excess kurtosis is 0) indicating moderate tails compared to heavy-tailed distributions. • The sum of independent normal variables is also normally distributed. • Plays a central role in statistics due to the Central Limit Theorem.
R Commands	Random Generation: <code>rnorm(n, mean = μ, sd = σ)</code> Density: <code>dnorm(x, mean = μ, sd = σ)</code> CDF: <code>pnorm(x, mean = μ, sd = σ)</code> Quantiles: <code>qnorm(prob, mean = μ, sd = σ)</code>

Table 9: Normal Distribution

3.4 Students' t Distribution

Student's t-Distribution	
What it models	Continuous data where the population standard deviation is unknown, commonly used in hypothesis testing and confidence interval estimation for small sample sizes.
Parameters	ν (degrees of freedom, shape parameter).
Range of Values	$(-\infty, \infty)$.
Probability Density Function (PDF)	$f(x) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\Gamma\left(\frac{\nu}{2}\right)} \left(1 + \frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}, \quad x \in \mathbb{R}.$
Expectation and Variance	$E[X] = \begin{cases} 0, & \text{for } \nu > 1, \\ \text{undefined}, & \text{for } \nu \leq 1. \end{cases}$ $\text{Var}(X) = \begin{cases} \frac{\nu}{\nu-2}, & \text{for } \nu > 2, \\ \infty, & \text{for } 1 < \nu \leq 2, \\ \text{undefined}, & \text{for } \nu \leq 1. \end{cases}$
Important Characteristics	• The distribution is symmetric around 0. • As $\nu \rightarrow \infty$, the t-distribution approaches the standard normal distribution. • The tails are heavier than the normal distribution, making it more robust to outliers. • The mean is defined only for $\nu > 1$, and the variance is finite only for $\nu > 2$. • Commonly used in t-tests and Bayesian statistics.
R Commands	Random Generation: <code>rt(n, df = ν)</code> Density: <code>dt(x, df = ν)</code> CDF: <code>pt(x, df = ν)</code> Quantiles: <code>qt(prob, df = ν)</code>

Table 10: Student's t-Distribution

Remark. The Student's t-distribution with one degree of freedom ($\nu = 1$) is exactly the Cauchy distribution with location parameter 0 and scale parameter 1.

3.5 Beta Distribution

Beta Distribution	
What it models	Continuous data in the range $[0, 1]$, often used to model proportions, probabilities, and Bayesian priors.
Parameters	$\alpha > 0$ (shape parameter), $\beta > 0$ (shape parameter).
Range of Values	$[0, 1]$.
Probability Density Function (PDF)	$f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}, \quad 0 \leq x \leq 1,$ where $B(\alpha, \beta) = \int_0^1 t^{\alpha-1}(1-t)^{\beta-1}dt$ is the Beta function.
Expectation and Variance	$E[X] = \frac{\alpha}{\alpha + \beta}, \quad \text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}.$
Important Characteristics	• The distribution is bounded between 0 and 1. • The shape is flexible, allowing uniform, unimodal, and U-shaped distributions depending on α and β. • If $\alpha = \beta = 1$, the distribution is uniform on $[0, 1]$. • Used in Bayesian statistics as a conjugate prior for binomial and Bernoulli distributions. • Often applied in modeling probabilities, proportions, and reliability analysis.
R Commands	Random Generation: <code>rbeta(n, shape1 = α, shape2 = β)</code> Density: <code>dbeta(x, shape1 = α, shape2 = β)</code> CDF: <code>pbeta(x, shape1 = α, shape2 = β)</code> Quantiles: <code>qbeta(prob, shape1 = α, shape2 = β)</code>

Table 11: Beta Distribution

Shape and Skewness: α and β control the skewness and modality of the distribution.

- If $\alpha > \beta$, the distribution is skewed to the right (i.e., it favors values closer to 1).
- If $\alpha < \beta$, the distribution is skewed to the left (i.e., it favors values closer to 0).
- If $\alpha = \beta$, the distribution is symmetric around 0.5.

Example 3.4. Modeling conversion rates in digital marketing

In digital marketing, one of the key performance indicators is the *conversion rate*—the proportion of website visitors who take a desired action (e.g., make a purchase). Since a conversion rate is a proportion, its values naturally lie in the interval $[0, 1]$. When there is uncertainty or variability about this rate, it is common to model it with a *Beta distribution*.

Suppose a company launches a new online advertising campaign, and based on historical data and market research, they believe that the conversion rate is typically around 30% but with some uncertainty.

To capture this belief, the conversion rate X is modeled as a Beta random variable:

$$X \sim \text{Beta}(\alpha, \beta),$$

with parameters $\alpha = 3$ and $\beta = 7$.

The probability density function (pdf) of the Beta distribution is given by

$$p(x; \alpha, \beta) = \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}, \quad 0 \leq x \leq 1.$$

For our parameters:

- $\alpha = 3$ and $\beta = 7$,
- The **expected conversion rate** is

$$E[X] = \frac{\alpha}{\alpha + \beta} = \frac{3}{3 + 7} = 0.3.$$

- The **variance** is

$$\text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)} = \frac{3 \cdot 7}{10^2 \cdot 11} \approx 0.0191.$$

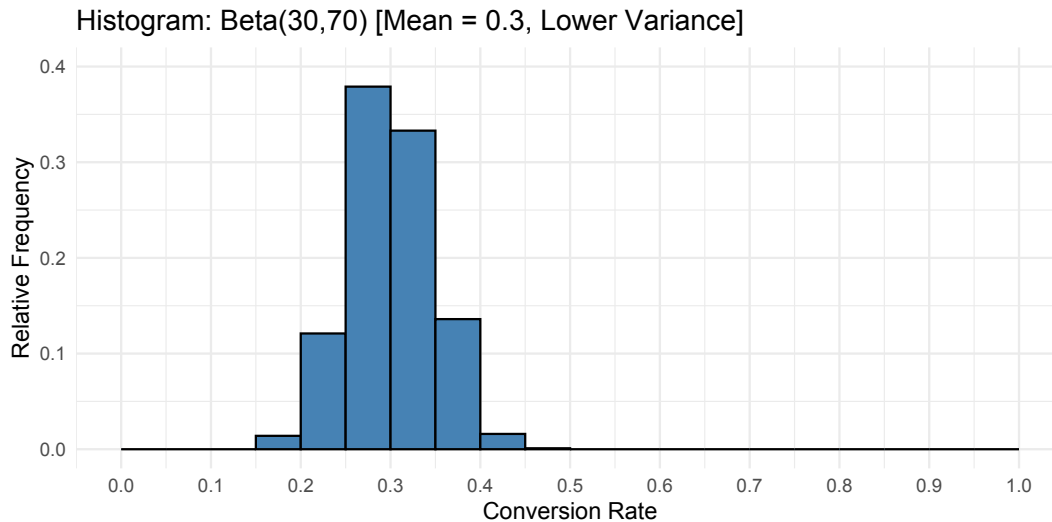
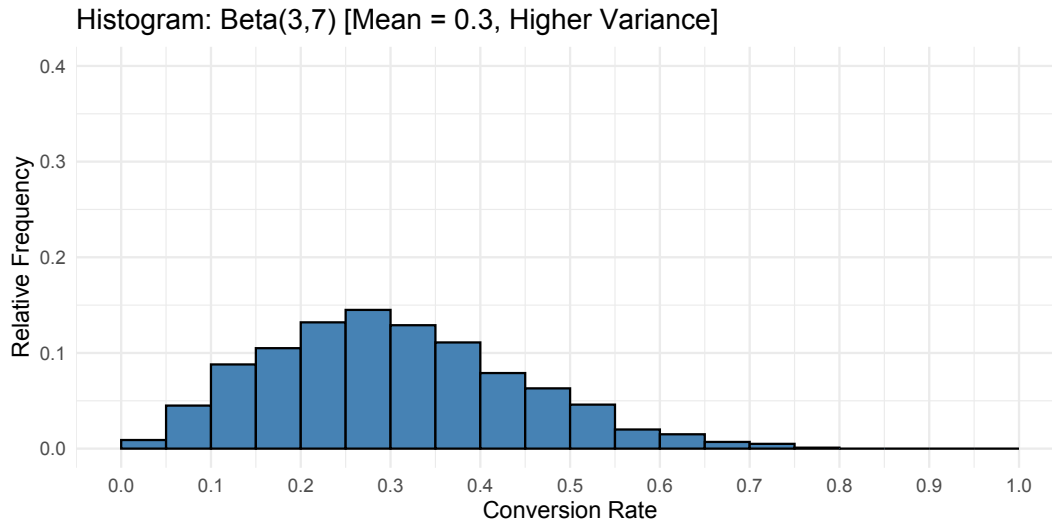


Figure 5: Data simulated from Exponential distribution

□

Modeling conversion rates with the Beta distribution is useful not only as a descriptive model but also as a prior in Bayesian analyses. It helps marketers quantify uncertainty and update their beliefs as more data become available.

In Bayesian statistics, the Beta distribution is commonly used as a conjugate prior for binomial proportions. In this context, $\alpha - 1$ can be interpreted as the prior number of successes, and $\beta - 1$ as the prior number of failures. This interpretation provides an intuitive way to update beliefs about a probability as new data are observed.

This will be a main discussion in the next chapters.

3.6 Gamma Distribution

The Gamma distribution is flexible and widely used to model waiting times and other positive-valued variables.

Gamma Distribution	
What it models	Continuous waiting times for the occurrence of k independent events in a Poisson process, generalizing the Exponential distribution. Often used in reliability analysis and Bayesian statistics.
Parameters	$\alpha > 0$ (shape parameter), $\beta > 0$ (rate parameter).
Range of Values	$[0, \infty)$.
Probability Density Function (PDF)	$f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \quad x > 0,$ where $\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t} dt$ is the Gamma function.
Expectation and Variance	$E[X] = \frac{\alpha}{\beta}, \quad \text{Var}(X) = \frac{\alpha}{\beta^2}.$
Important Characteristics	• The distribution is always non-negative and right-skewed. • If $\alpha = 1$, the Gamma distribution reduces to the Exponential distribution. • The sum of independent Gamma-distributed variables with the same rate parameter is also Gamma-distributed. • Used in Bayesian analysis as a conjugate prior for Poisson and Exponential likelihoods. • Common applications include modeling waiting times, rainfall distributions, and insurance claims.
R Commands	Random Generation: <code>rgamma(n, shape = α, rate = β)</code> Density: <code>dgamma(x, shape = α, rate = β)</code> CDF: <code>pgamma(x, shape = α, rate = β)</code> Quantiles: <code>qgamma(prob, shape = α, rate = β)</code>

Table 12: Gamma Distribution

The general gamma distribution is a two-parameter family of continuous probability distributions that generalizes the exponential and Erlang distributions. It is defined by a shape parameter and a rate (or scale) parameter.

Suppose that

$$X_1, X_2, \dots, X_n \sim \text{Exponential}(\lambda)$$

are i.i.d. Then their sum

$$T = X_1 + X_2 + \dots + X_n$$

follows a Gamma distribution:

$$T \sim \text{Gamma}(n, \lambda),$$

with shape parameter n and rate parameter λ . This is also known as **Erlang(n, λ) distribution**.

If instead the X_i are independent but not identically distributed, for example,

$$X_i \sim \text{Exponential}(\lambda_i) \quad (i = 1, 2, \dots, n),$$

then the sum

$$T = \sum_{i=1}^n X_i$$

follows a **hypoexponential distribution** (also known as the generalized Erlang distribution). Its Laplace transform is given by

$$\mathcal{L}_T(s) = \prod_{i=1}^n \frac{\lambda_i}{\lambda_i + s}.$$

In this case, the probability density function can be expressed via a partial fraction decomposition, though a simple closed form exists only in specific cases (such as for $n = 2$).

Example 3.5. Modeling total service time in a bank

Consider a busy bank where each customer's service time is assumed to be exponentially distributed with an average service time of 5 minutes. Let the service time for each customer be represented by an independent random variable X_i such that:

$$X_i \sim \text{Exponential}(\lambda),$$

with the rate parameter $\lambda = \frac{1}{5}$ per minute.

Suppose we are interested in the total time T required to serve 4 consecutive customers. Since the sum of independent exponential random variables follows a Gamma distribution, we have:

$$T = X_1 + X_2 + X_3 + X_4 \sim \text{Gamma}(\alpha, \beta),$$

where:

- $\alpha = 4$ is the shape parameter (the number of customers),
- $\beta = \frac{1}{5}$ is the rate parameter (the rate of service).

The probability density function (pdf) of T is:

$$p_T(t) = \frac{\beta^\alpha}{\Gamma(\alpha)} t^{\alpha-1} e^{-\beta t} = \frac{\left(\frac{1}{5}\right)^4}{\Gamma(4)} t^3 e^{-t/5}, \quad t \geq 0.$$

We can calculate the following

- **Expected total service time**

$$E[T] = \frac{\alpha}{\beta} = \frac{4}{1/5} = 20 \text{ minutes.}$$

- **Variance**

$$\text{Var}(T) = \frac{\alpha}{\beta^2} = \frac{4}{(1/5)^2} = 100 \text{ minutes}^2.$$

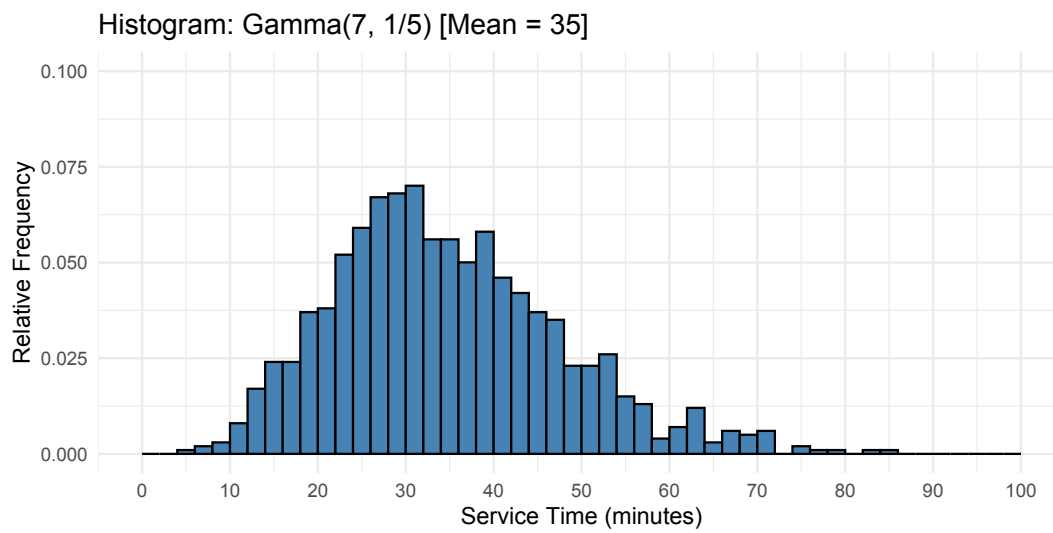
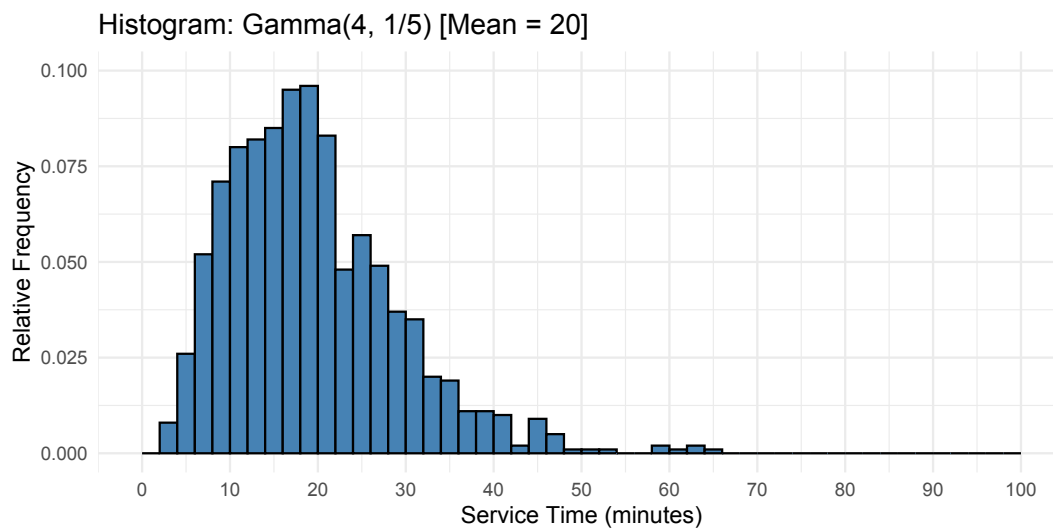


Figure 6: Data simulated from Gamma distribution

□

3.7 Logistic distribution

Logistic Distribution	
What it models	Often used to model binary outcomes (as in logistic regression), growth processes, diffusion phenomena, and saturation effects via an S-shaped (sigmoidal) CDF.
Parameters	μ (location parameter) and $s > 0$ (scale parameter).
Range of Values	$(-\infty, \infty)$.
PDF	$f(x; \mu, s) = \frac{e^{-(x-\mu)/s}}{s(1 + e^{-(x-\mu)/s})^2}, \quad x \in \mathbb{R}.$
CDF	$F(x; \mu, s) = \frac{1}{1 + e^{-(x-\mu)/s}}, \quad x \in \mathbb{R}.$
Mean	μ (by symmetry).
Variance	$\frac{s^2\pi^2}{3}.$
Skewness	0 (symmetric distribution).
Excess Kurtosis	≈ 1.2 (total kurtosis ≈ 4.2).
Tail Behavior	Heavier tails than the normal distribution.
R Commands	Random Generation: <code>rlogis(n, location = μ, scale = s)</code> Density: <code>dlogis(x, location = μ, scale = s)</code> CDF: <code>plogis(x, location = μ, scale = s)</code> Quantiles: <code>qlogis(prob, location = μ, scale = s)</code>

Table 13: Logistic Distribution

The logistic distribution is symmetric about its mean and exhibits heavier tails than the normal distribution. Its S-shaped cumulative distribution function makes it especially useful for modeling binary outcomes, which is the foundation of logistic regression. The distribution's location parameter μ sets its center, while the scale parameter s determines its spread. It is used to model processes that saturate over time, such as population growth or the diffusion of innovations.

Example 3.6. Application in logistic regression

Logistic regression is a method for modeling a binary outcome variable using one or more predictor variables. It is based on the logistic function, which is the CDF of the logistic distribution. For a binary response Y and predictors $X_1, X_2, \dots, X_k$, the model is expressed as:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 x_1 + \dots + \beta_k x_k,$$

where $p = P(Y = 1 \mid X)$. This can be rewritten as:

$$p = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_k x_k)}}.$$

The S-shaped curve of this function makes it ideal for modeling probabilities, ensuring that predicted values lie between 0 and 1. It ensures that small changes in the predictor values around the decision boundary can lead to significant changes in the predicted probability, which is particularly useful in classification tasks.

The Pima Indians Diabetes dataset is a well-known dataset that contains clinical measurements such as body mass index (BMI), blood pressure, and age, along with a binary indicator of diabetes diagnosis.

Logistic regression is used to estimate the probability of a positive diabetes diagnosis based on clinical features.

□

The logistic distribution also finds applications in modeling processes that exhibit saturation, rapid growth followed by stabilization, or S-shaped curves.

- **Population growth data** - many countries' population growth over time follows a logistic pattern, where initial exponential growth slows down as the population reaches a carrying capacity. A notable dataset is the *United Nations World Population Prospects* dataset, which provides historical and projected population figures for various countries. By fitting a logistic growth curve to these data, one can model the saturation effect due to limited resources.
- **Dose-response modeling** - In pharmacology and toxicology, dose-response curves often exhibit an S-shaped (sigmoidal) form. The logistic function is used to estimate the dose at which 50% of the maximal effect is observed, thereby modeling the saturation of the response at high doses. Thus the model captures the relationship between drug concentration and biological response.
- **Epidemic outbreak data** - the cumulative number of cases during an epidemic often follows a logistic growth curve. Early in the COVID-19 pandemic, researchers used logistic models to describe the cumulative number of cases as the infection spread rapidly and then slowed as interventions took effect. Popular datasets are those compiled by the *Johns Hopkins University Center for Systems Science and Engineering (CSSE)*

Example 3.7. Population growth In many real-life situations, processes exhibit saturation behavior, where growth starts slowly, increases rapidly during an exponential phase, and then levels off as it approaches a limit. A classic example is the modeling of population growth in a confined environment, where the number of individuals eventually reaches a carrying capacity. Such processes can be modeled using the logistic distribution.

To illustrate the density and the cdf, we simulate 100 observations from a logistic distribution with location parameter $\mu = 0$ and scale parameter $s = 1$.

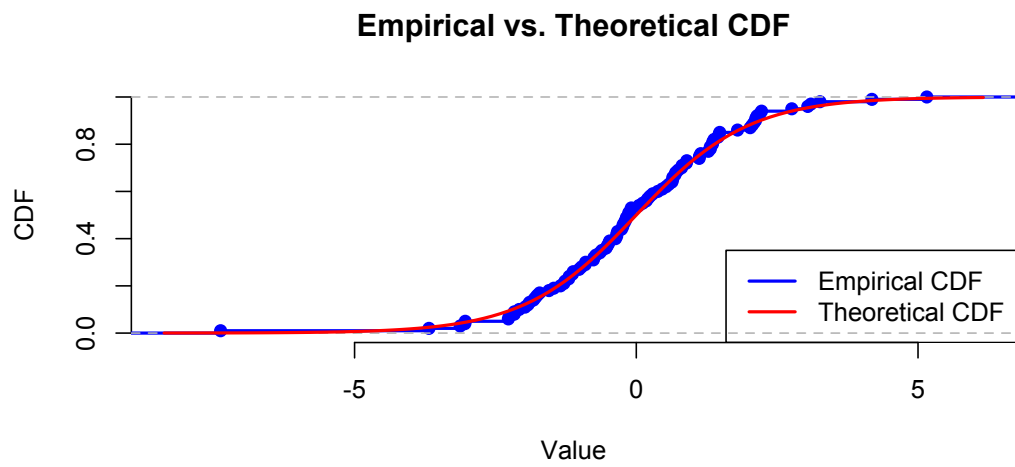
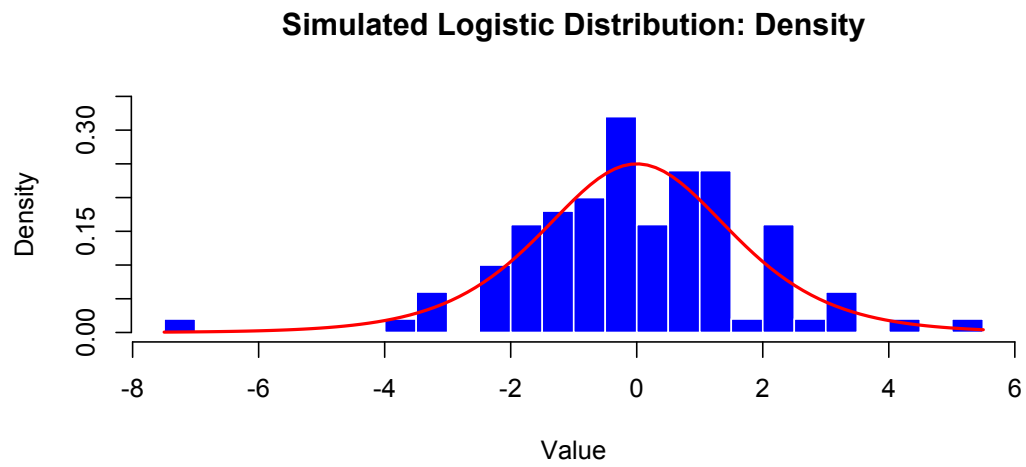


Figure 7: Data simulated from Logistic distribution

□

3.8 Weibull distribution

Weibull Distribution	
What it models	Widely used to model time-to-failure data in reliability engineering, as well as phenomena such as wind speed, material strength, and other lifetime data.
Parameters	Shape parameter k (sometimes denoted as β) and scale parameter λ (sometimes denoted as α).
Range of Values	$[0, \infty)$.
PDF	$f(x; k, \lambda) = \frac{k}{\lambda} \left(\frac{x}{\lambda}\right)^{k-1} e^{-(x/\lambda)^k}, \quad x \geq 0.$
CDF	$F(x; k, \lambda) = 1 - e^{-(x/\lambda)^k}, \quad x \geq 0.$
Expectation and Variance	$E[X] = \lambda \Gamma\left(1 + \frac{1}{k}\right), \quad \text{Var}(X) = \lambda^2 \left[\Gamma\left(1 + \frac{2}{k}\right) - \left(\Gamma\left(1 + \frac{1}{k}\right)\right)^2 \right].$
R Commands	Random Generation: <code>rweibull(n, shape = k, scale = lambda)</code> Density: <code>dweibull(x, shape = k, scale = lambda)</code> CDF: <code>pweibull(x, shape = k, scale = lambda)</code> Quantiles: <code>qweibull(prob, shape = k, scale = lambda)</code>

Table 14: Weibull Distribution

The Weibull distribution is a flexible lifetime distribution characterized by two parameters — a shape parameter and a scale parameter. It is used to model a wide range of failure behaviors. It is one of the most popular models in survival analysis. It is used to model survival times (or time-to-event data).

Depending on the shape parameter, it can model increasing, constant, or decreasing failure rates. When the shape parameter $k < 1$, it indicates a decreasing hazard rate (early failures); $k=1$ corresponds to the exponential distribution (constant hazard); and $k > 1$ implies an increasing hazard rate (wear-out failures).

The Weibull distribution's skewness, excess kurtosis, and tail behavior are all controlled by its shape parameter k . Lower values of k lead to a distribution with heavy tails, high skewness, and high excess kurtosis, while higher values of k result in a distribution that is more symmetric with lighter tails.

Some of many applications of Weibull distribution are:

- **Reliability engineering and life data analysis** - models time-to-failure of products and components using flexible hazard rates (increasing, constant, or decreasing), ideal for reliability testing.
- **Survival analysis** - captures various hazard functions to model survival times in medical research and actuarial studies.
- **Industrial engineering** - predicts maintenance schedules and warranty analysis by modeling the lifespan of mechanical and electronic systems.
- **Wind speed modeling** - describes wind speed distributions in renewable energy and meteorology, essential for assessing wind power potential.
- **Hydrology and environmental studies** - models phenomena like river flow rates and rainfall, accommodating both moderate and extreme events.
- **Material strength** - predicts the breaking strength of materials under stress, aiding in design and quality control.
- **Economics and finance** - Occasionally used to model time between events or economic variables when data exhibit non-constant hazard rates.

3.9 Laplace distribution

Laplace Distribution	
What it models	Continuous data with a sharp peak at the center and heavier tails than the normal distribution. Often used in robust statistics, signal processing, and econometrics.
Parameters	Location parameter μ and scale parameter $b > 0$.
Range of Values	$(-\infty, \infty)$.
Probability Density Function (PDF)	$f(x \mu, b) = \frac{1}{2b} \exp\left(-\frac{ x - \mu }{b}\right), \quad x \in \mathbb{R},$ where $\mu \in \mathbb{R}$ and $b > 0$.
Expectation and Variance	$E[X] = \mu, \quad \text{Var}(X) = 2b^2.$
Important Characteristics	• Symmetric about μ. • Exhibits heavier tails compared to the normal distribution. • Useful for modeling data with outliers or sharp peaks. • Also known as the Double Exponential distribution.
R Commands	Random Generation (using LaplacesDemon package): <code>rlaplace(n, location = μ, scale = b)</code> Density: <code>dlaplace(x, location = μ, scale = b)</code> CDF: <code>plaplace(x, location = μ, scale = b)</code> Quantiles: <code>qlaplace(prob, location = μ, scale = b)</code>

Table 15: Laplace Distribution

This distribution is a robust alternative to the normal distribution when the data exhibit sharper peaks (higher kurtosis) and heavier tails than a normal distribution. It is used in signal processing, financial modeling. It is used in a common technique in differential privacy - random noise (often drawn from a Laplace distribution) is added to the results of data queries. The overall information remains useful, but it's hard to tell if any one person's data was included.

The Laplace distribution is the prior for LASSO (Least Absolute Shrinkage and Selection Operator) in regression models because of its sparse property (it promotes coefficients that are exactly zero). In contrast, Ridge regression assumes normally distributed priors, leading to small but nonzero coefficients. We will study about priors in the section for Bayesian statistics.

3.10 Lognormal distribution

Lognormal Distribution	
What it models	Data that are positively skewed and multiplicative in nature, such as stock prices, incomes, or biological measurements.
Parameters	μ (mean of $\ln(X)$) and σ (standard deviation of $\ln(X)$).
Range of Values	$(0, \infty)$.
PDF	$f(x; \mu, \sigma) = \frac{1}{x \sigma \sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right), \quad x > 0.$
Expectation and Variance	$E[X] = e^{\mu + \frac{\sigma^2}{2}}, \quad \text{Var}(X) = (e^{\sigma^2} - 1) e^{2\mu + \sigma^2}.$
R Commands	Random Generation: <code>rlnorm(n, meanlog = μ, sdlog = σ)</code> Density: <code>dlnorm(x, meanlog = μ, sdlog = σ)</code> CDF: <code>plnorm(x, meanlog = μ, sdlog = σ)</code> Quantiles: <code>qlnorm(prob, meanlog = μ, sdlog = σ)</code>

Table 16: Lognormal Distribution

Let Z be a standard normal variable, and let μ and σ be two real numbers with $\sigma > 0$. Then $Z \sim N(\mu, \sigma^2)$, and the distribution of the random variable

$$X = e^{\mu + \sigma Z}$$

is called the *log-normal distribution* with parameters μ and σ . These parameters represent the expected value (or mean) and standard deviation of the variable's natural logarithm, $\ln(X)$, not the expectation and standard deviation of X itself.

Equivalently, if we assume that X follows a Lognormal distribution:

$$X \sim \text{Lognormal}(\mu, \sigma),$$

this implies that

$$\ln(X) \sim N(\mu, \sigma^2).$$

A random variable which is log-normally distributed takes only positive real values.

The Lognormal distribution is used to analyze data with multiplicative property, such as income, stock prices etc (they change by multiplicative factor over time). Income is considered multiplicative because changes in income are typically expressed in percentage terms rather than absolute differences. This means that, over time, incomes tend to grow by a certain rate (e.g., via compound interest or percentage raises), causing the effect of multiple factors to multiply rather than add. For example, if an individual's income increases by 10% each year, the total income after several years is the product of successive growth factors rather than the sum of fixed increments. Mathematically, if the income in one period is I_0 and it grows by a factor of g each period, then after n periods the income is given by:

$$I_n = I_0 \cdot g^n.$$

This multiplicative process is more naturally modeled on the logarithmic scale since:

$$\ln(I_n) = \ln(I_0) + n \ln(g),$$

which converts the multiplicative growth into an additive process. This is why many economic models and statistical analyses use the log of income.

Example 3.8. Modeling household income with a Lognormal distribution

Household incomes are often positively skewed; most families earn incomes near the median, but a few households earn significantly more, leading to a long right tail. Because the logarithm of income tends to be more symmetrically distributed, the Lognormal distribution is a natural model for income data.

Let us denote household income by X , then

$$X \sim \text{Lognormal}(\mu, \sigma),$$

The probability density function (pdf) of X is given by

$$p(x; \mu, \sigma) = \frac{1}{x \sigma \sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right), \quad x > 0.$$

- μ is the mean of $\ln(X)$ and roughly corresponds to the log of the median income.
- σ measures the variability of the logarithm of income. A larger σ indicates a wider spread and more pronounced right skew.

Consider a study where the logarithm of household incomes is approximately normally distributed with $\mu = 10$ and $\sigma = 0.8$. This corresponds to a median income of approximately e^{10} and a wide dispersion around that value. Also, μ and σ capture the central tendency and variability of the logarithms of incomes, which are often more symmetric and easier to model than the raw income data, which tend to be highly skewed.

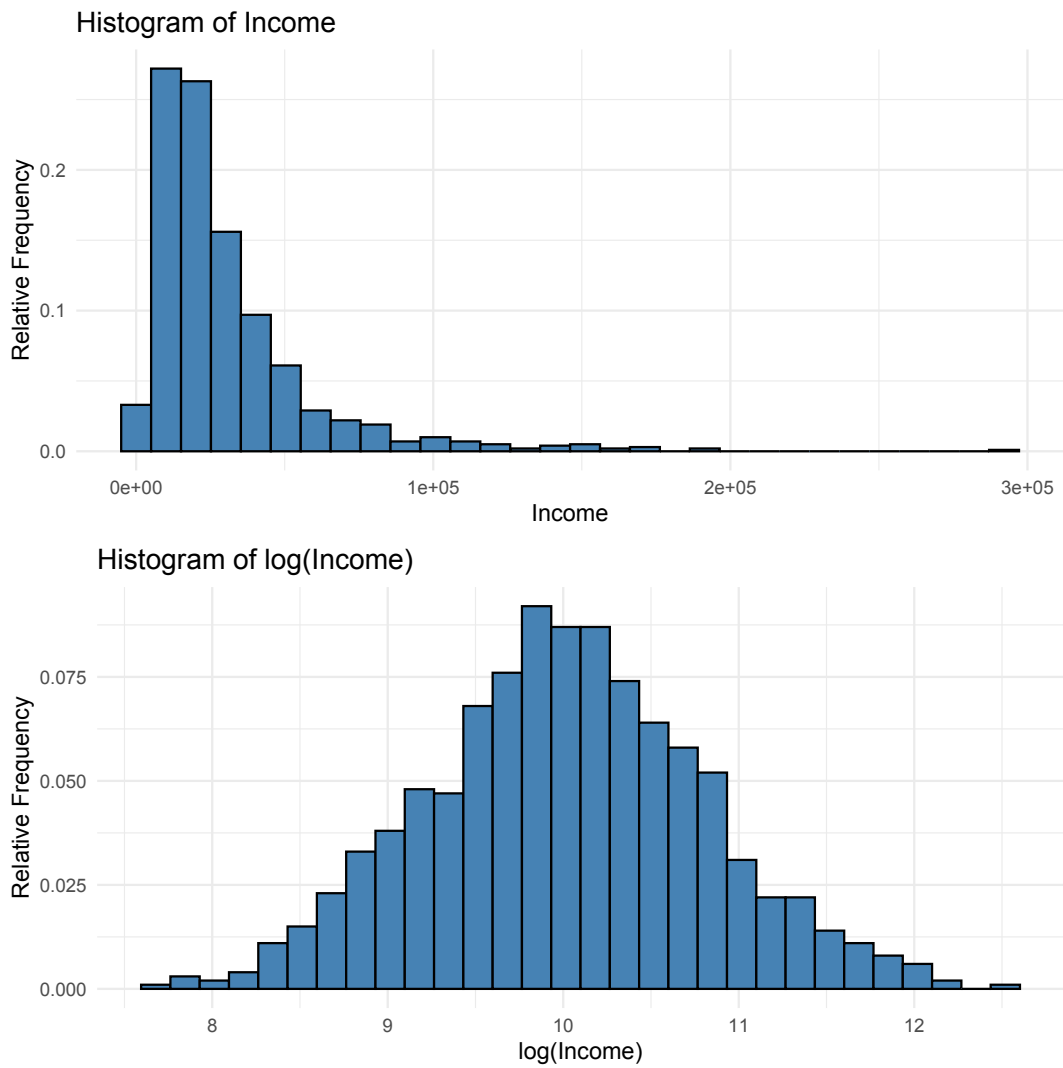


Figure 8: Data simulated from Lognormal distribution

□

Remark. Notice how the extreme values are transformed when we take logarithm (first and second histogram). This data transformation is often used when we want to deal with outliers, but we have to be careful - such a transformation is applicable if the data is multiplicative. The results on transformed data may not be interpretable.

3.11 Pareto distribution

Pareto Distribution	
What it models	Heavy-tailed phenomena where a small number of occurrences account for a large effect, such as income distribution, city sizes, or file sizes on the internet.
Parameters	$x_m > 0$ (scale or minimum value), $\alpha > 0$ (shape parameter).
Range of Values	$[x_m, \infty)$.
Probability Density Function (PDF)	$f(x) = \frac{\alpha x_m^\alpha}{x^{\alpha+1}}, \quad x \geq x_m,$ where x_m is the minimum possible value and α controls the tail heaviness.
Expectation and Variance	$E[X] = \frac{\alpha x_m}{\alpha - 1} \quad (\alpha > 1),$ $\text{Var}(X) = \frac{\alpha x_m^2}{(\alpha - 1)^2(\alpha - 2)} \quad (\alpha > 2).$
Important Characteristics	• Heavy-tailed behavior, indicating that extreme values occur more frequently. • Scale invariance, meaning the distribution maintains its form across different scales. • Mean and variance exist only for $\alpha > 1$ and $\alpha > 2$ respectively.
R Commands	Random Generation: <code>rpareto(n, xm, alpha)</code> (from packages like VGAM) Density: <code>dpareto(x, xm, alpha)</code> CDF: <code>ppareto(x, xm, alpha)</code> Quantiles: <code>qpareto(prob, xm, alpha)</code>

Table 17: Pareto Distribution

The Pareto distribution is a power-law probability distribution that is commonly used to model phenomena with heavy tails. It is especially useful in economics and the social sciences to describe the distribution of wealth, income, or city sizes, where a small number of occurrences account for a large portion of the effect. (Read below about the power law.)

Definition

A random variable X is said to follow a Pareto distribution with scale parameter $x_m > 0$ and shape parameter $\alpha > 0$ if its probability density function (pdf) is given by:

$$p(x; x_m, \alpha) = \frac{\alpha x_m^\alpha}{x^{\alpha+1}}, \quad x \geq x_m,$$

and $p(x) = 0$ for $x < x_m$.

Parameters

- **x_m :** The minimum (or threshold) value of X . It is the smallest value that X can take.
- **α :** The shape parameter, which controls the thickness of the tail. A smaller α indicates a heavier tail, meaning that extreme values are more probable.
- **Range of Values:** $X \in [x_m, \infty)$.
- **Expectation:** The mean of X exists only if $\alpha > 1$ and is given by

$$E[X] = \frac{\alpha x_m}{\alpha - 1}.$$

If $\alpha \leq 1$, the mean is infinite. This means the probability of extremely large values is very high, the tails are very heavy and this influences the mean. See below the discussion for analyzes of heavy tailed distributions

- **Variance:** The variance exists only if $\alpha > 2$ and is given by

$$\text{Var}(X) = \frac{\alpha x_m^2}{(\alpha - 1)^2(\alpha - 2)}.$$

For $1 < \alpha \leq 2$, the variance is infinite.

The Pareto distribution is frequently applied in scenarios where a small number of occurrences account for a large portion of the effect. For example:

- **Wealth and income distribution** - A small percentage of the population often holds a large share of total wealth, a phenomenon well-modeled by a Pareto law.
- **City sizes** - The distribution of city populations tends to follow a Pareto distribution, where a few cities are extremely large while most are much smaller.
- **Natural disasters** - The magnitudes of events such as earthquakes or floods can be modeled using Pareto tails, capturing the high likelihood of extreme events.
- **File sizes on the Internet** - In network traffic, a small number of files (e.g., videos or high-resolution images) may account for most of the data volume, which follows a Pareto pattern.
- **Insurance claims** - In insurance, a few large claims often constitute a significant proportion of total losses, making the Pareto distribution suitable for modeling risk.

Example 3.9. Analyzing wealth using Pareto distribution.

In many economies, wealth is distributed unevenly: a small proportion of the population holds a large share of the total wealth, while the majority has much less. In our example, we simulate wealth data using the following parameters:

- Minimum wealth, $x_m = 1000$ (e.g., \$1,000).
- Shape parameter, $\alpha = 1.5$, which indicates a heavy tail, meaning there is a non-negligible probability of very high wealth values.

The simulated data represent the wealth of 1000 individuals drawn from this Pareto model.

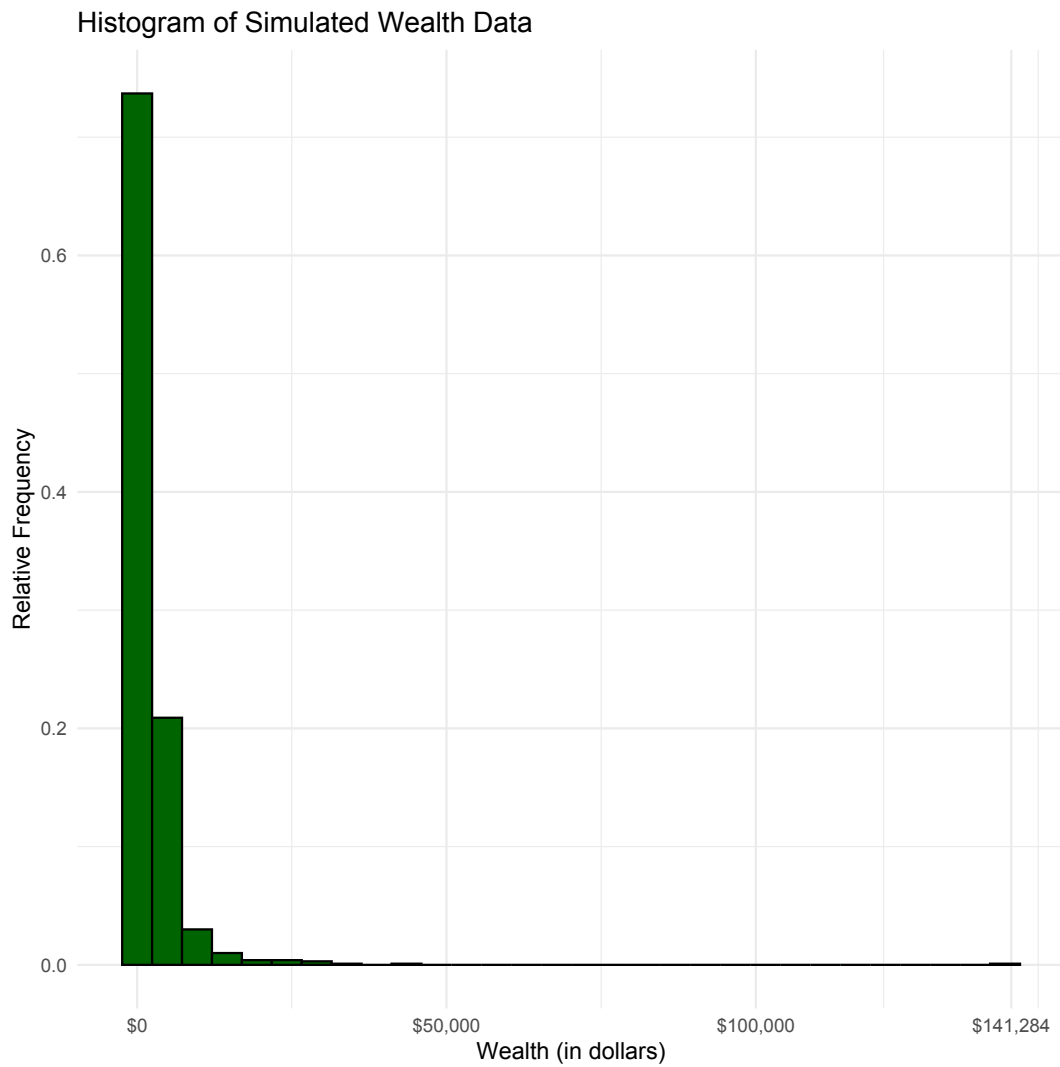


Figure 9: Data simulated from Pareto distribution

□

In this distribution a better vizualisation is needed for the tails.

3.12 Cauchy distribution (aka Lorentz Distribution)

Cauchy Distribution	
What it models	A heavy-tailed distribution often used in physics to model resonance phenomena and in statistics as an example of a distribution without a defined mean or variance.
Parameters	x_0 (location parameter) and $\gamma > 0$ (scale parameter).
Range of Values	$(-\infty, \infty)$.
Probability Density Function (PDF)	$f(x; x_0, \gamma) = \frac{1}{\pi\gamma \left[1 + \left(\frac{x-x_0}{\gamma} \right)^2 \right]}, \quad x \in \mathbb{R},$ where x_0 is the location and γ controls the half-width at half-maximum.
Expectation and Variance	• Mean: Undefined. • Variance: Undefined. • Median and Mode: x_0.
Important Characteristics	• Exhibits heavy-tailed behavior, with frequent extreme values. • Lacks a defined mean and variance, challenging traditional statistical inference. • Serves as a classic counterexample for the Law of Large Numbers. • Can be derived as the ratio of two independent standard normal random variables.
R Commands	Random Generation: <code>rcauchy(n, location = x_0, scale = γ)</code> Density: <code>dcauchy(x, location = x_0, scale = γ)</code> CDF: <code>pcauchy(x, location = x_0, scale = γ)</code> Quantiles: <code>qcauchy(prob, location = x_0, scale = γ)</code>

Table 18: Cauchy Distribution

The **Cauchy distribution** is a continuous probability distribution known for its heavy tails and the fact that its mean and variance are undefined. In many applications—such as physics, where it is also known as the Lorentz distribution—it provides a useful model for resonant phenomena. A particularly striking feature of the Cauchy distribution is that, despite being derived from well-behaved normal distributions, its heavy tail leads to unusual statistical properties.

The **standard Cauchy distribution** is defined by the probability density function (PDF):

$$p(x) = \frac{1}{\pi} \cdot \frac{1}{1 + x^2}, \quad x \in \mathbb{R}.$$

Its cumulative distribution function (CDF) is given by:

$$F(x) = \frac{1}{\pi} \arctan(x) + \frac{1}{2}.$$

A key property of the standard Cauchy distribution is that it can be obtained as the ratio of two independent standard normal random variables. That is, if

$$U \sim N(0, 1) \quad \text{and} \quad V \sim N(0, 1),$$

are independent, then the random variable

$$X = \frac{U}{V}$$

follows a standard Cauchy distribution.

The Cauchy distribution can be generalized by introducing a location parameter x_0 and a scale parameter $\gamma > 0$. The **generalized (or shifted and scaled) Cauchy distribution** has the following PDF:

$$p(x; x_0, \gamma) = \frac{1}{\pi\gamma} \cdot \frac{1}{1 + \left(\frac{x-x_0}{\gamma}\right)^2}, \quad x \in \mathbb{R}.$$

Its CDF is given by:

$$F(x; x_0, \gamma) = \frac{1}{\pi} \arctan\left(\frac{x - x_0}{\gamma}\right) + \frac{1}{2}.$$

Here the location parameter x_0 determines the peak (or “center”) of the distribution, where the density has the highest value. The scale parameter γ determines the half-width at half-maximum; that is, the density at the points $x_0 \pm \gamma$ is exactly half of its maximum value (attained at x_0). This roughly indicates the spread of the data.

The general Cauchy distribution is related to the standard version through a location-scale transformation. Specifically, if

$$Y \sim \text{Cauchy}(0, 1)$$

(as obtained from the ratio of two independent standard normals), then

$$X = x_0 + \gamma Y$$

follows a Cauchy distribution with parameters x_0 (location) and γ (scale).

The Cauchy distribution is a heavy-tailed distribution with undefined mean and variance. It is well-suited for modeling phenomena with significant extreme variability (heavy tails) and resonant behaviors, especially in physics and engineering.

Example 3.10. Data simulated from Cauchy distribution.

The following plot represents the relative frequency of 10,000 values from a Cauchy distribution with location 0 and scale 1 - only the between -25 and 25 are represented. The legend includes the minimum and maximum values of the raw simulated data, as well as the theoretical probability that a Cauchy-distributed variable exceeds 25.

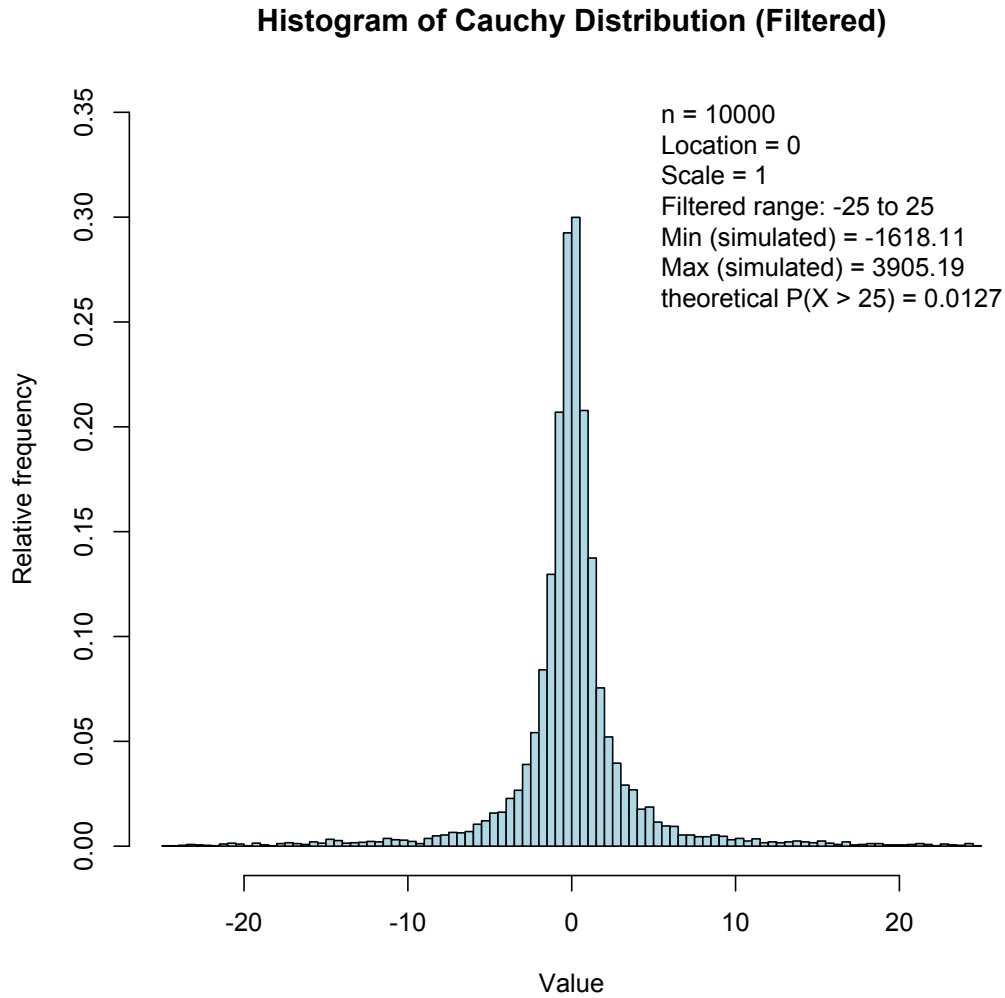


Figure 10: Data simulated from Cauchy distribution

□

Remark. The Student's t-distribution with 1 degree of freedom is exactly the standard Cauchy distribution. That is, if the t-distribution has degrees of freedom $\nu = 1$, its probability density function becomes

$$p(x) = \frac{1}{\pi(1 + x^2)},$$

which is identical to the PDF of the standard Cauchy distribution.

This equivalence arises because the t-distribution is derived from the ratio of a standard normal variable and the square root of an independent chi-squared variable (scaled appropriately). When $\nu = 1$, the chi-squared variable has a heavy tail, resulting in the t-distribution having the heavy-tailed behavior characteristic of the Cauchy distribution.

3.13 How to analyze heavy tailed distributions

When dealing with heavy-tailed distributions (Pareto, Cauchy and other distributions), the tail may be so heavy that the mean and/or the variance may be infinite. Thus many standard statistical measures and tests can not be applied.

- In this cases mean and standard deviation can not be used to describe the data. We have to choose measures which are less influenced by extreme values as median, quantiles, trimmed means, median absolute deviation (MAD) etc.
- Many standard statistical methods are based on means and variances (Central limit theorem, confidence intervals, estimators especially with method of moments, parametric tests etc.). This methods can not be used when mean and variance are infinite or nonexistent. Alternative methods should be used, specifically designed for analysis of heavy tails. This part of statistics is called extreme value theory. One example is Hill estimator for the parameter α in Pareto distribution - we can focus on analysis of the tail and the extreme values.
- Logarithmic transformations, when suitable, can transform the data and decrease the extreme values, they can reveal some structures and some standards methods may be applicable.
- Also we should consider some non-parametric and simulation methods. (Discuss Kurtosis)

3.14 Bimodal distributions are their analyzes

A **bimodal distribution** is a probability distribution with two distinct local maxima (modes) in its probability density function (PDF) or frequency histogram. These two peaks often suggest that the data may be drawn from two different underlying subpopulations or processes.

Characteristics

- Two Peaks - the defining feature is the presence of two separate modes;
- Heterogeneity: Bimodality often indicates that the dataset comprises two groups with different characteristics;
- Variability - depending on the separation and height of the peaks, the overall spread and shape of the distribution may differ from standard unimodal distributions.
- Skewness and kurtosis - the individual modes may have different skewness or kurtosis properties.

What are we allowed to do?

- We can use histograms or kernel density estimates to visually inspect the data for bimodality;
- If bimodality is observed, explore whether it reflects genuine subgroups in the data. Domain knowledge is important;
- We can use mixture models (e.g., Gaussian mixture models) to formally model the two components.
- When we want to summarize central tendency, we can use more robust statistics and consider medians or quantiles if outliers or mixed subpopulations are present.

What to be careful about?

- We should not assume normality and we should not automatically apply standard parametric tests which assume unimodal, symmetric data;
- We should ensure that the observed bimodality is not due to data collection errors or improper binning,
- There may be an overlap - in cases of slight bimodality, be cautious about splitting the data without a strong theoretical or empirical justification.

Important Features

- Mode Separation - the distance between the two peaks can provide insights into the degree of heterogeneity.
- Relative Peak Heights - differences in peak heights may indicate the relative size or importance of the subpopulations.
- How the distribution tails behave beyond the modes can be important for risk assessment and extreme value analysis.

How to analyze bimodal data?

- Graphical Methods - histograms, kernel density plots, and box plots are useful for initial exploration;
- Statistical Tests - formal tests like Hartigan's Dip Test or Silverman's Test can be used to assess multimodality;
- Calculate Bimodality coefficient (BC): The following formula is used $BC = (\text{skewness}^2 + 1) / \text{kurtosis}$. For a uniform distribution, BC is about 0.555. Thus, if the computed BC exceeds this value, it may suggest the presence of bimodality.
- Mixture Models - fit a mixture model (e.g., Gaussian mixture model) to estimate the parameters of the underlying distributions.
- Clustering- we can use clustering techniques (such as k-means or hierarchical clustering) to identify and separate subpopulations.

Example 3.11. Old Faithful dataset. Analyze the data with some of the methods mentioned above.

Old Faithful Geyser: Eruption length vs. waiting time

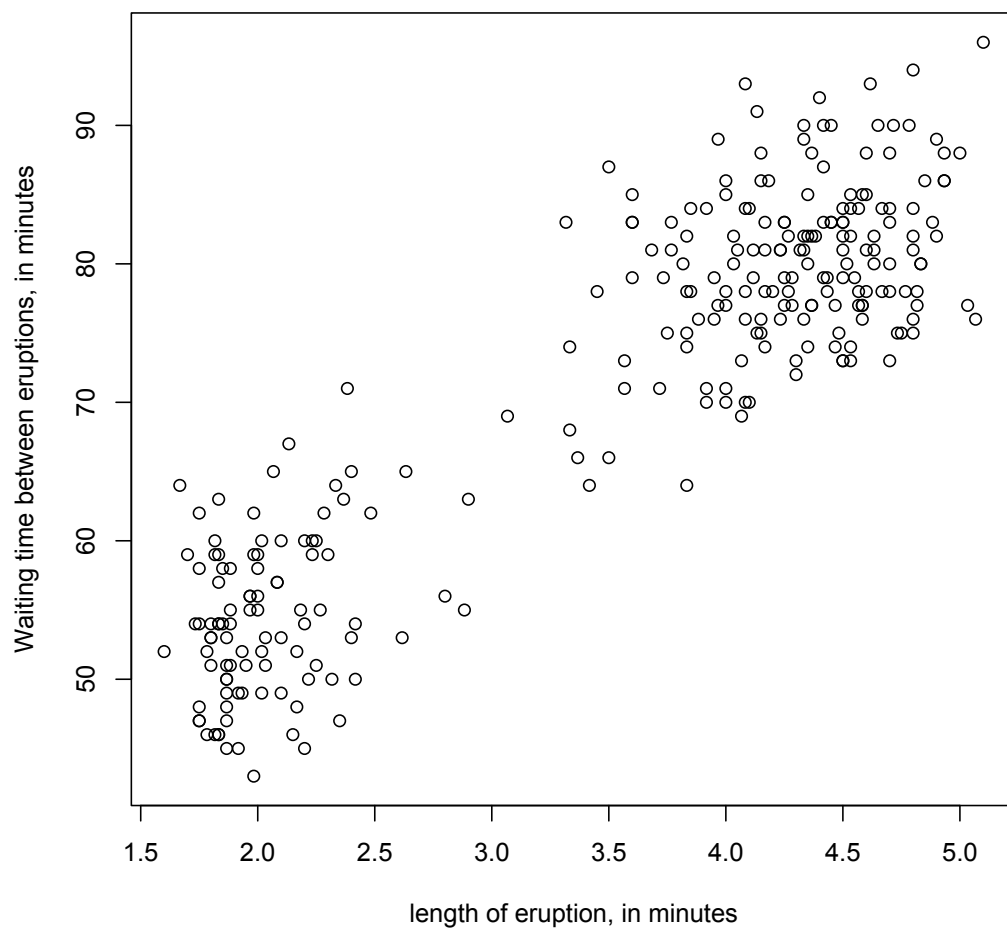


Figure 11: Summary of Old Faithful dataset

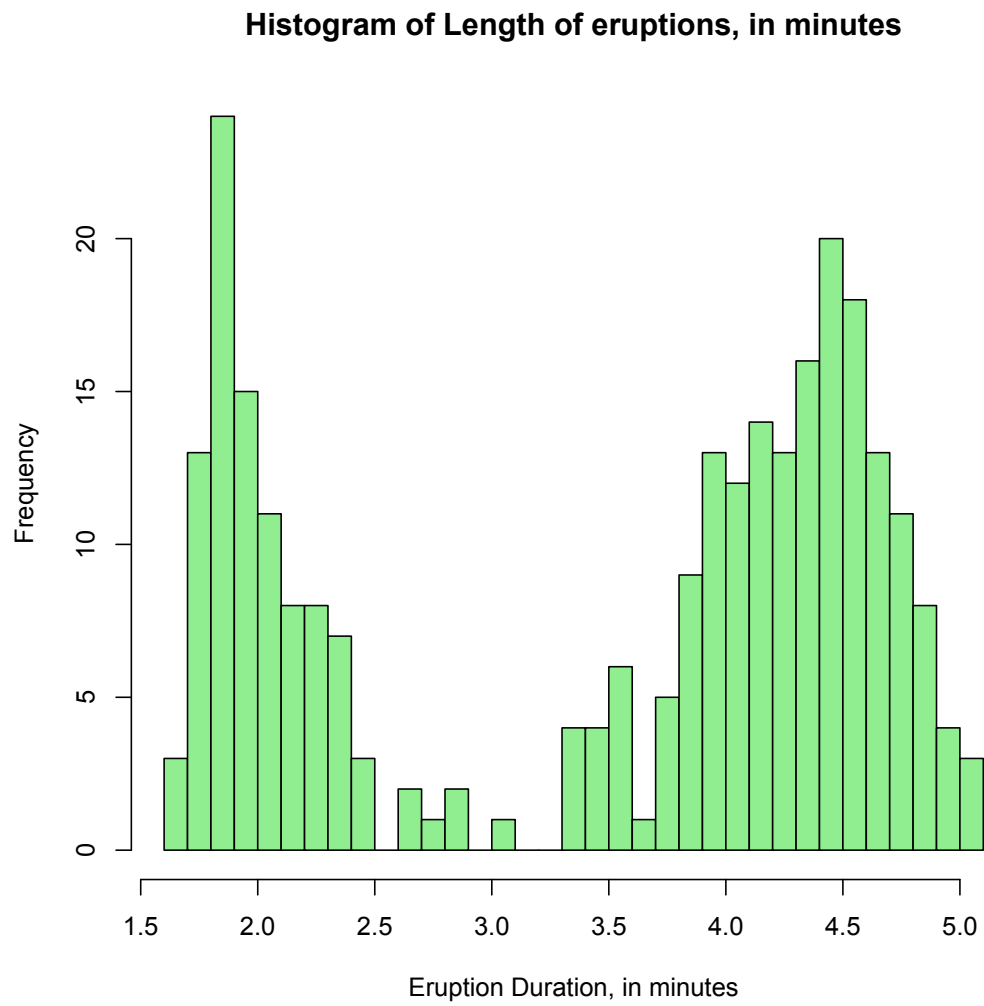


Figure 12: Length of eruptions

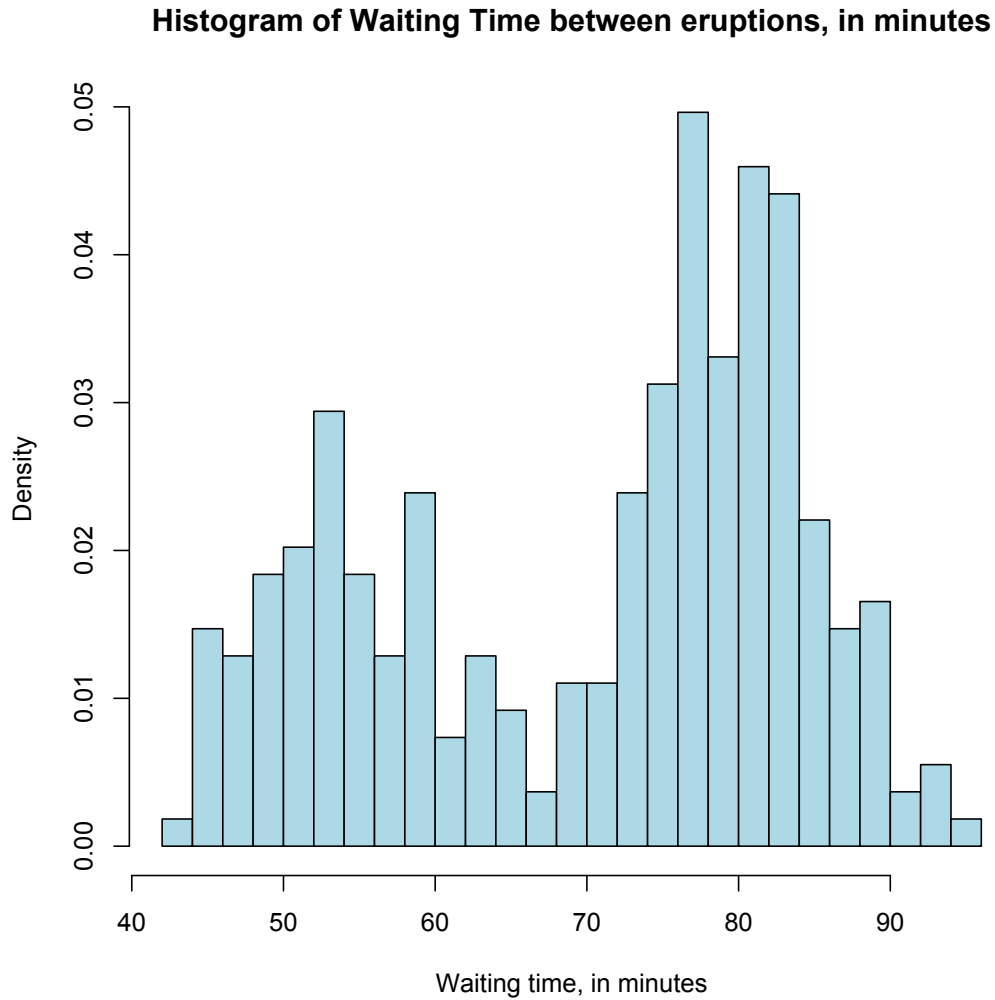


Figure 13: waiting time between eruptions

□

3.15 Mixture models and applications

3.16 Circular statistics

3.17 Convolutions and compound distributions

4 Families of distributions

Next we present several important families of distributions.

4.1 Exponential family

This is a very broad class of distributions which have a probability density (or mass) functions in a specific canonical form. Many properties of these distributions, such as the mean, variance, and moment generating

functions etc. can be derived from this form, making it easier to perform analytical calculations. This family includes many well-known distributions such as Normal (Gaussian), Exponential, Gamma, Bernoulli and Binomial, Poisson and Beta distributions. In Bayesian context these distributions have so called conjugate prior and we will use them in the Chapter for Bayesian statistics.

4.2 Location-Scale Family

Distributions in this family can be transformed into one another via linear transformation. This family includes: Normal, Cauchy Distribution, Student's t and Logistic Distribution. The key idea is that if X is a random variable from a standard distribution, then the linearly transformed r.v. $Y = \mu + \sigma X$ will have the same shape but with a different center and spread.

These distributions are used when we want to capture the effects of shifting and rescaling. Data from these distributions can be standardized by subtracting the mean and dividing by the standard deviation - it will be transformed in standard form of the same distribution. This is common in statistical data analyses.

4.3 Stable distributions

Definition 4.1. Let f and g be two functions defined on $\mathbb{R}$ (or a subset of $\mathbb{R}$). The *convolution* of f and g , denoted by $f * g$, is defined by

$$(f * g)(x) = \int_{-\infty}^{\infty} f(x - y) g(y) dy,$$

provided the integral exists for all x .

In the context of probability theory, if p_X and p_Y are the probability density functions (pdfs) of independent random variables X and Y respectively, then the pdf of the sum $Z = X + Y$ is given by the convolution (when defined) $p_Z = p_X * p_Y$:

$$p_Z(z) = \int_{-\infty}^{\infty} p_X(z - y) p_Y(y) dy.$$

Definition 4.2 (Stable Distribution). • A probability distribution is called **stable** if a linear combination of two independent copies of a random variable with that distribution has the same distribution, up to location and scale parameters.

- A random variable X is said to have a **stable distribution** if for any positive integers n and any independent copies $X_1, X_2, \dots, X_n$ of X , there exist constants $a_n > 0$ and $b_n \in \mathbb{R}$ such that

$$X_1 + X_2 + \dots + X_n \stackrel{d}{=} a_n X + b_n,$$

where $\stackrel{d}{=}$ denotes equality in distribution.

In other words, if we denote by $S_n = X_1 + X_2 + \dots + X_n$ the sum of n independent copies of X , then there exist constants a_n and b_n such that

$$S_n \sim a_n X + b_n.$$

Some examples:

- In Financial/Insurance modeling - when we model the total return of a portfolio, convolution can be used to combine the probability distributions of independent asset returns. The resulting distribution describes the overall risk and return of the portfolio;

- In environmental sciences - when we model the cumulative effects of daily measurements. For example, if daily rainfall amounts are modeled as random variables, the convolution of their distributions gives insight into monthly or yearly totals;
- In Reliability engineering - the time until failure of a system made up of several independent components can be modeled by convolving the lifetime distributions of individual components.

This stability property under convolution is significant because it implies that the sum of independent stable random variables (after appropriate normalization) will have the same type of distribution, which is a key idea behind the Central Limit Theorem (when the variance is finite) and more generally in the theory of heavy-tailed phenomena.

The normal distribution is a famous example of a stable distribution with finite variance. However, stable distributions also include heavy-tailed distributions such as the Cauchy distribution, which is stable with the additional property that its variance is infinite. Another distribution in this family is Levy distribution, which is used in Physics, Finance, Analyzes of animal foraging, signal processing, etc.

4.4 Extreme value distributions

Extreme value distributions arise in the study of the extreme values (maximum or minimum) of large samples of random variables. More specifically these distributions arise in the study of first and last order statistics.

The key idea of extreme value theory is that the maximum (or minimum) of a large number of independent, identically distributed (i.i.d.) random variables — when properly normalized — will converge in distribution to one of three types of distributions. Which type of distribution applies depends on the tail behavior.

Type I Gumbel distribution - The Gumbel distribution typically appears as the limiting distribution for the maximum of samples drawn from distributions with exponentially decaying tails (light-tailed distributions), such as the normal or exponential distributions.

Type II Fréchet distribution The Fréchet distribution arises as the limit for the maximum of samples from heavy-tailed distributions, such as the Pareto distribution. It is characterized by a power-law decay in the tail - if we move further into the tail of a probability distribution, the probabilities decrease at a rate proportional to a power of the variable, i.e. for large x $p(x) \sim Cx^{-\alpha}$, where C is a constant $\alpha > 0$.

Such Power-law tails are observed in many real-world phenomena - we have observed already the case for wealth distribution. Another examples are

- Earthquake Magnitudes: Larger earthquakes, while rare, occur more frequently than would be predicted by exponential decay;
- Many real networks have many nodes with small degree and a few nodes with extremely large degree. This can be modeled with a power-law tail:

$$P(X \geq k) \sim Ck^{-\alpha}$$

which means that the probability that a node has at least k links decays as a power of k . Types of datasets: a web graph, social network, citation network, GitHub dependency network, Internet traffic (the number of links on web pages).

- Word Frequencies and Zipf's Law: In natural language, word frequencies have been observed to follow a power-law pattern known as **Zipf's law**:

$$f(r) \propto \frac{1}{r^s},$$

where r is the rank of a word, $f(r)$ is its frequency and s is often chosen to be 1. Thus, a few words are extremely common, while most words are rare. This affects search engines, text compression, and language modeling: common words must be downweighted because they carry little discriminative information, while rare words often carry important meaning. In large language models, this creates a long-tail problem: frequent tokens dominate training, whereas rare words, technical terms, names, and low-resource languages are harder to learn. This motivates subword tokenization, embeddings that share information across contexts, and careful balancing or upsampling of underrepresented data sources.

Type III Weibull distribution The Weibull extreme value distribution (in the context of extremes) appears when the underlying distribution has a finite upper endpoint (or lower endpoint for minima). It models the extreme values of distributions with a bounded tail.

The applications are broad:

- Modeling extreme losses and gains (risk management, finance, insurance);
- Modeling extreme weather events (floods, hurricanes) - estimating the probability of very rare events is important for planning and risk assessment;
- Predicting the failure times of production components which are subject to stress (reliability engineering).

Machine learning with heavy-tailed and extreme-value data. When data contain heavy tails or extreme values, standard machine learning methods must be used carefully, because models optimized for average error often describe the center of the distribution well but fail in the tail. In such problems, the goal is often not only prediction of the mean, but estimation of rare-event probabilities, high quantiles, exceedance risk, or tail behavior. Standard approaches include log-transformations, robust regression, quantile regression, anomaly detection methods, and hybrid models that combine machine learning for covariates with extreme-value models such as GEV or generalized Pareto distributions for the tail.

Important practical issues are tail scarcity and imbalance: extreme observations are rare, so models may ignore them unless the loss function, sampling strategy, or evaluation metric gives them sufficient importance. Tail-weighted losses, cost-sensitive learning, oversampling of extreme cases, or separate modeling of exceedances over a threshold are often used. Evaluation should include tail-specific metrics, such as high-quantile error, exceedance classification accuracy, or calibration of rare-event probabilities, not only RMSE or overall accuracy.

One should be careful not to treat all extremes as errors or outliers, because in heavy-tailed data they may be genuine observations. At the same time, individual extremes have high influence, so robust validation, sensitivity analysis, and comparison with simpler statistical tail models are important. Random train-test splitting can also be misleading in temporal or spatial extremes; block or time-aware validation is usually more appropriate.

Clustering and classification with heavy-tailed data. For classification, the main difficulty is that extreme cases are often rare, so the problem becomes an imbalanced-learning problem. Standard classifiers may achieve high overall accuracy while missing the rare but important tail events. Therefore, cost-sensitive classification, class weighting, oversampling of rare events, threshold tuning, anomaly-detection methods, and evaluation by precision, recall, F1-score, ROC/PR curves, and tail-specific error rates are commonly used. In some applications, labels are defined through exceedances, for example $X > u$, and the task becomes classification of whether an observation belongs to the extreme-risk region.

For clustering, methods such as k -means should be used carefully, because means and Euclidean distances are strongly affected by extreme observations. More robust alternatives include clustering after log-transformation, clustering with robust distances, k -medoids instead of k -means, density-based methods such as DBSCAN/HDBSCAN, or model-based clustering with heavy-tailed distributions such as Student- t

mixtures. In multivariate extreme-value analysis, one may also cluster observations according to their tail behavior, for example by grouping observations that become extreme together.

The main warning is not to automatically remove extreme observations as outliers. In this setting, they may be the most informative part of the data. Instead, one should decide whether the goal is to model the bulk of the data, the tail, or both separately.

There are different methods for statistical inference when working with extreme value problems. One of them is block maxima method - the data is divided into blocks and the maximum in each block is analyzed.

Example 4.3. Extreme flood levels of a river. We can be interested in the annual maximum river discharge to assess flood risk. We can collect hydrological data - yearly the maximum water level (or discharge) is recorded. The data can be divided in blocks (here we will use a year as a block) and the yearly annual maxima can be analyzed with the extreme value theory.

Studies have shown that the Gumbel distribution is often a good model, because river flows typically have exponentially decaying tails. We have to fit appropriate distribution and then it can be used to estimate the probability of very high flood levels. Based on this model we can define what a "100-year flood" is - this is a flood level expected to be exceeded on average once every 100 years. This is then used for infrastructure planning (dams, flood control systems, insurance companies calculate premiums etc.)

□

Example 4.4. Simulation of yearly rainfall as generalized Pareto distribution

We simulate daily rainfall measurements over 100 years, with 100 days per year using a Generalized Pareto Distribution (GPD) with heavy tails. We want to analyze the extreme values of this dataset by extracting the annual minimum and maximum rainfall amounts. We compare their empirical distributions to the theoretical distributions of the corresponding first and last order statistics.

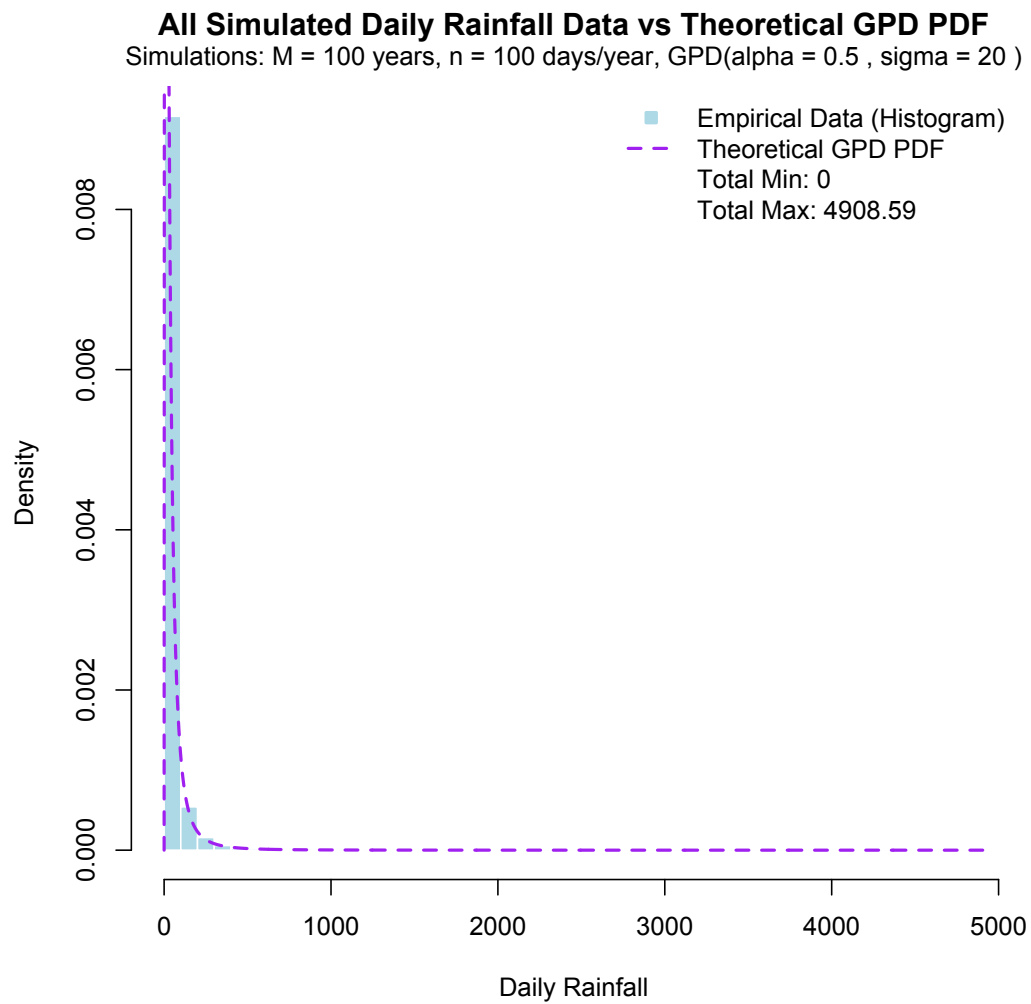


Figure 14: Total Daily Rainfall data simulated from Generalized Pareto distribution

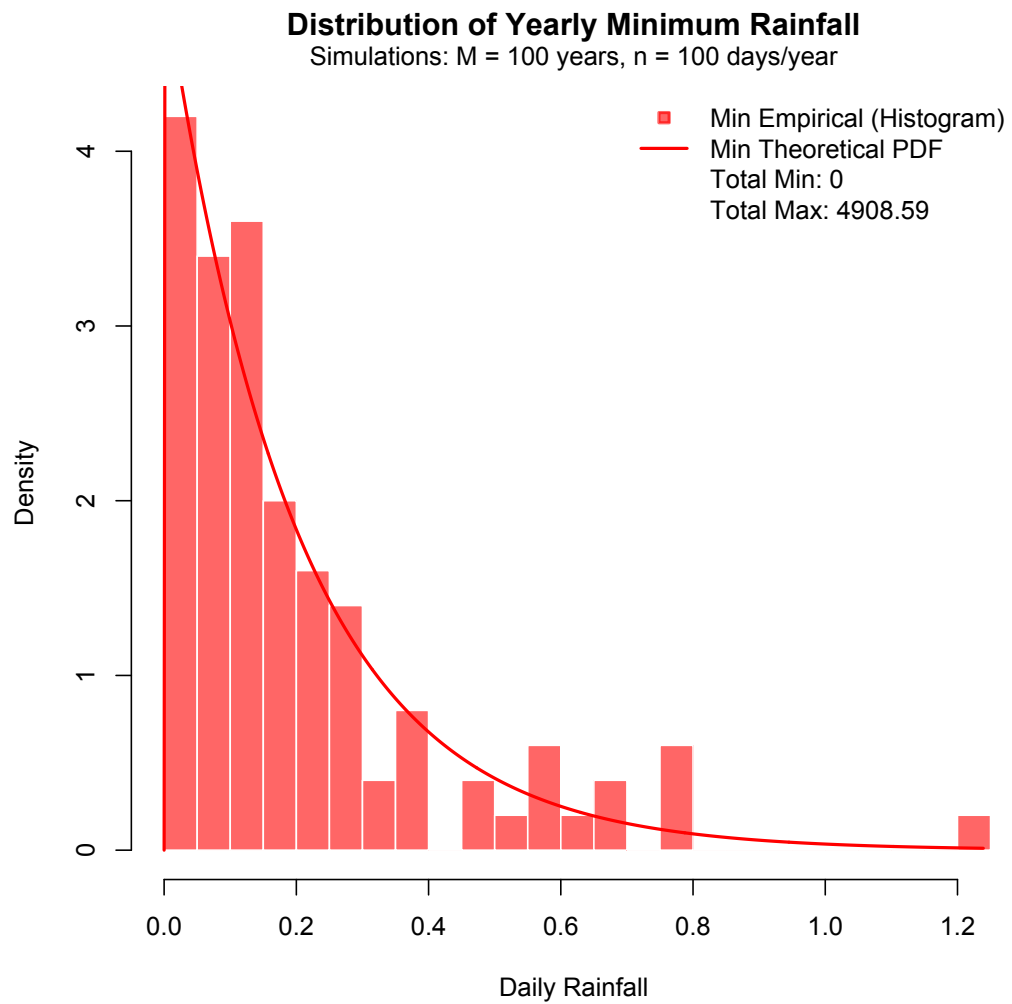


Figure 15: Yearly Minimum Rainfall

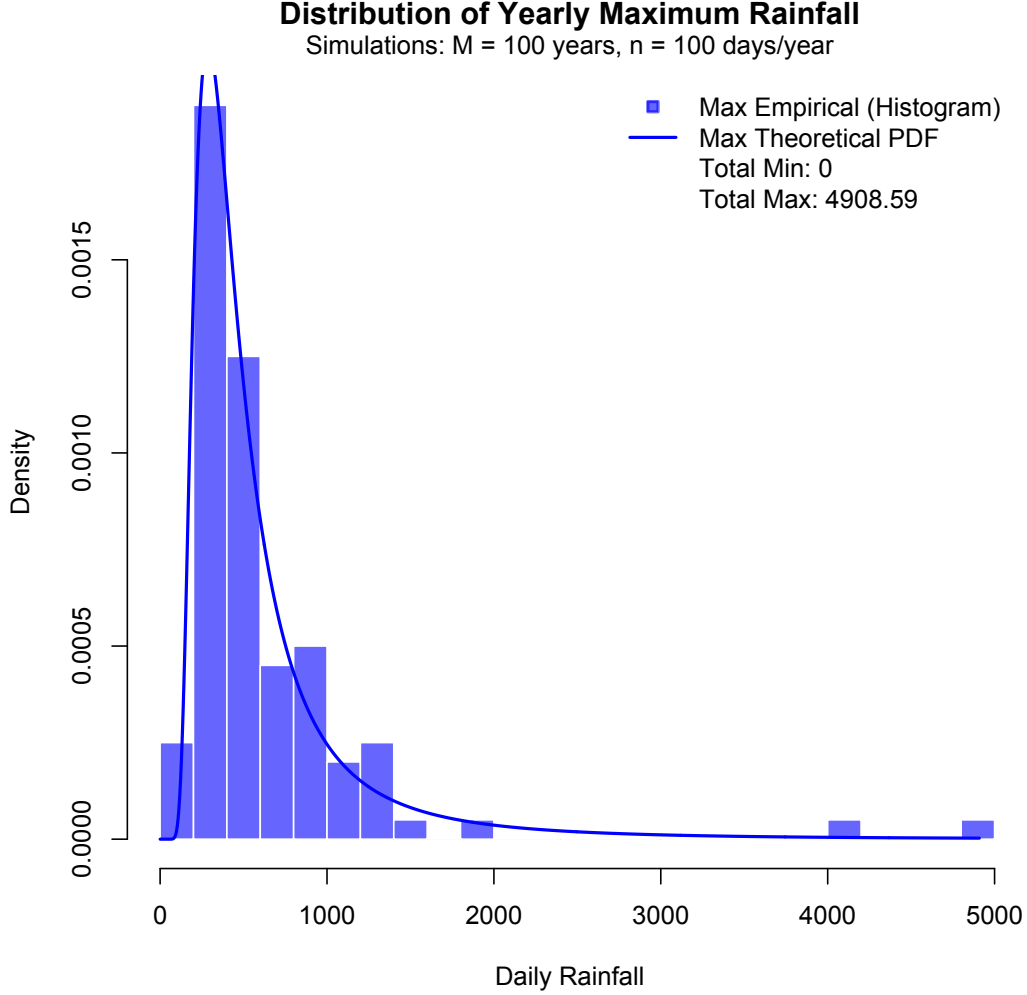


Figure 16: Yearly Maximum Rainfall

□

(Change the Min and Max plot with a legend which would also include min and max values from the min and max (not total) sample).

We note that the distribution of the first and last ordered statistics are theoretically known, if the underlying distribution of the original data is known and has some properties.

Theorem 4.5 (Distribution of Order Statistics). Let $X_1, X_2, \dots, X_n$ be independent and identically distributed random variables with common probability density function $f(x)$ and cumulative distribution function $F(x)$. Let

$$X_{(1)} \leq X_{(2)} \leq \dots \leq X_{(n)}$$

denote the order statistics. Then, the probability density function (pdf) of the k -th order statistic $X_{(k)}$ is given by

$$p_{X_{(k)}}(x) = \frac{n!}{(k-1)!(n-k)!} [F(x)]^{k-1} [1 - F(x)]^{n-k} f(x), \quad x \in \mathbb{R},$$

and its cumulative distribution function (cdf) is

$$F_{X_{(k)}}(x) = \sum_{j=k}^n \binom{n}{j} [F(x)]^j [1 - F(x)]^{n-j}.$$

Special Cases:

- **Minimum order statistic** - for $k = 1$, the first order statistic $X_{(1)}$ (the minimum) has pdf

$$p_{X_{(1)}}(x) = n [1 - F(x)]^{n-1} f(x),$$

and cdf

$$F_{X_{(1)}}(x) = 1 - [1 - F(x)]^n.$$

- **Maximum order statistic** - for $k = n$, the last order statistic $X_{(n)}$ (the maximum) has pdf

$$p_{X_{(n)}}(x) = n [F(x)]^{n-1} f(x),$$

and cdf

$$F_{X_{(n)}}(x) = [F(x)]^n.$$

We have used this theorem to plot the theoretical pdf's in the previous plots. We can also use the theorem when we want to work with the median random variable ($n+1$ -th order statistic for an odd sample size) which is used as an estimator of central tendency and it is less sensitive to outliers than the mean. Order statistics are also used to estimate quantiles - essential in summarizing the distribution of data and in nonparametric statistical inference and methods (rank tests, sign tests) etc. Sometimes we may want to know something about the intermediate order statistics - we may want to gain knowledge about extreme events, apart from the minimum and maximum.

5 Solved exercises

Exercise 5.1. Sum of two independent exponential random variables

Suppose the time until a customer is served at a counter is modeled by an exponential random variable X with parameter λ , so that its probability density function (pdf) is

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

Assume that the service times for two independent customers are given by X and Y (both $\sim \text{Exponential}(\lambda)$). Find the pdf of the total service time $Z = X + Y$.

Solution.

Since X and Y are independent, the pdf of the sum $Z = X + Y$ is given by the convolution of $p_X(x)$ and $f_Y(y)$:

$$p_Z(z) = \int_0^z p_X(z-y) p_Y(y) dy.$$

If we substitute the exponential densities, we have:

$$p_Z(z) = \int_0^z \lambda e^{-\lambda(z-y)} \cdot \lambda e^{-\lambda y} dy = \lambda^2 e^{-\lambda z} \int_0^z dy = \lambda^2 z e^{-\lambda z}, \quad z \geq 0.$$

This is the pdf of a Gamma distribution with shape parameter $k = 2$ and rate parameter λ .

Thus we conclude that the total waiting time for two consecutive events (each having an exponential waiting time) follows a Gamma distribution. Such a result is widely used in queueing theory, reliability analysis, and various fields where waiting times or lifetimes are modeled. ■

Exercise 5.2 (Sum of uniform random variables on $[0, 1]$.] Let X and Y be independent random variables, each uniformly distributed on the interval $[0, 1]$. Their probability density functions (pdf) are

$$f_X(x) = 1 \quad \text{and} \quad f_Y(y) = 1, \quad \text{for } 0 \leq x, y \leq 1.$$

Find the pdf of the sum $Z = X + Y$ using convolution.

Solution. Since X and Y are both supported on $[0, 1]$, the sum $Z = X + Y$ is supported on $[0, 2]$. The convolution integral becomes

$$f_Z(z) = \int_{\max(0, z-1)}^{\min(z, 1)} f_X(z-y)f_Y(y) dy.$$

Since $f_X(x) = 1$ and $f_Y(y) = 1$, we have

$$f_Z(z) = \int_{\max(0, z-1)}^{\min(z, 1)} 1 dy = \min(z, 1) - \max(0, z-1).$$

We now break the analysis into two cases:

Case 1: $0 \leq z \leq 1$ Here, $\max(0, z-1) = 0$ and $\min(z, 1) = z$. Therefore,

$$f_Z(z) = z - 0 = z.$$

Case 2: $1 < z \leq 2$ Now, $\max(0, z-1) = z-1$ (since $z-1 > 0$) and $\min(z, 1) = 1$. Thus,

$$f_Z(z) = 1 - (z-1) = 2 - z.$$

Thus, the pdf of Z is given by

$$f_Z(z) = \begin{cases} z, & 0 \leq z \leq 1, \\ 2 - z, & 1 < z \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

■

Exercise 5.3 (Sum of non-identical uniform random variables). Let X be uniformly distributed on $[0, 2]$ with pdf

$$f_X(x) = \frac{1}{2}, \quad 0 \leq x \leq 2,$$

and let Y be uniformly distributed on $[0, 3]$ with pdf

$$f_Y(y) = \frac{1}{3}, \quad 0 \leq y \leq 3.$$

Compute the pdf of the sum $Z = X + Y$ using convolution.

Solution. The variable X is supported on $[0, 2]$ and Y on $[0, 3]$, so their sum $Z = X + Y$ is supported on $[0, 5]$. The convolution integral becomes

$$f_Z(z) = \int_{y=\max(0, z-2)}^{\min(z, 3)} f_X(z-y)f_Y(y) dy.$$

Substitute $f_X(x) = \frac{1}{2}$ and $f_Y(y) = \frac{1}{3}$ to obtain

$$f_Z(z) = \frac{1}{6} \int_{y=\max(0, z-2)}^{\min(z, 3)} dy = \frac{1}{6} (\min(z, 3) - \max(0, z-2)).$$

We now consider three cases based on the value of z :

Case 1: $0 \leq z \leq 2$ Here, $z - 2 \leq 0$, so $\max(0, z - 2) = 0$ and $\min(z, 3) = z$. Thus,

$$f_Z(z) = \frac{1}{6}(z - 0) = \frac{z}{6}.$$

Case 2: $2 < z \leq 3$ Now, $z - 2 > 0$ and $z \leq 3$ so $\max(0, z - 2) = z - 2$ and $\min(z, 3) = z$. Therefore,

$$f_Z(z) = \frac{1}{6}(z - (z - 2)) = \frac{1}{6} \cdot 2 = \frac{1}{3}.$$

Case 3: $3 < z \leq 5$ In this case, $z > 3$ so $\min(z, 3) = 3$ and $\max(0, z - 2) = z - 2$. Thus,

$$f_Z(z) = \frac{1}{6}(3 - (z - 2)) = \frac{1}{6}(5 - z).$$

Summarizing, the pdf of Z is

$$f_Z(z) = \begin{cases} \frac{z}{6}, & 0 \leq z \leq 2, \\ \frac{1}{3}, & 2 < z \leq 3, \\ \frac{5-z}{6}, & 3 < z \leq 5, \\ 0, & \text{otherwise.} \end{cases}$$

■

Exercise 5.4 (Sum of two exponential random variables with different rates). Let X and Y be independent random variables with exponential distributions:

$$X \sim \text{Exp}(\lambda_1), \quad f_X(x) = \lambda_1 e^{-\lambda_1 x}, \quad x \geq 0,$$

$$Y \sim \text{Exp}(\lambda_2), \quad f_Y(y) = \lambda_2 e^{-\lambda_2 y}, \quad y \geq 0,$$

with $\lambda_1 \neq \lambda_2$. Find the probability density function of $Z = X + Y$ using convolution.

Solution. The pdf of Z is given by:

$$f_Z(z) = \int_0^z f_X(z - y) f_Y(y) dy = \int_0^z \lambda_1 e^{-\lambda_1(z-y)} \lambda_2 e^{-\lambda_2 y} dy.$$

i.e.

$$f_Z(z) = \lambda_1 \lambda_2 e^{-\lambda_1 z} \int_0^z e^{-(\lambda_2 - \lambda_1)y} dy.$$

If we assume $\lambda_2 \neq \lambda_1$ the integral is evaluated as:

$$\int_0^z e^{-(\lambda_2 - \lambda_1)y} dy = \frac{1 - e^{-(\lambda_2 - \lambda_1)z}}{\lambda_2 - \lambda_1}.$$

Thus, the pdf of Z is:

$$f_Z(z) = \frac{\lambda_1 \lambda_2}{\lambda_2 - \lambda_1} (e^{-\lambda_1 z} - e^{-\lambda_2 z}), \quad z \geq 0.$$

■

Exercise 5.5. Stability of the Cauchy distribution

Let X and Y be two independent standard Cauchy random variables. Show that the sum

$$Z = X + Y$$

is also Cauchy distributed and determine its parameters.

Solution. Let us first define characteristic function of a random variable.

Let X be a real-valued random variable with probability distribution $F_X(x)$. The *characteristic function* of X is defined as

$$\varphi_X(t) = \mathbb{E}[e^{itX}] = \int_{-\infty}^{\infty} e^{itx} dF_X(x), \quad t \in \mathbb{R}.$$

The characteristic function has the following properties:

- **Normalization:** $\varphi_X(0) = 1$.
- **Symmetry:** $\varphi_X(-t) = \overline{\varphi_X(t)}$, where $\overline{\varphi_X(t)}$ denotes the complex conjugate of $\varphi_X(t)$.
- **Uniqueness:** The characteristic function uniquely determines the probability distribution of X .
- **Uniform Continuity:** $\varphi_X(t)$ is uniformly continuous in t .
- **Independence and Convolution:** If X and Y are independent random variables, then the characteristic function of their sum is the product of their individual characteristic functions:

$$\varphi_{X+Y}(t) = \varphi_X(t) \varphi_Y(t).$$

- **Inversion Formula:** Under suitable conditions, the probability distribution F_X can be recovered from $\varphi_X(t)$ via the Fourier inversion formula.
- **Moment Generation:** If X has moments, they can be obtained by differentiating $\varphi_X(t)$ at $t = 0$. In particular, if $\mathbb{E}[|X|^n] < \infty$, then

$$\varphi_X^{(n)}(0) = i^n \mathbb{E}[X^n].$$

A standard Cauchy random variable X has the probability density function

$$f_X(x) = \frac{1}{\pi(1+x^2)}, \quad x \in \mathbb{R},$$

and its characteristic function is given by

$$\varphi_X(t) = \mathbb{E}[e^{itX}] = e^{-|t|}.$$

Since X and Y are independent and identically distributed, the characteristic function of Y is also

$$\varphi_Y(t) = e^{-|t|}.$$

The characteristic function of the sum $Z = X + Y$ is the product of the individual characteristic functions:

$$\varphi_Z(t) = \varphi_X(t) \varphi_Y(t) = e^{-|t|} \cdot e^{-|t|} = e^{-2|t|}.$$

We know that Cauchy distribution with location parameter 0 and scale parameter γ has the characteristic function

$$\varphi(t) = e^{-\gamma|t|}.$$

Thus, $e^{-2|t|}$ is the characteristic function of a Cauchy distribution with location 0 and scale parameter 2.

Therefore, we conclude that

$$Z = X + Y \sim \text{Cauchy}(0, 2).$$

This example illustrates the *stability property* of the Cauchy distribution: the sum of two independent standard Cauchy random variables is itself Cauchy, up to a scaling factor. ■

Exercise 5.6. Independent Binomials Prove that, for any two Binomial random variables with the same success probability:

$$X \sim \text{Bin}(n_1, p) \quad \text{and} \quad Y \sim \text{Bin}(n_2, p),$$

the sum of these random variables is also Binomial:

$$X + Y \sim \text{Bin}(n_1 + n_2, p).$$

Remark. . This result does not hold if the two distributions have different success probabilities.

Exercise 5.7. Independent Poissons Prove that, for any two Poisson random variables:

$$X \sim \text{Poi}(\lambda_1) \quad \text{and} \quad Y \sim \text{Poi}(\lambda_2),$$

the sum is also Poisson, i.e.

$$X + Y \sim \text{Poi}(\lambda_1 + \lambda_2).$$

This property holds regardless of whether λ_1 and λ_2 are equal.

Exercise 5.8. Independent Normals Prove that, for any two Normal random variables

$$X \sim N(\mu_1, \sigma_1^2) \quad \text{and} \quad Y \sim N(\mu_2, \sigma_2^2),$$

the sum is also normal, i.e.

$$X + Y \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2).$$

6 Projects

Exercise 6.1. Collect data on city sizes in USA from <https://simplemaps.com/data/us-cities>. Visualize and statistically analyze the data.

Exercise 6.2. Collect data from World Inequality database <https://wid.world/research-tools/> and see if Pareto distribution is suitable for analyzing.

Exercise 6.3. Collect data on Drug-response and see the applicability of logistic distribution. Details for analysis can be found here <https://uncsrp.github.io/Data-Analysis-Training-Modules/dose-response-modeling.html>

Exercise 6.4. Collect data on population growth and analyze it. Possible source <https://ourworldindata.org/un-population-2024-revision>

Exercise 6.5. Research on Weibull distribution for modeling of wind speeds. Check <https://datacatalog.worldbank.org/search/dataset/0038643> and other resources.

Exercise 6.6. Research and implement techniques like Laplace and Gaussian mechanism with data for differential privacy. Include noise distribution analysis, sensitivity analysis, trade-off curve (error vs privacy parameter), visualisation, statistical tests etc.

7 Discrete Multivariate Distributions

(The following section is not represented with all the details)

Definition 7.1. Joint PMF for Discrete d -Dimensional Random Vectors

Let $\mathbf{X} = (X_1, X_2, \dots, X_d)$ be a d -dimensional discrete random vector defined on a probability space $(\Omega, \mathcal{F}, P)$, where each component X_i takes values in a discrete set $S_i \subseteq \mathbb{R}$. The **joint probability mass function (PMF)** of $\mathbf{X}$ is a function:

$$p_{\mathbf{X}}(x_1, x_2, \dots, x_d) = P(X_1 = x_1, X_2 = x_2, \dots, X_d = x_d),$$

defined for all $(x_1, x_2, \dots, x_d) \in S_1 \times S_2 \times \dots \times S_d$, and satisfying:

1. $p_{\mathbf{X}}(x_1, x_2, \dots, x_d) \geq 0$ for all $(x_1, x_2, \dots, x_d)$.
2. $\sum_{(x_1, x_2, \dots, x_d) \in S_1 \times S_2 \times \dots \times S_d} p_{\mathbf{X}}(x_1, x_2, \dots, x_d) = 1$.

Definition 7.2. Marginal PMF

Let $\mathbf{X} = (X_1, X_2, \dots, X_d)$ be a discrete random vector with **joint PMF** $p_{\mathbf{X}}(x_1, x_2, \dots, x_d)$.

Let us consider arbitrary partition of variables into two subsets:

$$\mathbf{X}_1 = (X_{i_1}, X_{i_2}, \dots, X_{i_k}), \quad \mathbf{X}_2 = (X_{i_{k+1}}, X_{i_{k+2}}, \dots, X_{i_{d-k}}),$$

where $\mathbf{X}_1$ consists of k variables and $\mathbf{X}_2$ consists of the remaining $d - k$ variables,

- The **marginal PMF** of X_i is obtained by summing over all other variables:

$$p_{X_i}(x_{i_1}, \dots, x_{i_k}) = \sum_{x_{i_j} \in S_{i_j}, j \in \{k+1, \dots, d-k\}} p_{\mathbf{X}}(x_1, x_2, \dots, x_d), \quad x_{i_l} \in S_{i_l}.$$

For example, the marginal of a r.v. X_i is given by

$$p_{X_i}(x_i) = \sum_{\substack{x_j \in S_j \\ j \neq i}} p_{\mathbf{X}}(x_1, x_2, \dots, x_d), \quad x_i \in S_i.$$

- The **conditional PMF** of $\mathbf{X}_1$ given $\mathbf{X}_2 = \mathbf{x}_2$ is:

$$\begin{aligned} p_{\mathbf{X}_1|\mathbf{X}_2}(\mathbf{x}_1 | \mathbf{x}_2) &= p_{\mathbf{X}_1|\mathbf{X}_2}(x_{i_1}, \dots, x_{i_k} | x_{i_{k+1}}, \dots, x_{i_d}) \\ &= \frac{p(x_1, x_2, \dots, x_d)}{p_{\mathbf{X}_2}(x_{i_{k+1}}, \dots, x_{i_d})} = \frac{p_{\mathbf{X}}(\mathbf{x}_1, \mathbf{x}_2)}{p_{\mathbf{X}_2}(\mathbf{x}_2)}, \end{aligned}$$

where: - $f_{\mathbf{X}}(\mathbf{x}_1, \mathbf{x}_2)$ is the joint density of all variables.

For example, given that $X_j = x_j$, we can calculate:

$$p(x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_d | x_j) = \frac{p(x_1, x_2, \dots, x_d)}{p_{X_j}(x_j)}, \quad \text{for } p_{X_j}(x_j) > 0.$$

Sometimes, as in the equation above we can omit the index, but we have in mind which is the corresponding density.

Exercise 7.3. Consider a company that interviews candidates for a job and evaluates them based on two criteria: candidate's **performance in the interview** given with r.v. X (Good = 1, Bad = 0) and Candidate's **hiring decision** represented by r.v. Y (Hired = 1, Not Hired = 0)

The probabilities of these outcomes are given in the following **joint probability mass function (PMF)** table:

$P(X, Y)$	$Y = 1$ (Hired)	$Y = 0$ (Not Hired)
$X = 1$ (Good Interview)	0.4	0.2
$X = 0$ (Bad Interview)	0.1	0.3

- i) Compute the **marginal PMF** of X and Y .
- ii) Compute the **conditional probabilities**:
 - $P(Y = 1 \mid X = 1)$ and $P(Y = 1 \mid X = 0)$.
 - $P(X = 1 \mid Y = 1)$ and $P(X = 1 \mid Y = 0)$.
- iii) Interpret the results: What do the marginal and conditional probabilities tell us about the relationship between interview performance and hiring decisions?

Solution.

- i) **Compute the Marginal PMFs**

The **marginal PMF of X** is obtained by summing over all values of Y :

$$P(X = 1) = P(1, 1) + P(1, 0) = 0.4 + 0.2 = 0.6$$

$$P(X = 0) = P(0, 1) + P(0, 0) = 0.1 + 0.3 = 0.4$$

The **marginal PMF of Y** is obtained by summing over all values of X :

$$P(Y = 1) = P(1, 1) + P(0, 1) = 0.4 + 0.1 = 0.5$$

$$P(Y = 0) = P(1, 0) + P(0, 0) = 0.2 + 0.3 = 0.5$$

- ii) **Conditional probability of hiring given interview performance:**

$$P(Y = 1 \mid X = 1) = \frac{P(1, 1)}{P(X = 1)} = \frac{0.4}{0.6} = 0.67$$

$$P(Y = 1 \mid X = 0) = \frac{P(0, 1)}{P(X = 0)} = \frac{0.1}{0.4} = 0.25$$

Conditional probability of interview performance given hiring decision:

$$P(X = 1 \mid Y = 1) = \frac{P(1, 1)}{P(Y = 1)} = \frac{0.4}{0.5} = 0.8$$

$$P(X = 1 \mid Y = 0) = \frac{P(1, 0)}{P(Y = 0)} = \frac{0.2}{0.5} = 0.4$$

- iii) **Interpret the results**

- The marginal probabilities indicate that 60% of candidates performed well, and 50% were hired.

- The conditional probability $P(Y = 1 \mid X = 1) = 0.67$ suggests that candidates who performed well had a 67% chance of being hired.
- The conditional probability $P(Y = 1 \mid X = 0) = 0.25$ shows that even with a poor interview, there is still a 25% chance of getting hired.
- The conditional probability $P(X = 1 \mid Y = 1) = 0.8$ implies that 80% of hired candidates performed well in the interview.

$P(X, Y)$	$Y = 1$ (Hired)	$Y = 0$ (Not Hired)	$P(X)$ (Marginal)
$X = 1$ (Good Interview)	0.4	0.2	0.6
$X = 0$ (Bad Interview)	0.1	0.3	0.4
$P(Y)$ (Marginal)	0.5	0.5	1.0

Table 19: Joint PMF with Marginal Distributions

■

Remark. For a joint probability mass function (PMF) $P(X, Y)$, the marginal probability of X is computed by summing over all possible values of Y :

$$P(X = x) = \sum_y P(X = x, Y = y).$$

This follows the **law of total probability**, which states that if Y is a discrete random variable partitioning the sample space, then:

$$P(X = x) = \sum_y P(X = x \mid Y = y)P(Y = y).$$

Marginal probability actually tells us the overall likelihood of a single variable, without considering other variables.

Definition 7.4. Independence of Discrete Random Variables

The discrete random variables $X_1, X_2, \dots, X_d$ are said to be **independent** if and only if their **joint PMF** factorizes into the product of their **marginal PMFs**:

$$p_{\mathbf{X}}(x_1, x_2, \dots, x_d) = \prod_{i=1}^d p_{X_i}(x_i),$$

for all $(x_1, x_2, \dots, x_d) \in S_1 \times S_2 \times \dots \times S_d$.

Exercise 7.5. In the previous example, determine whether X and Y are independent.

Solution. To determine whether the random variables X (Interview Performance) and Y (Hiring Decision) are **independent**, we have to check if:

$$P(X = x, Y = y) = P(X = x)P(Y = y) \quad \text{for all } x, y.$$

$$P(X = 1)P(Y = 1) = (0.6)(0.5) = 0.3 \neq P(1, 1) = 0.4.$$

$$P(X = 1)P(Y = 0) = (0.6)(0.5) = 0.3 \neq P(1, 0) = 0.2.$$

$$P(X = 0)P(Y = 1) = (0.4)(0.5) = 0.2 \neq P(0, 1) = 0.1.$$

$$P(X = 0)P(Y = 0) = (0.4)(0.5) = 0.2 \neq P(0, 0) = 0.3.$$

We conclude that the variables X (Interview Performance) and Y (Hiring Decision) are **not independent**. This means that **interview performance influences the hiring decision**, and the hiring decision does not occur independently of the candidate's performance. ■

Example 7.6. Splitting web server requests into humans and bots

Let N be the number of requests to a web server per day, with

$$N \sim \text{Poi}(\lambda).$$

Each request comes independently from a human (with probability p) or from a bot (with probability $1 - p$). Define:

X = number of requests from humans per day,

Y = number of requests from bots per day.

Given N requests in total, the number of human requests follows a Binomial distribution:

$$X \mid N \sim \text{Bin}(N, p),$$

and similarly, since every request is either human or bot, we have

$$Y \mid N \sim \text{Bin}(N, 1 - p).$$

We wish to calculate the probability of getting exactly i human requests and j bot requests, i.e., $P(X = i, Y = j)$.

Using the chain rule, we have:

$$P(X = i, Y = j) = P(X = i, Y = j \mid X + Y = i + j) P(X + Y = i + j).$$

Since $X + Y = N$ and given $N = i + j$ the number of human requests is distributed as

$$X \mid (X + Y = i + j) \sim \text{Bin}(i + j, p),$$

we can write:

$$P(X = i, Y = j \mid X + Y = i + j) = \binom{i + j}{i} p^i (1 - p)^j.$$

Also, since $X + Y = N$ and $N \sim \text{Poi}(\lambda)$, we have:

$$P(X + Y = i + j) = P(N = i + j) = e^{-\lambda} \frac{\lambda^{i+j}}{(i + j)!}.$$

Thus, combining these results:

$$P(X = i, Y = j) = \binom{i + j}{i} p^i (1 - p)^j e^{-\lambda} \frac{\lambda^{i+j}}{(i + j)!}.$$

We can additionally show that this joint probability can be factorized into the product of two independent Poisson probabilities:

$$X \sim \text{Poi}(p\lambda) \quad \text{and} \quad Y \sim \text{Poi}((1 - p)\lambda).$$

□

Definition 7.7. Multinomial distribution

The multivariate version of a Binomial is called a **Multinomial distribution**. Consider drawing a ball from an urn which contains balls of k different colors, labeled “color 1, color 2, ..., color k ”. Let p_j be the probability of drawing a ball of color j ,

$$p = (p_1, p_2, \dots, p_k),$$

with $p_j \geq 0$ for each j and

$$\sum_{j=1}^k p_j = 1.$$

Suppose we perform n independent draws with replacement. Define $X = (X_1, X_2, \dots, X_k)$, where X_j is the number of times that color j appears, so that

$$n = \sum_{j=1}^k X_j.$$

We say that X has a Multinomial(n, p) distribution, written $X \sim \text{Multinomial}(n, p)$. The probability mass function is given by

$$p(x_1, \dots, x_k) = \binom{n}{x_1, x_2, \dots, x_k} p_1^{x_1} p_2^{x_2} \cdots p_k^{x_k},$$

where the multinomial coefficient is defined as

$$\binom{n}{x_1, x_2, \dots, x_k} = \frac{n!}{x_1! x_2! \cdots x_k!}.$$

The marginal distribution of X_j is Binomial(n, p_j).

Definition 7.8. Expectation of a Function of a Discrete Random Vector

Let $g : S_1 \times S_2 \times \cdots \times S_d \rightarrow \mathbb{R}$ be a real-valued function of a discrete random vector $\mathbf{X} = (X_1, X_2, \dots, X_d)$. The **expectation** of $g(\mathbf{X})$ is given by:

$$E[g(\mathbf{X})] = \sum_{\substack{x_i \in S_i \\ i=1, \dots, d}} g(x_1, x_2, \dots, x_d) p_{\mathbf{X}}(x_1, x_2, \dots, x_d).$$

Definition 7.9. Covariance Between Two Discrete Random Variables

Let X, Y be two discrete random variables with expectations $E[X]$ and $E[Y]$. The **covariance** between X and Y is defined as:

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])].$$

Equivalently, it can be computed as:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y].$$

Definition 7.10. Correlation Coefficient

The **correlation coefficient** (or Pearson’s correlation coefficient) between two discrete random variables X and Y is given by:

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y},$$

where $\sigma_X = \sqrt{\text{Var}(X)}$ and $\sigma_Y = \sqrt{\text{Var}(Y)}$.

Feature	Covariance $\text{Cov}(X, Y)$	Correlation $\rho_{X,Y}$
Measures	Direction of relationship	Direction and strength of relationship
Range	$(-\infty, \infty)$ (unbounded)	$[-1, 1]$ (bounded)
Units	Depends on the units of X and Y	Unitless (standardized)
Comparability	Hard to compare across datasets	Easily comparable across different datasets
Effect of Scaling	Affected by scaling (e.g., multiplying X or Y by a constant changes covariance)	Not affected by scaling (remains the same when X or Y is multiplied by a constant)

Table 20: Comparison of Covariance and Correlation

Definition 7.11. Sample Correlation Coefficient

Let $(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$ be a sample of size n from a bivariate distribution. The **sample correlation coefficient** r is defined as:

$$r = \frac{S_{XY}}{S_X S_Y}$$

where:

- S_{XY} is the **sample covariance**, given by:

$$S_{XY} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})$$

- S_X and S_Y are the **sample standard deviations**, given by:

$$S_X = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2}, \quad S_Y = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2}$$

- $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ is the sample mean of X ,
- $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$ is the sample mean of Y .

Example 7.12. To illustrate the meaning of correlation and covariance, we consider the following example - Let $X \sim \mathcal{N}(5, 2)$ and $Y = 2X + e$, where $e \sim \mathcal{N}(0, 1)$. Obviously there is a linear dependence between X and Y . Next we simulate 1000 values for both random variables.

The first figure represents the relative frequencies of both random variables and their standardized versions

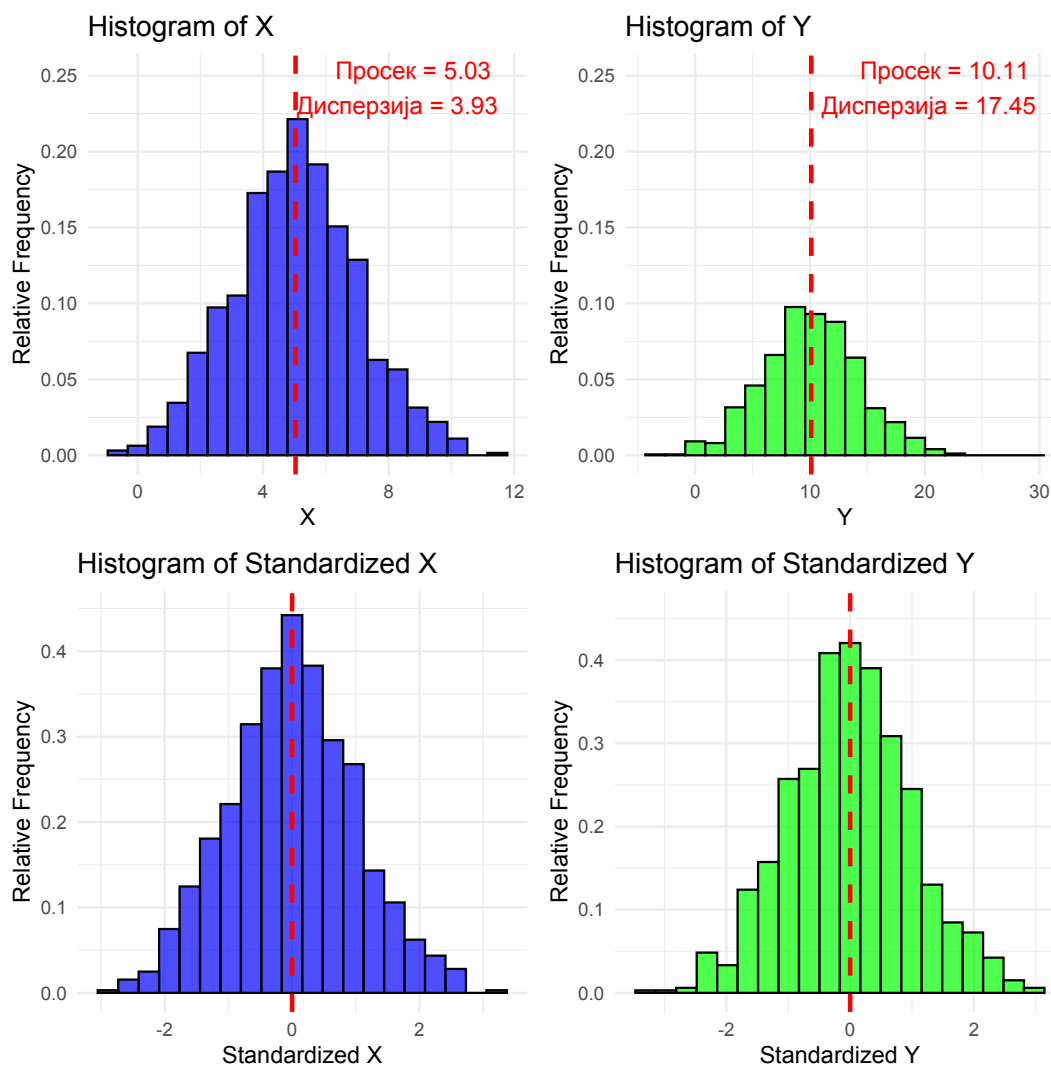


Figure 17

The second figure represents the scatter plot of (X,Y) and the corresponding standardized data

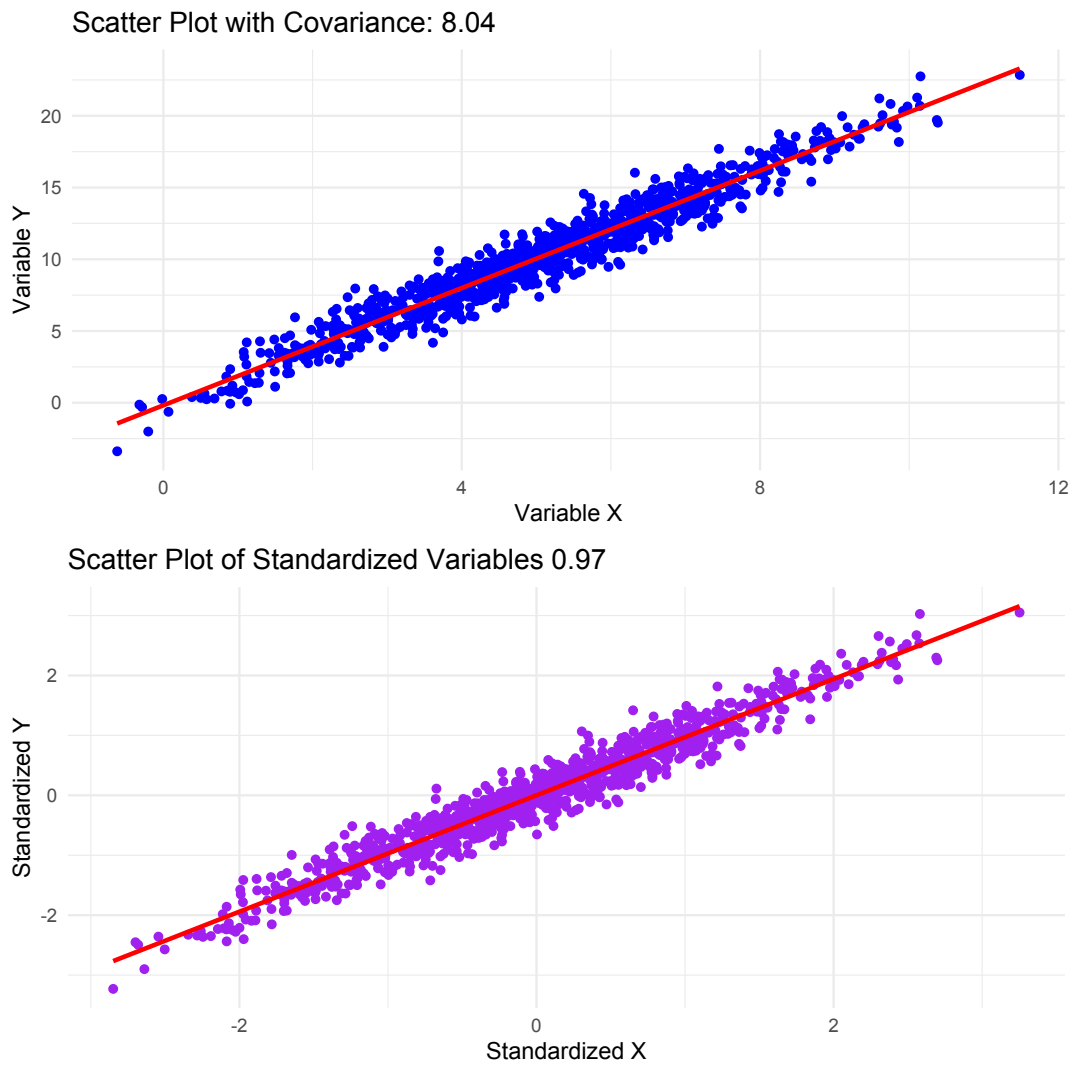


Figure 18

The third figure represents the relative frequencies of the deviation product and the scaled product with their means, which are corresponding estimated covariance and correlation coefficient of the data.

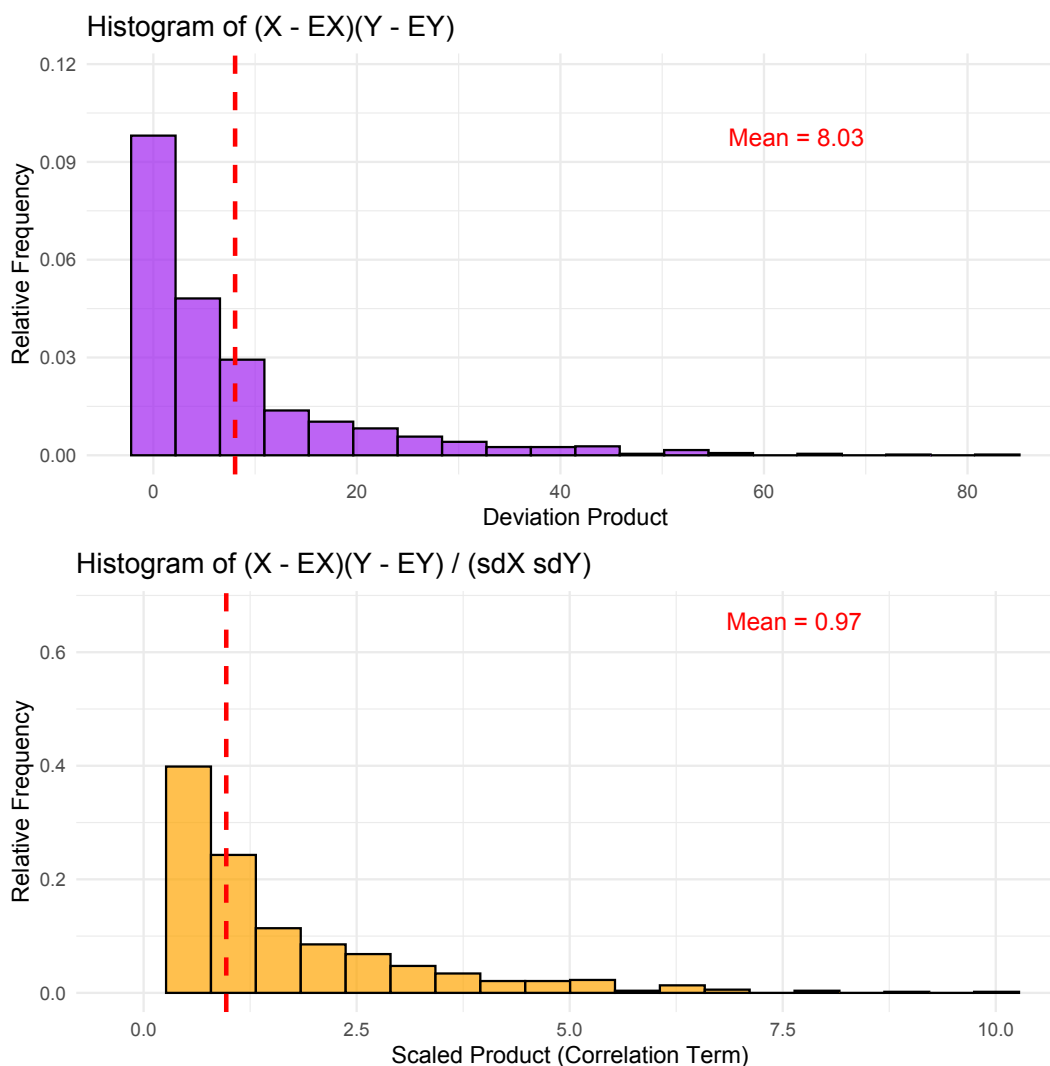


Figure 19

□

Remark. Check details of correlation vs. independence.

8 Continuous Multivariate Distributions

(The following section is not represented with all the details)

A **continuous multivariate distribution** describes the probability structure of a vector of continuous random variables. Instead of assigning probabilities to specific values, continuous distributions assign probability to **regions** via the **joint probability density function (PDF)**.

Definition 8.1 (Joint Probability Density Function (PDF)). Let $\mathbf{X} = (X_1, X_2, \dots, X_d)$ be a d -dimensional continuous random vector. The **joint probability density function (PDF)** is a function:

$$f_{\mathbf{X}}(x_1, x_2, \dots, x_d)$$

such that for any region $R \subseteq \mathbb{R}^d$, the probability that $\mathbf{X}$ falls within R is:

$$P((X_1, X_2, \dots, X_d) \in R) = \int_R f_{\mathbf{X}}(x_1, x_2, \dots, x_d) dx_1 dx_2 \dots dx_d.$$

Theorem 8.2 (Properties of Joint PDFs). For any valid joint PDF:

1. $f_{\mathbf{X}}(x_1, x_2, \dots, x_d) \geq 0$ for all $(x_1, x_2, \dots, x_d)$.
2. The total probability integrates to 1:

$$\int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} f_{\mathbf{X}}(x_1, x_2, \dots, x_d) dx_1 \dots dx_d = 1.$$

Example 8.3 (Bivariate Uniform Distribution). A simple example of a continuous multivariate distribution is the **bivariate uniform distribution** over the unit square $[0, 1] \times [0, 1]$, where:

$$f_{X,Y}(x, y) = \begin{cases} 1, & 0 \leq x \leq 1, \quad 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

This means that any point inside the square is equally likely. □

Example 8.4 (Bivariate Normal Distribution). A widely used example is the **bivariate normal distribution**, given by:

$$f_{X,Y}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)} \left[\frac{(x-\mu_X)^2}{\sigma_X^2} - \frac{2\rho(x-\mu_X)(y-\mu_Y)}{\sigma_X\sigma_Y} + \frac{(y-\mu_Y)^2}{\sigma_Y^2} \right]\right).$$

The parameters μ_X, μ_Y (means), σ_X, σ_Y (standard deviations), and ρ (correlation) fully define the distribution. □

Definition 8.5. Marginal and conditional probability density function (PDF)

Let $\mathbf{X} = (X_1, X_2, \dots, X_d)$ be a d -dimensional continuous random vector with joint probability density function $f_{\mathbf{X}}(x_1, x_2, \dots, x_d)$. The **marginal PDF** of a subset of variables is obtained by integrating out the unwanted variables.

Let us consider arbitrary partition of variables into two subsets:

$$\mathbf{X}_1 = (X_{i_1}, X_{i_2}, \dots, X_{i_k}), \quad \mathbf{X}_2 = (X_{i_{k+1}}, X_{i_{k+2}}, \dots, X_{i_{d-k}}),$$

where $\mathbf{X}_1$ consists of k variables and $\mathbf{X}_2$ consists of the remaining $d - k$ variables,

- $f_{\mathbf{X}_2}(\mathbf{x}_2)$ is the **marginal density** of $\mathbf{X}_2$, obtained by integrating out $\mathbf{X}_1$:

$$f_{\mathbf{X}_2}(\mathbf{x}_2) = f_{\mathbf{X}_2}(x_{i_{k+1}}, \dots, x_{i_{d-k}}) = \int_{\mathbb{R}^k} f_{\mathbf{X}}(\mathbf{x}_1, \mathbf{x}_2) d\mathbf{x}_1 = \int_{\mathbb{R}^k} f_{\mathbf{X}}(x_1, x_2, \dots, x_d) dx_{i_1} \dots dx_{i_k}$$

For example, the marginal PDF of X_1 is:

$$f_{X_1}(x_1) = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} f_{\mathbf{X}}(x_1, x_2, \dots, x_d) dx_2 \dots dx_d.$$

- The **conditional density function** of $\mathbf{X}_1$ given $\mathbf{X}_2 = \mathbf{x}_2$ is:

$$\begin{aligned} f_{\mathbf{X}_1|\mathbf{X}_2}(\mathbf{x}_1 | \mathbf{x}_2) &= f_{\mathbf{X}_1|\mathbf{X}_2}(x_{i_1}, \dots, x_{i_k} | x_{i_{k+1}}, \dots, x_{i_{d-k}}) \\ &= \frac{f(x_1, x_2, \dots, x_d)}{f_{\mathbf{X}_2}(x_{i_{k+1}}, \dots, x_{i_{d-k}})} = \frac{f_{\mathbf{X}}(\mathbf{x}_1, \mathbf{x}_2)}{f_{\mathbf{X}_2}(\mathbf{x}_2)}, \end{aligned}$$

where $f_{\mathbf{X}}(\mathbf{x}_1, \mathbf{x}_2)$ is the joint density of all variables.

Remark. Marginal and conditional probability distributions have the properties as any other distribution (non-negative density that integrates to 1 etc.)

Example 8.6 (Marginals of a Bivariate Normal Distribution). For a bivariate normal $(X, Y) \sim \mathcal{N}(\mu, \Sigma)$, the marginal distributions are:

$$X \sim \mathcal{N}(\mu_X, \sigma_X^2), \quad Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2).$$

□

Remark. One of the most useful multidimensional distributions is multivariate normal, especially the independent multivariate normal (where individual components of the vector are independent between each other). See for details in standard probability text books. For some definitions and examples of joint distributions and multiple random variables check https://www.probabilitycourse.com/chapter5/5_1_0_joint_distributions.php Check the theorem for correlation vs. independence.

Example 8.7. To understand the meaning of joint, marginal and conditional distributions lets some visualisations (taken from https://en.wikipedia.org/wiki/Multivariate_normal_distribution) and <https://stats.stackexchange.com/questions/656358/visualisation-of-a-conditional-distribution>.

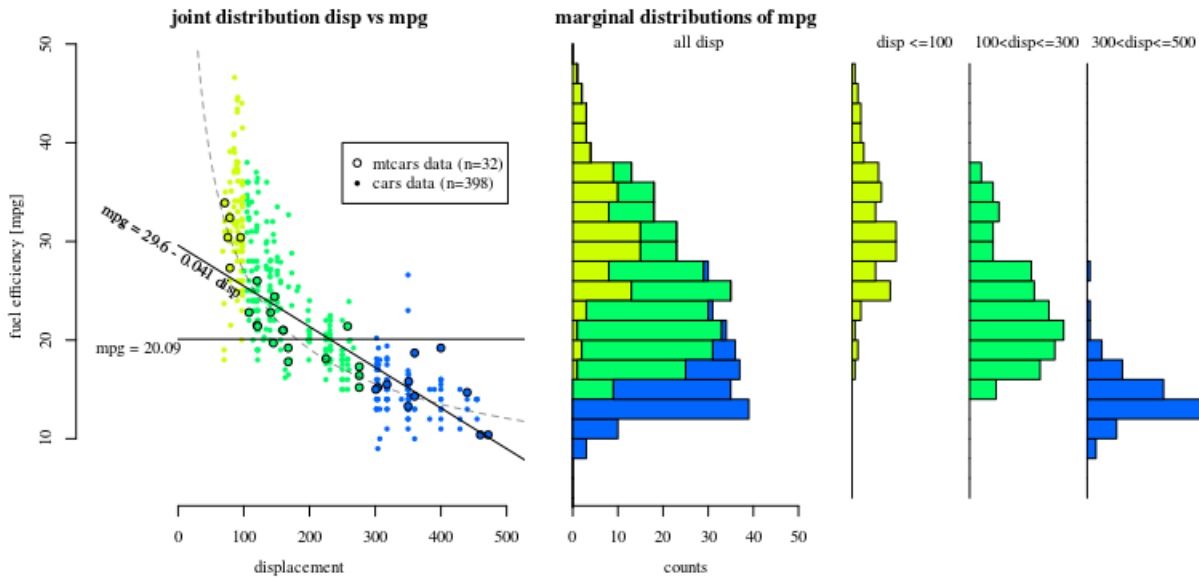


Figure 20: Joint, marginal and conditional distributios

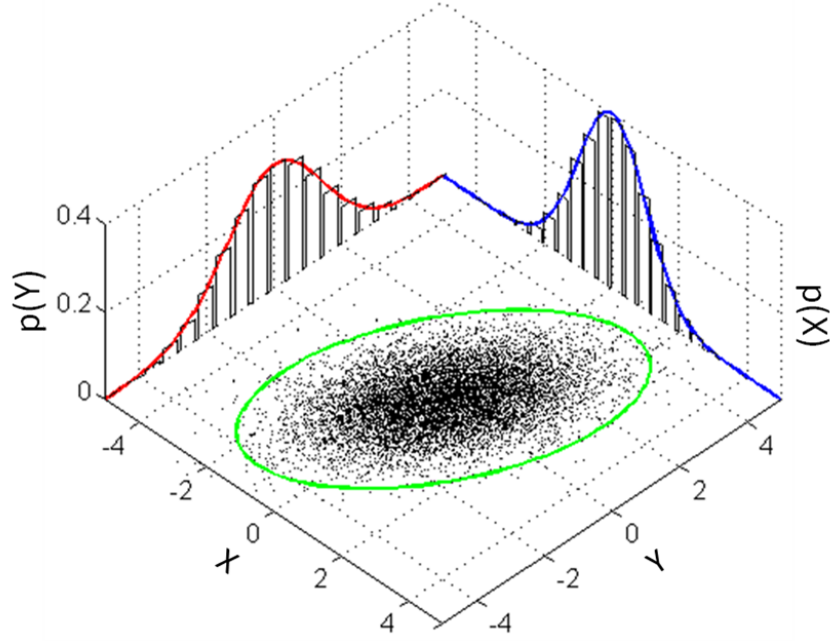


Figure 21: Multivariate normal distribution

Contour plots of a multinormal distribution display the regions where the density is constant. In the bivariate case, these iso-density curves are ellipses centered at the mean. Their shape and orientation are determined by the covariance matrix, with the eigenvalues indicating the spread in each direction and the eigenvectors indicating the axes of the ellipses. In higher dimensions, these contours become ellipsoids.

Their orientation and shape are determined by the covariance matrix: The center of the ellipses is the mean μ . The axes of the ellipses align with the eigenvectors of the covariance matrix Σ . The lengths of the semi-axes are related to the square roots of the eigenvalues of Σ . These contour plots visually represent how the density "spreads out" in different directions and how the correlation between variables affects the shape of the distribution.

For a bivariate normal distribution, the iso-density ellipses are circles when $\sigma_1^2 = \sigma_2^2$ and $\rho = 0$ (covariance matrix is proportional to the identity matrix).

If the variables are only uncorrelated (i.e. the correlation $\rho = 0$) but have different variances, then the covariance matrix is diagonal but not proportional to the identity. In that case, the iso-density contours are ellipses with axes aligned with the coordinate axes, and their radii are proportional to the standard deviations.

□

Let us consider the case when we have two random variables (discrete or continuous). We will give the law of total probability and see its connection to the definition of marginal distributions.

Theorem 8.8. Law of total probability for random variables

Let X and Y be two random variables defined on the same probability space. The marginal probability distribution of X is obtained by summing or integrating over all possible values of Y , depending on whether Y is discrete or continuous.

- **Let X and Y be both discrete**

Suppose X takes values in a countable set $\mathcal{X} \subseteq \mathbb{R}$ and Y takes values in a countable set $\mathcal{Y} \subseteq \mathbb{R}$. The probability mass function (PMF) of X is given by:

$$p_X(x) = \sum_{y \in \mathcal{Y}} P(X = x \mid Y = y)p_Y(y), \quad x \in \mathcal{X}.$$

- **Let X be continuous and Y be discrete**

Suppose X takes values in $\mathcal{X} \subseteq \mathbb{R}$ (which may be an interval), while Y takes values in a countable set $\mathcal{Y} \subseteq \mathbb{R}$. The probability density function (PDF) of X is given by:

$$f_X(x) = \sum_{y \in \mathcal{Y}} f_{X|Y}(x | y) p_Y(y), \quad x \in \mathcal{X}.$$

- **Let X be discrete and Y be continuous**

Suppose X takes values in a countable set $\mathcal{X} \subseteq \mathbb{R}$ and Y takes values in an interval $\mathcal{Y} \subseteq \mathbb{R}$. The probability mass function (PMF) of X is given by:

$$p_X(x) = \int_{\mathcal{Y}} P(X = x | Y = y) f_Y(y) dy, \quad x \in \mathcal{X}.$$

- **Let X and Y be both continuous**

Suppose X takes values in $\mathcal{X} \subseteq \mathbb{R}$ (which may be an interval), and Y takes values in $\mathcal{Y} \subseteq \mathbb{R}$ (also an interval). The probability density function (PDF) of X is given by:

$$f_X(x) = \int_{\mathcal{Y}} f_{X|Y}(x | y) f_Y(y) dy, \quad x \in \mathcal{X}.$$

Example 8.9. Suppose a student takes an exam where the grade X depends on the difficulty level Y of the exam. The difficulty level Y can be:

- Easy ($Y = 1$) with probability $P(Y = 1) = 0.6$,
- Hard ($Y = 2$) with probability $P(Y = 2) = 0.4$.

The probability mass function (PMF) of the grade X given Y is:

$$P(X = x | Y = 1) = \begin{cases} 0.5, & x = A \\ 0.3, & x = B \\ 0.2, & x = C \end{cases}$$

$$P(X = x | Y = 2) = \begin{cases} 0.2, & x = A \\ 0.5, & x = B \\ 0.3, & x = C \end{cases}$$

Find the marginal probability mass function $P(X = x)$. □

Solution. Using the law of total probability:

$$P(X = x) = \sum_{y \in \mathcal{Y}} P(X = x | Y = y) P(Y = y),$$

we compute:

$$P(X = A) = (0.5 \times 0.6) + (0.2 \times 0.4) = 0.3 + 0.08 = 0.38.$$

$$P(X = B) = (0.3 \times 0.6) + (0.5 \times 0.4) = 0.18 + 0.2 = 0.38.$$

$$P(X = C) = (0.2 \times 0.6) + (0.3 \times 0.4) = 0.12 + 0.12 = 0.24.$$

Thus, the marginal PMF of X is:

$$P(X = A) = 0.38, \quad P(X = B) = 0.38, \quad P(X = C) = 0.24.$$

■

Example 8.10. Let X be the time (in minutes) a customer spends in a store, which depends on whether they are a *first-time* visitor ($Y = 1$) or a *returning* customer ($Y = 2$). The probability mass function of Y is:

$$P(Y = 1) = 0.3, \quad P(Y = 2) = 0.7.$$

The conditional probability density functions of X given Y are:

$$f_{X|Y}(x | 1) = \frac{1}{10}e^{-x/10}, \quad x > 0,$$

$$f_{X|Y}(x | 2) = \frac{1}{5}e^{-x/5}, \quad x > 0.$$

Find the marginal density function $f_X(x)$. □

Solution. Using the law of total probability for continuous X :

$$f_X(x) = \sum_{y \in \mathcal{Y}} f_{X|Y}(x | y)P(Y = y),$$

we compute:

$$\begin{aligned} f_X(x) &= f_{X|Y}(x | 1)P(Y = 1) + f_{X|Y}(x | 2)P(Y = 2). \\ &= \left(\frac{1}{10}e^{-x/10} \times 0.3 \right) + \left(\frac{1}{5}e^{-x/5} \times 0.7 \right). \\ &= 0.03e^{-x/10} + 0.14e^{-x/5}, \quad x > 0. \end{aligned}$$

■

Example 8.11. A machine can fail on any given day with probability $P(X = 1 | Y = y)$, where Y is the temperature of the factory (a continuous variable). The probability density function of Y is:

$$f_Y(y) = \frac{1}{10}, \quad 20 \leq y \leq 30.$$

The probability of failure given temperature y is:

$$P(X = 1 | Y = y) = \frac{y - 20}{10}, \quad 20 \leq y \leq 30.$$

Find $P(X = 1)$. □

Solution. Using the law of total probability:

$$\begin{aligned} P(X = 1) &= \int_{\mathcal{Y}} P(X = 1 | Y = y)f_Y(y) dy. \\ &= \int_{20}^{30} \left(\frac{y - 20}{10} \right) \times \frac{1}{10} dy. \\ &= \frac{1}{100} \int_{20}^{30} (y - 20) dy. \\ &= \frac{1}{100} \left[\frac{(y - 20)^2}{2} \right]_{20}^{30}. \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{100} \left(\frac{(30-20)^2}{2} - \frac{(20-20)^2}{2} \right) \\
&= \frac{1}{100} \times \frac{100}{2} = \frac{50}{100} = 0.5.
\end{aligned}$$

Thus, the probability of failure $P(X = 1)$ is 0.5. ■

Example 8.12. Let X and Y have the joint density function:

$$f_{X,Y}(x, y) = 2e^{-x}e^{-y}, \quad x > 0, y > 0.$$

Find the marginal density function $f_X(x)$. □

Solution. Using the law of total probability for continuous variables:

$$\begin{aligned}
f_X(x) &= \int_{\mathbb{R}} f_{X,Y}(x, y) dy. \\
&= \int_0^{\infty} 2e^{-x}e^{-y} dy. \\
&= 2e^{-x} \int_0^{\infty} e^{-y} dy. \\
&= 2e^{-x} [-e^{-y}]_0^{\infty}.
\end{aligned}$$

$$= 2e^{-x}(1 - 0) = 2e^{-x}, \quad x > 0.$$

Thus, the marginal density function of X is:

$$f_X(x) = 2e^{-x}, \quad x > 0.$$
■

Definition 8.13. Chain rule for random vectors and independence

Let $X = (X_1, X_2, \dots, X_d)$ be a random vector defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

• Discrete random variables

Let X_i be discrete random variables suitably defined, with a joint probability mass function (pmf)

$$p(x_1, x_2, \dots, x_d) = \mathbb{P}(X_1 = x_1, X_2 = x_2, \dots, X_d = x_d).$$

The chain rule (or product rule) for probabilities states that:

$$p(x_1, x_2, \dots, x_d) = p(x_1) p(x_2 | x_1) p(x_3 | x_1, x_2) \cdots p(x_d | x_1, \dots, x_{d-1}),$$

or more compactly,

$$p(x_1, x_2, \dots, x_d) = \prod_{i=1}^d p(x_i | x_1, x_2, \dots, x_{i-1}).$$

If for any $(x_1, x_2, \dots, x_n)$ $p(x_1, x_2, \dots, x_d) = \prod_{i=1}^d p_{X_i}(x_i)$

then $X_1, X_2, \dots, X_n$ are independent r.v.s. Here p_{X_i} are the corresponding marginal distributions.

- **Continuous Random Variables**

If X_i be continuous random variables suitably defined, $X = (X_1, \dots, X_d)$ has joint probability density function (pdf) $f(x_1, x_2, \dots, x_d)$ (with respect to an appropriate product measure, such as the Lebesgue measure)

$$\mathbb{P}(X \in A) = \int_A f(x_1, x_2, \dots, x_d) dx_1 dx_2 \cdots dx_d,$$

for any measurable set $A \subset \mathbb{R}^d$. In this case, the chain rule for densities is analogous:

$$f(x_1, x_2, \dots, x_d) = f(x_1) f(x_2 | x_1) f(x_3 | x_1, x_2) \cdots f(x_d | x_1, \dots, x_{d-1}),$$

or equivalently,

$$f(x_1, x_2, \dots, x_d) = \prod_{i=1}^d f(x_i | x_1, x_2, \dots, x_{i-1}),$$

where the conditional density is defined by

$$f(x_i | x_1, \dots, x_{i-1}) = \frac{f(x_1, x_2, \dots, x_i)}{f(x_1, x_2, \dots, x_{i-1})}, \quad \text{provided } f(x_1, x_2, \dots, x_{i-1}) > 0.$$

If for any $(x_1, x_2, \dots, x_n)$ $f(x_1, x_2, \dots, x_d) = \prod_{i=1}^d f_{X_i}(x_i)$

then $X_1, X_2, \dots, X_n$ are independent r.v.s. Here $f_{X_i}(x_i)$ are the corresponding marginals

Why does this work for densities? The chain rule for densities follows from the same principle as for discrete probabilities. In the discrete case, probabilities satisfy

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \mathbb{P}(B | A).$$

For continuous random variables, the density $f(x_1, \dots, x_d)$ is the Radon–Nikodym derivative of the joint probability measure with respect to the product measure. The conditional density is defined as

$$f(x_i | x_1, \dots, x_{i-1}) = \frac{f(x_1, \dots, x_i)}{f(x_1, \dots, x_{i-1})},$$

whenever $f(x_1, \dots, x_{i-1})$ is positive. Thus, by applying the definition iteratively, we obtain

$$f(x_1, x_2, \dots, x_d) = f(x_1) f(x_2 | x_1) \cdots f(x_d | x_1, \dots, x_{d-1}).$$

In other words, the fundamental property of conditional probability is preserved under differentiation (with respect to a suitable measure), which is why the chain rule holds for both probabilities and densities.

Remark. For independent random variables, the conditional PMF is equal to the marginal PMF. In other words, knowing the value of Y does not provide any information about X

$$p_{X|Y}(x_i | y_i) = p_X(x_i)$$

9 Likelihood Function

The likelihood function is one of the most fundamental concepts in statistical inference. It quantifies how probable the observed data are for different values of the unknown parameters. Unlike a probability function, which expresses the probability of an event given a parameter, the likelihood function treats the data as fixed and considers the parameter as a variable.

Likelihood-based methods, such as *Maximum Likelihood Estimation (MLE)*, are widely used in parameter estimation, model selection, and Bayesian inference.

10 Definition of the Likelihood Function

Definition 10.1 (Likelihood Function). Let $\mathbf{X} = (X_1, X_2, \dots, X_n)$ be a random sample from a probability distribution that depends on an unknown parameter θ . The likelihood function of θ , given the observed data $\mathbf{x} = (x_1, \dots, x_n)$, is defined as:

$$L(\theta \mid x_1, \dots, x_n) = f_{\mathbf{X}}(x_1, \dots, x_n \mid \theta),$$

or shortly written as

$$L(\theta \mid \mathbf{x}) = f_{\mathbf{X}}(\mathbf{x} \mid \theta),$$

where:

- $f_{\mathbf{X}}(\mathbf{x} \mid \theta)$ is the joint probability mass function (PMF) if $\mathbf{X}$ is discrete, or the joint probability density function (PDF) if $\mathbf{X}$ is continuous.
- $L(\theta \mid \mathbf{x})$ is interpreted as a function of θ , treating $\mathbf{x}$ as fixed.
- For a discrete random variable X with probability mass function (PMF) $p(X \mid \theta)$, the likelihood function for a sample $(x_1, x_2, \dots, x_n)$ is:

$$L(\theta \mid \mathbf{x}) = \prod_{i=1}^n p(x_i \mid \theta).$$

- For a continuous random variable X with probability density function (PDF) $f(X \mid \theta)$, the likelihood function is:

$$L(\theta \mid \mathbf{x}) = \prod_{i=1}^n f(x_i \mid \theta).$$

We have used here the independence of $X_1, \dots, X_n$.

Example 10.2 (Bernoulli Likelihood). Suppose $X_1, X_2, \dots, X_n$ are independent Bernoulli trials with unknown probability p :

$$P(X_i = 1) = p, \quad P(X_i = 0) = 1 - p.$$

Then the likelihood function is:

$$L(p \mid \mathbf{x}) = \prod_{i=1}^n p^{x_i} (1 - p)^{1-x_i} = p^{\sum_{i=1}^n x_i} (1 - p)^{n - \sum_{i=1}^n x_i}.$$

□

Example 10.3 (Normal Likelihood). Suppose $X_1, X_2, \dots, X_n$ are independent normal random variables with mean μ and variance σ^2 :

$$X_i \sim \mathcal{N}(\mu, \sigma^2).$$

The likelihood function is:

$$L(\mu, \sigma^2 \mid \mathbf{x}) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right).$$

□

Definition 10.4 (Log-Likelihood Function). The log-likelihood function is the natural logarithm of the likelihood function:

$$\ell(\theta \mid \mathbf{x}) = \log L(\theta \mid \mathbf{x}).$$

Example 10.5 (Log-Likelihood for Bernoulli Model). For the Bernoulli model:

$$\ell(p \mid \mathbf{x}) = \sum_{i=1}^n x_i \log p + (n - \sum_{i=1}^n x_i) \log(1 - p).$$

□

Definition 10.6 (Maximum Likelihood Estimation (MLE)). The maximum likelihood estimator $\hat{\theta}$ of θ is the value that maximizes the likelihood function:

$$\hat{\theta} = \arg \max_{\theta} L(\theta \mid \mathbf{x}).$$

Equivalently, since the logarithm is a monotonic function:

$$\hat{\theta} = \arg \max_{\theta} \ell(\theta \mid \mathbf{x}).$$

- **Invariance Property:** If $\hat{\theta}$ maximizes $L(\theta)$, then any one-to-one transformation $g(\theta)$ also satisfies:

$$g(\hat{\theta}) = \arg \max_{\theta} L(\theta).$$

- **Likelihood Principle:** The likelihood contains all information in the sample about θ .
- **Sufficiency:** If $S(\mathbf{x})$ is a sufficient statistic, then $L(\theta \mid \mathbf{x}) = g(S(\mathbf{x}), \theta)$.

Remark. Probability function $P(X = x \mid \theta)$ treats θ as fixed and x as variable. Likelihood function $L(\theta \mid x)$ treats x as fixed and θ as variable. In this notation

$$L(\theta \mid \mathbf{x}) = f(\mathbf{x} \mid \theta).$$

the conditioning on θ only means that PMF or the PDF depend on theta. It could be also written as $f_{\theta}(\mathbf{x})$. The likelihood function is a function of the unknown parameter θ , and we use it to estimate the value of this parameter. It tells us which values of θ make the observed data most probable.

Example 10.7. Likelihood Function and Maximum Likelihood Estimator (MLE) for Normal Distribution with Known Variance

Let $X_1, X_2, \dots, X_n$ be a random sample from a normal distribution with unknown mean μ and known variance σ^2 :

$$X_i \sim \mathcal{N}(\mu, \sigma^2), \quad i = 1, 2, \dots, n$$

The joint probability density function (PDF) of the sample is given by:

$$f(x_1, x_2, \dots, x_n \mid \mu) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

Thus, the likelihood function is:

$$L(\mu \mid x_1, \dots, x_n) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$$

Taking the natural logarithm, the log-likelihood function is:

$$\log L(\mu) = -\frac{n}{2} \log(2\pi\sigma^2) - \sum_{i=1}^n \frac{(X_i - \mu)^2}{2\sigma^2}$$

To find the MLE of μ , we differentiate $\log L(\mu)$ with respect to μ :

$$\frac{d}{d\mu} \log L(\mu) = \sum_{i=1}^n \frac{x_i - \mu}{\sigma^2}$$

We set the derivative to be equal to zero:

$$\sum_{i=1}^n \frac{x_i - \mu}{\sigma^2} = 0$$

and solve for μ :

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n x_i \quad (\text{estimation})$$

Thus, the **MLE of μ** is the sample mean:

$$\hat{\mu} = \bar{X} \quad (\text{estimator})$$

This means the best estimate for μ in the likelihood sense is simply the sample mean.

□

Example 10.8. Let us illustrate the meaning of the likelihood function. Consider a sample of 3 points from normal distribution $N(\mu, \sigma_0^2) = N(3, 4)$ with known variance. On the plot below 3 different densities are represented, with $\mu = 1, \mu = 3$ and $\mu = 8$. Discuss what happens with the values of the likelihood function.

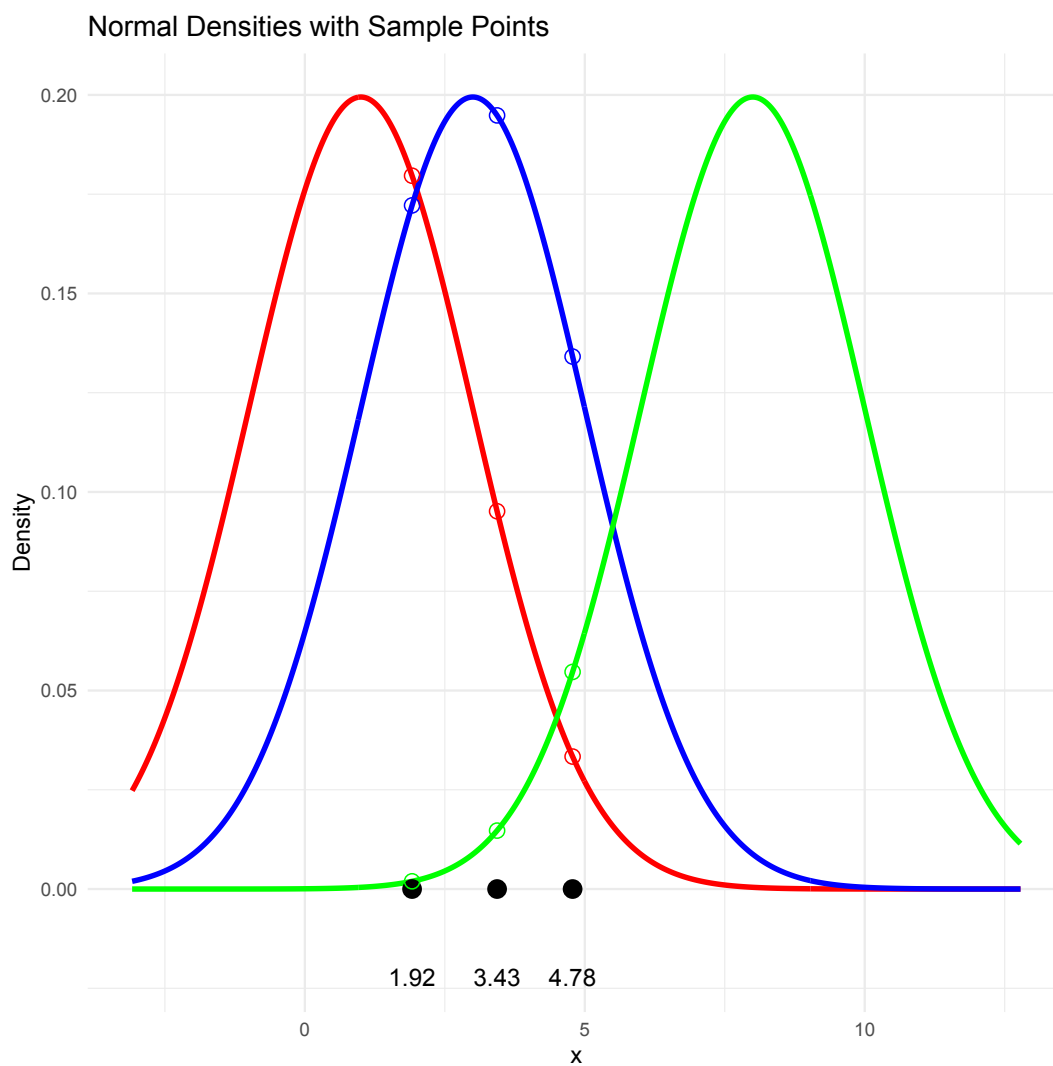


Figure 22: Illustrating meaning of Likelihood function

□

The following table gives the formula for the likelihood function and ML estimators (when they exist), for some distributions

Distribution	Likelihood Function (given data $\mathbf{x}$)	MLE
Bernoulli $p \in (0, 1)$	$L(p) = \prod_{i=1}^n p^{x_i} (1-p)^{1-x_i}$	$\hat{p} = \frac{1}{n} \sum_{i=1}^n x_i$
Binomial ($\text{Bin}(N, p)$) $p \in (0, 1), N \in \mathbb{N}$	$L(p) = \prod_{i=1}^n \binom{N}{x_i} p^{x_i} (1-p)^{N-x_i}$	$\hat{p} = \frac{1}{nN} \sum_{i=1}^n x_i$
Geometric $p \in (0, 1)$	$L(p) = \prod_{i=1}^n p (1-p)^{x_i-1}$	$\hat{p} = \frac{n}{\sum_{i=1}^n x_i}$
Poisson $\lambda \in (0, \infty)$	$L(\lambda) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!}$	$\hat{\lambda} = \frac{1}{n} \sum_{i=1}^n x_i$
Negative Binomial (with r known) $p \in (0, 1), r \in \mathbb{N}$	$L(p) = \prod_{i=1}^n \binom{x_i+r-1}{x_i} p^r (1-p)^{x_i}$	$\hat{p} = \frac{nr}{nr + \sum_{i=1}^n x_i}$
Discrete Uniform $a, b \in \mathbb{Z}, a < b$	$L(a, b) = \left(\frac{1}{b-a+1} \right)^n$ (with $a \leq \min_i x_i, b \geq \max_i x_i$)	$\hat{a} = \min_{1 \leq i \leq n} x_i, \hat{b} = \max_{1 \leq i \leq n} x_i$
Continuous Uniform $a, b \in \mathbb{R}, a < b$	$L(a, b) = \prod_{i=1}^n \frac{1}{b-a}$ (with $a \leq \min_i x_i, b \geq \max_i x_i$)	$\hat{a} = \min_{1 \leq i \leq n} x_i, \hat{b} = \max_{1 \leq i \leq n} x_i$
Exponential $\lambda > 0$	$L(\lambda) = \prod_{i=1}^n \lambda \exp(-\lambda x_i)$	$\hat{\lambda} = \frac{n}{\sum_{i=1}^n x_i}$
Normal $\mu \in \mathbb{R}, \sigma > 0$	$L(\mu, \sigma) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right)$	$\hat{\mu} = \bar{x}, \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$
Gamma (known shape α) $\alpha > 0$ (known), $\beta > 0$	$L(\beta) = \prod_{i=1}^n \frac{1}{\Gamma(\alpha) \beta^\alpha} x_i^{\alpha-1} \exp\left(-\frac{x_i}{\beta}\right)$	$\hat{\beta} = \frac{1}{n\alpha} \sum_{i=1}^n x_i$
Beta $\alpha > 0, \beta > 0$	$L(\alpha, \beta) = \prod_{i=1}^n \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\beta)} x_i^{\alpha-1} (1-x_i)^{\beta-1}, \quad x_i \in (0, 1)$	Closed-form MLEs not available (use numerical methods)

Table 21: Likelihood Functions and MLEs for Common Distributions

Example 10.9. Multinomial Likelihood and MLE (Multiple observations, advanced)

Suppose we have m independent observations from multinomial distribution $MN(n, k, p_1, \dots, p_k)$ (n -number of independent draws with replacement, from k different objects, with corresponding probabilities p_i to be drawn). For the i th observation, we have a multinomial count vector

$$X^{(i)} = (X_1^{(i)}, X_2^{(i)}, \dots, X_k^{(i)})$$

obtained from n trials, so that for an observation $x^{(i)} = (x_1^{(i)}, x_2^{(i)}, \dots, x_k^{(i)})$

$$\sum_{j=1}^k x_j^{(i)} = n.$$

Let the probability vector be

$$p = (p_1, p_2, \dots, p_k),$$

with $p_j \geq 0$ and $\sum_{j=1}^k p_j = 1$.

For each observation i , the probability mass function is

$$f(x^{(i)} | p) = \frac{n!}{x_1^{(i)}! x_2^{(i)}! \dots x_k^{(i)}!} \prod_{j=1}^k p_j^{x_j^{(i)}}.$$

Let $\mathbb{X}$ be the matrix of m observations

$$\mathbb{X} = \begin{pmatrix} x_1^{(1)} & x_2^{(1)} & \dots & x_j^{(1)} & \dots & x_k^{(1)} \\ \vdots & \vdots & \ddots & \vdots & & \vdots \\ x_1^{(i)} & x_2^{(i)} & \dots & x_j^{(i)} & \dots & x_k^{(i)} \\ \vdots & \vdots & & \vdots & \ddots & \vdots \\ x_1^{(m)} & x_2^{(m)} & \dots & x_j^{(m)} & \dots & x_k^{(m)} \end{pmatrix}$$

Since the m observations are independent, the joint likelihood function is

$$L(p|\mathbb{X}) = \prod_{i=1}^m \frac{n!}{x_1^{(i)}! x_2^{(i)}! \dots x_k^{(i)}!} \prod_{j=1}^k p_j^{x_j^{(i)}}.$$

We can rewrite this as

$$L(p|\mathbb{X}) = \left(\prod_{i=1}^m \frac{n!}{x_1^{(i)}! x_2^{(i)}! \dots x_k^{(i)}!} \right) \prod_{j=1}^k p_j^{\sum_{i=1}^m x_j^{(i)}}.$$

Let

$$T_j = \sum_{i=1}^m x_j^{(i)}$$

and $N = m \cdot n$. Then, the likelihood (ignoring the constant multiplicative factors that do not depend on p) becomes

$$L(p) \propto \prod_{j=1}^k p_j^{T_j}.$$

Taking the logarithm of the likelihood (and ignoring constants) gives

$$\ell(p) = \log L(p) = \sum_{j=1}^k T_j \log p_j + \text{constant}.$$

(Finding MLE - advanced, check the solution.) We maximize $\ell(p)$ subject to the constraint

$$\sum_{j=1}^k p_j = 1.$$

To find the maximum we need to form the Lagrangian (which includes the constraint and a parameter λ):

$$\mathcal{L}(p, \lambda) = \sum_{j=1}^k T_j \log p_j + \lambda \left(1 - \sum_{j=1}^k p_j \right).$$

Taking the derivative with respect to p_j and setting it to zero, we obtain:

$$\frac{\partial \mathcal{L}}{\partial p_j} = \frac{T_j}{p_j} - \lambda = 0 \implies p_j = \frac{T_j}{\lambda}.$$

Enforcing the constraint,

$$\sum_{j=1}^k p_j = \sum_{j=1}^k \frac{T_j}{\lambda} = \frac{1}{\lambda} \sum_{j=1}^k T_j = \frac{N}{\lambda} = 1,$$

which gives $\lambda = N$.

Thus, the maximum likelihood estimator for p_j is

$$\hat{p}_j = \frac{T_j}{N} = \frac{\sum_{i=1}^m X_j^{(i)}}{\sum_{i=1}^m n_i}, \quad j = 1, 2, \dots, k.$$

Remark. Understand here what are the observations from one experiment, solve the exercise with specific data.

□

Example 10.10. Stanford CS109 - Linear transform plus noise: Maximum Likelihood Estimation

Assume that a model is given by

$$Y = \theta X + Z,$$

where $Z \sim N(0, \sigma^2)$ and X is drawn from an unknown distribution. This implies that, conditional on X , the random variable Y follows a normal distribution:

$$Y \mid X \sim N(\theta X, \sigma^2).$$

(Understand what happens when X is fixed, i.e. $X = x_0$. Make simulations for this example - see the Project section.)

The conditional probability density function (PDF) of Y given X and θ is

$$f(Y = y \mid X = x_i, \theta) = \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{(y - \theta x_i)^2}{2\sigma^2}\right).$$

For a set of independent observations $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, the joint likelihood function (ignoring factors that do not depend on θ) is

$$L(\theta|x, y) = \prod_{i=1}^n f_{\theta}(x_i, y_i) = \prod_{i=1}^n f_{\theta}(y_i|x_i, \theta) f(x_i) \propto \prod_{i=1}^n f(y_i \mid x_i, \theta) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{(y_i - \theta x_i)^2}{2\sigma^2}\right).$$

Since the distribution of X does not depend on θ , any factors involving $f(x_i)$ can be ignored when maximizing with respect to θ . (Check the notation for errors.)

To obtain the Log-likelihood function we take the natural logarithm of $L(\theta)$ which yields the log-

likelihood:

$$\begin{aligned}
\ell(\theta|x, y) &= \log L(\theta|x, y) \\
&= \log \prod_{i=1}^n \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{(y_i - \theta x_i)^2}{2\sigma^2}\right) \\
&= \sum_{i=1}^n \log \left(\frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{(y_i - \theta x_i)^2}{2\sigma^2}\right) \right) \\
&= \sum_{i=1}^n \left[\log \frac{1}{\sqrt{2\pi} \sigma} - \frac{(y_i - \theta x_i)^2}{2\sigma^2} \right] \\
&= n \log \frac{1}{\sqrt{2\pi} \sigma} - \frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \theta x_i)^2.
\end{aligned}$$

Notice that the term $n \log \frac{1}{\sqrt{2\pi} \sigma}$ does not depend on θ .

Next we maximize with respect to θ . Since we are interested in maximizing $\ell(\theta)$ with respect to θ , and since the only θ -dependent part is

$$-\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \theta x_i)^2,$$

maximizing the log-likelihood is equivalent to minimizing the sum of squared errors:

$$\hat{\theta} = \arg \max_{\theta} \ell(\theta) = \arg \min_{\theta} \sum_{i=1}^n (y_i - \theta x_i)^2.$$

Thus, the maximum likelihood estimate (MLE) for θ is given by the least squares criterion:

$$\hat{\theta} = \arg \min_{\theta} \sum_{i=1}^n (y_i - \theta x_i)^2.$$

Remark. You will see this approach in linear regression.

□

11 Projects

Exercise 11.1. See Example 10.10 Assume that a model is given by

$$Y = \theta X + Z,$$

where $Z \sim N(0, \sigma^2)$ and X is drawn from an unknown distribution. This implies that, conditional on X , the random variable Y follows a normal distribution:

$$Y | X \sim N(\theta X, \sigma^2).$$

Simulate the model for different distributions of X . Find a way to make a good visualisation of the joint distribution of (X, Y) , the marginals and conditionals. Visualise a conditional histogram of $Y|_{X=x_0}$

12 Bayesian statistics

12.1 Motivation

Frequentist approach - no probabilistic statements (for estimations, confidence intervals ...) problem with multiple hypothesis testing, no use of information in previous research ...

13 Introduction

Let us see again the application of Bayesian theorem on the following example.

Example 13.1. ELISA is a method for HIV testing. The ELISA's sensitivity (true positive rate = probability that the test will be positive if the person has HIV = one minus the false negative rate) is 0.93. The true negative of the test (one minus the false positive rate) is 0.99. If the prevalence of HIV in the general population is estimated to be 1.48 in 1000 American adults, calculate the probability that the person with a positive result on the test indeed has HIV.

We will use the following notation H_1 - person has HIV, $P(H_1) = \frac{1.48}{1000} = 0.00148$,
 H_2 - person does not have HIV, $P(H_2) = 1 - P(H_1) = 0.99852$,
 A - the ELISA test is positive
 $A|H_1$ - true positive, $P(A|H_1) = 0.93$
 $\bar{A}|H_2$ - true negative, $P(\bar{A}|H_2) = 0.99$
Then, the probability for a positive test is

$$P(A) = P(H_1)P(A|H_1) + P(H_2)P(A|H_2) = 0.0113616$$

and thus $P(H_1|A) = \frac{P(A|H_1)P(H_1)}{P(A)} = 0.12$

We can also calculate $P(H_1|\bar{A}) = 0.0001$.

This means that, even if the test is positive, there is a small probability for HIV (for low risk people from the general population). The probability that a person does not have HIV, given a positive test is $1 - 0.12 = 0.88$. Also, the probability that a person has HIV given a negative test is 0.0001. What would we expect as a result here, if the person has a higher risk of HIV?

What if we ran a second test, after the first positive test? What is then the conditional probability that a person has HIV? We are looking for the probability of the event A_2 - second test is positive. We use the updated probabilities and the assumptions that the two tests are independent (not necessarily true). After observing one positive test, now the probability that that person has HIV is 0.12, i.e. we updated the probability of H_1

$$P(H_1|A_2) = \frac{P(A_2|H_1)P(H_1)}{P(A_2)} = \frac{0.93 \cdot 0.12}{0.93 \cdot 0.12 + (1 - 0.99) \cdot (1 - 0.12)} = 0.93$$

We have used Bayes updating of the the probability. We have used the prior information, that the first test was positive and thus the probability that a person has a HIV after second positive test has increased significantly. $\square$

Lets extend the discussion on the previous example. Let us assume that we test 200 people chosen randomly. What is then the expected number of positive tests among the 200? What is the expected number of true positives/true negatives?

In order to answer these questions, we will introduce random variables. Let $Y \sim B(1, \theta)$ be a Bernoulli r.v. which is 1 when the test is positive and 0 otherwise. Then $P(Y = 1) = P(A) = 0.0113616$ and

$P(Y = 0) = P(\bar{A}) = 0.9886384$ and let X be a random variable with the following conditional distributions $X|_{Y=1} = X_A \sim B(1, 0.12)$ and $X_{Y=0} = X_{\bar{A}} \sim B(1, 0.001)$. Then we would obtain the following $P(X = 1|Y = 1) = P(H_1|A) = 0.12$ etc.

If we have tested 200 people, then that is a sample from Y i.e. from the distribution. With this notation we can answer the given questions. Read also about conditional expectation.

Connect the previous example with the next one

Example 13.2. We have a friend Peter, who loves betting on the outcome of a coin flip. Peter has two coins, one of which is defect, i.e. biased towards heads with a probability of 0.7. Peter could choose any of the coins with equal probability.

- Calculate the probability that we observe head.
- Given that we observed head, what is the probability that he has chosen the biased coin?
- If we already observed a head, what is the probability that in the next flip we will again observe head (build the tree)

Let us denote with Y the random variable which is 0 when we have chosen the unbiased coin and 1 otherwise. Let X be the random variable which is 1 when we observe head and 0 otherwise. The distribution of X depends on the choice of the coin. Thus, if we have chosen the unbiased coin, i.e. when $Y = 0$ the conditional distribution of $X_{Y=0}$ is the uniform on $\{0, 1\}$. We have

$$Y \sim \begin{pmatrix} 0 & 1 \\ 1/2 & 1/2 \end{pmatrix}, \quad X_{Y=0} \sim \begin{pmatrix} 0 & 1 \\ 1/2 & 1/2 \end{pmatrix}, \quad X_{Y=1} \sim \begin{pmatrix} 0 & 1 \\ 0.3 & 0.7 \end{pmatrix}$$

Let us denote with θ the probability to observe a head. We can see θ as a random variable itself with a probability distribution

$$\theta \sim \begin{pmatrix} 0.5 & 0.7 \\ 1/2 & 1/2 \end{pmatrix}.$$

We will use the notation $X \sim B(1, \theta)$. We emphasize again that θ is now a random variable and we can condition X on θ . Thus $P(X = 1|\theta = 0.7)$ is the probability that we have observed head when $\theta = 0.7$, which is actually the case when we have chosen the faulty (biased) coin. $\square$

Example 13.3. Suppose now that Peter has a coin with an unknown probability for heads

- What would be the classical statistical approach to find the probability of heads?
- What if we know that all values of θ on $[0, 1]$ are possible with equal probability? What is then the probability of observing a head?
- What if the distribution of θ is some other density on $[0, 1]$. Take as an examples the following densities on $[0, 1]$ - $p(\theta) = \theta + 0.5$ and $p(\theta) = 2\theta$. Illustrate the densities, calculate $P(X = 1)$ and discuss the differences. How does the distribution of θ influence the probability in both cases?
- What if we know nothing about the distribution of θ but Peter gives us few trials with the coin? Would it make it easier for us to bet? How? What if we got 4 times tail and one head? Are the densities in c) good choice for the model?

Solution.

- Classical statistical approach would include sampling with the coin, obtaining estimators using the Bernoulli distribution and performing some statistical tests.

- b) If all values from $[0, 1]$ are equally possible, that means that assume that θ is a random variable with uniform distribution. Now when we want to calculate $P(X = 1)$ we need to calculate conditional probabilities $P(X = 1|\theta)$ for all possible values for θ . Since θ in this case is continuous random variable, we actually have to use the Law of total probability using the density of θ

$$P(X = 1) = \int_0^1 P(X = 1|\theta)p(\theta)d\theta = \int_0^1 \theta \cdot 1d\theta = \frac{\theta^2}{2} \Big|_0^1 = \frac{1}{2}$$

- c) Both densities are represented on the graph. As before, we will use the law of total probability. When $p(\theta)$ we have

$$P(X = 1) = \int_0^1 P(X = 1|\theta)p(\theta)d\theta = \int_0^1 \theta \cdot (\theta + 0.5)d\theta = \left(\frac{\theta^3}{3} + \frac{\theta^2}{4}\right) \Big|_0^1 = \frac{7}{12}.$$

When $p(\theta) = 2\theta$ we have

$$P(X = 1) = \int_0^1 P(X = 1|\theta)p(\theta)d\theta = \int_0^1 2\theta^2d\theta = \frac{2\theta^3}{3} \Big|_0^1 = \frac{2}{3}.$$

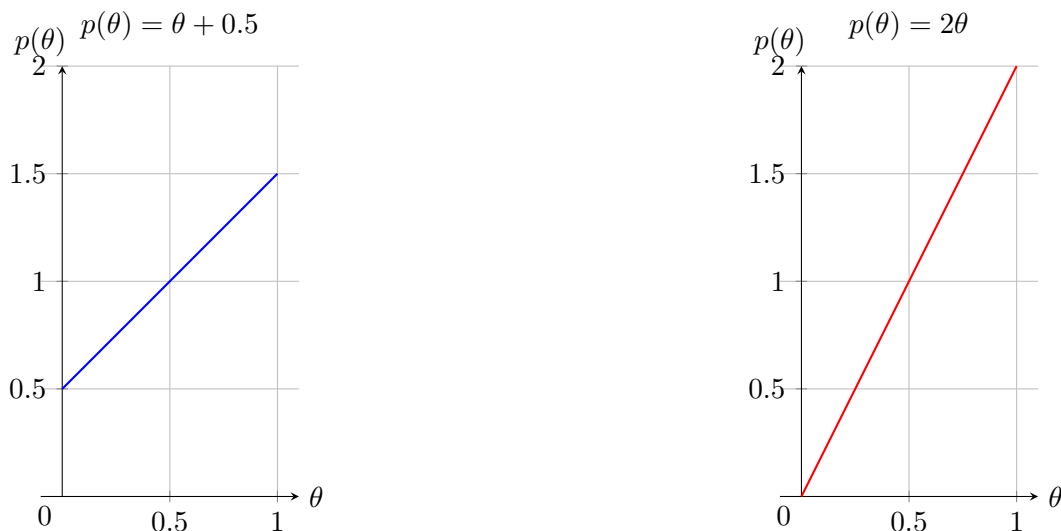


Figure 23: Two different density functions for θ .

In both cases the density is such that the values in the interval $[\frac{1}{2}, 1]$ are more probable than those values in $[0, \frac{1}{2}]$. The first density is "closer" to the uniform case (all values on $[0, 1]$ are probable with a small preference to those greater than $\frac{1}{2}$). Thus the total probability $P(X = 1)$ is higher, but close to $\frac{1}{2}$. The second density prefers larger values significantly, which increases the total probability.

- d) We will discuss this in the next part.



Example 13.4. A nontransparent jar contains thousands of beads, red and white, but we do not know the fraction of red beads. We choose at random, with no replacement few beads from the jar, and it turns out that most of them are red. What of the following describes your opinion about the conditional probability of the next bead being red, given the previous draws?

- a) The probability decreases, since we already removed red beads from finitely many that are in the jar.
- b) The probability increases, because initially we did not expect that there were so many red beads in the jar.

□

In the rest of the discussion we will use the following notations:

- θ - unobservable population parameter of interest (single parameter or vector of parameters);
- $p(\theta)$ - distribution of the parameter θ , usual notation for the prior distribution (our belief of the distribution of the parameter θ);
- y, D - observed data ex. $y = (y_1, y_2, \dots, y_n)$;
- $\tilde{y}$ - unknown but potentially observable data;
- $p(\theta|y), p(\tilde{y}|y)$ - conditional probabilities, conditioned on the observed data y ;
- $p(\theta, y)$ - joint probability distribution;
- $p(y|\theta)$ - The same notation is often used to denote different things - likelihood and sampling distribution (data distribution) which depends on the parameter θ . We have to be careful about this. For given θ $p(y|\theta)$ is distribution of our data (meaning it describes our model distribution, which depends on θ). But if we are given data the same notation will be used for likelihood function which is then function of θ . Thus, if we want to discuss the distribution of the data given the model, sometimes a more appropriate notation for the sampling distribution is $p_\theta(y)$;
- $p(\theta|y)$ - posterior distribution of θ (our belief in the distribution after seeing the data);
- $p(y)$ - "evidence", total probability of the data calculated across all the possible values of θ . As we have seen before, in the case of discrete θ this is given with $p(y) = \sum_{\theta} p(\theta)p(y|\theta)$. In the case of continuous θ this becomes $p(y) = \int_{\theta} p(\theta)p(y|\theta)d\theta$

$$\underbrace{p(\theta|y)}_{\text{posterior dist.}} = \frac{\overbrace{p(\theta, y)}^{\text{joint prob.}}}{p(y)} = \frac{\overbrace{p(\theta)}^{\text{prior dist.}} \cdot \overbrace{p(y|\theta)}^{\text{likelihood}}}{\underbrace{p(y)}_{\text{"evidence"}}}$$

The evidence here is a constant, also called normalization factor (normalization constant). We remark that if we integrate the numerator, with respect to all possible values of θ , we will obtain the denominator. Thus $\int_{\theta} p(\theta|y)d\theta = 1$. This, together with the positivity of $p(\theta|y)$ insurance that it is a proper density function (or in the discrete case, proper law of probability). This is important for our discussion.

We will use some theoretical results without discussion like Fubini's theorem, property of **exchangeability** of data - the probability distribution of $p(y_1, \dots, y_n)$ is invariant up to permutation, the order of updating does not affect the final posterior distribution. Read more details in "A First Course of Bayesian Statistical Methods" by Peter D. Hoff.

14 Data from Bernoulli distribution

Let us consider a case where our data are modeled with a Bernoulli random variable, i.e. let $Y \sim \mathcal{B}(1, \theta)$, where θ is an unknown parameter. Then

$$Y \sim \begin{pmatrix} 0 & 1 \\ 1 - \theta & \theta \end{pmatrix} \quad \text{or} \quad p(y = 1|\theta) = \theta \quad \text{and} \quad p(y = 0|\theta) = 1 - \theta.$$

We can write the probability law in the following form

$$p(y|\theta) = \theta^y(1 - \theta)^{1-y} \quad (1)$$

We remark again about different aspects of this formula. When θ is known or fixed, this becomes probability distribution $p_\theta(y)$. When data is given of fixed, this is the likelihood - it is a function of θ , not a probability distribution. In the following we will usually mean of the latter, i.e. this will be the notation for the likelihood.

If $y = (y_1, y_2, \dots, y_n)$ then

$$p(y|\theta) = p(y_1, y_2, \dots, y_n|\theta) = \prod_{i=1}^n \theta^{y_i} (1 - \theta)^{1-y_i} = \theta^{\sum_{i=1}^n y_i} (1 - \theta)^{n - \sum_{i=1}^n y_i} \quad (2)$$

The unknown parameter is probability parameter with a domain $[0, 1]$. Thus, we have to chose prior for θ which is a density on $[0, 1]$. There are different choices and we will discuss why **Beta distribution** is a good one.

Let $\theta \sim \text{Beta}(\alpha, \beta)$. The density of the distribution is given by

$$p(\theta) = \frac{\theta^{\alpha-1}(1 - \theta)^{\beta-1}}{B(\alpha, \beta)} \quad \alpha, \beta > 0, \theta \in [0, 1] \quad (3)$$

Here, the term $B(\alpha, \beta)$ is the so called beta function, defined by the integral

$$B(\alpha, \beta) = \int_0^1 t^{\alpha-1}(1 - t)^{\beta-1} dt.$$

It can also be expressed in terms of the gamma function:

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}.$$

The gamma function for a real number $x > 0$ is defined by

$$\Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt.$$

This integral converges for all $x > 0$. A key property of the gamma function is that for any positive integer n ,

$$\Gamma(n) = (n - 1)!.$$

and for each positive real number x , it satisfies the relation

$$\Gamma(x + 1) = x \Gamma(x).$$

In equation (3) $B(\alpha, \beta)$ is a normalizing constant, which depends only on the parameters α and β . Let us find the posterior for θ , given the observed data. We have

$$p(\theta|y) = \frac{p(\theta)p(y|\theta)}{p(y)} \quad (4)$$

$$= \frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{\underbrace{B(\alpha, \beta)}_{p(\theta)}} \cdot \underbrace{\theta^{\sum_{i=1}^n y_i} (1-\theta)^{n-\sum_{i=1}^n y_i}}_{p(y|\theta)} \underbrace{\frac{1}{\int_0^1 p(\theta)p(y|\theta)d\theta}}_{p(y)} \quad (5)$$

$$= \frac{\Gamma(\alpha + \beta)}{\underbrace{\Gamma(\alpha)\Gamma(\beta) \int_0^1 p(\theta)p(y, \theta)d\theta}_C} \cdot \theta^{\overbrace{\alpha + \sum_{i=1}^n y_i - 1}^{\alpha_1}} \cdot (1-\theta)^{\overbrace{\beta + n - \sum_{i=1}^n y_i - 1}^{\beta_1}} \quad (6)$$

$$= C \cdot \theta^{\alpha_1-1} \cdot (1-\theta)^{\beta_1-1} \sim \text{Beta}(\alpha_1, \beta_1)$$

where $\alpha_1 = \alpha + \sum_{i=1}^n y_i$ and $\beta_1 = \beta + n - \sum_{i=1}^n y_i$.

We know that the posterior is a proper density function. It has the form of Beta distribution with parameters α_1, β_1 . Thus we can conclude that the normalising constant is in fact $C = \frac{1}{B(\alpha_1, \beta_1)}$ (check the beta function in R).

Example 14.1. Before we analyse further, let us illustrate the previous transformation through the example Example (13.3). Peter has a coin with unknown probability of heads ($y = 1$) θ . Let us assume as a prior we have taken $\theta \sim \text{Beta}(6, 8)$, i.e.

$$p(\theta) = \frac{\Gamma(14)}{\Gamma(6)\Gamma(8)} \theta^5 (1-\theta)^7$$

We will discuss the choice of the prior later. Suppose we flipped the coin $n=6$ times, which gave us the sample $y = \{0, 1, 0, 1, 0, 0, 1, 0, 0, 0\}$. The likelihood is given by

$$p(y|\theta) = p(y=1|\theta)^3 \cdot p(y=0|\theta)^7 = \theta^3 \cdot (1-\theta)^7 \quad (7)$$

Then for the posterior distribution we have

$$p(y|\theta) = \frac{\Gamma(14)}{\Gamma(6)\Gamma(8) \int_0^1 p(\theta)p(y, \theta)d\theta} \cdot \theta^9 (1-\theta)^{15} \sim \text{Beta}(9, 15) \quad (8)$$

□

Let us denote $z = \sum_{i=1}^n y_i$ - this is the number of successes in our data ("1"s, number of heads, number of patients for whom the therapy worked etc.). Then $n - z$ is the number of failures ("0"s etc.). With this notation, the posterior distribution of θ is $\text{Beta}(\alpha + z, \beta + n - z)$. Notice how the likelihood and the posterior do not depend on the order of the data, but on z and n only. Thus this method would work if we had only the knowledge of z and n , and not the whole sample.

Example 14.2. In the previous example, we may not have the information for the whole sample. Peter may flip the coin and tell us that there were 3 successes (3 heads) between the 10 flips. Thus $z = \sum_{i=1}^n y_i = 3$, $n - z = 7$ and we may still calculate the same posterior. □

Let us discuss the chosen prior and the data and see how they changed the posterior.

Example 14.3. The following picture shows the prior and posterior densities

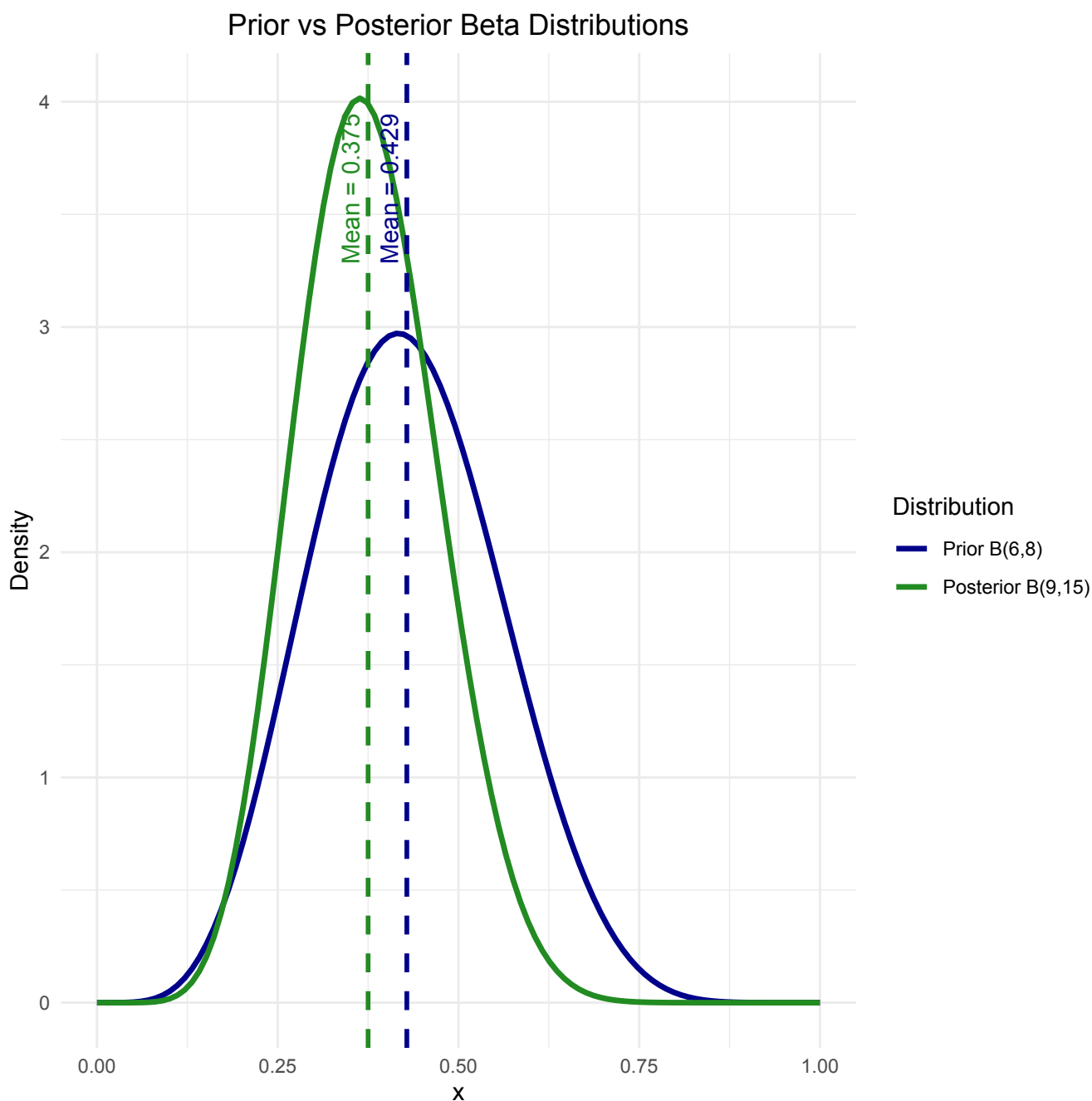


Figure 24: Prior vs posterior Beta distributions

We can see that the posterior distribution has different mean and variance. Discuss what this means. What does the prior reflect? How has the posterior changed with respect to the prior and how the data have influenced this change?

Let us suppose that when flipping the coin we have obtained a bigger sample with $n = 30$ and $z = \sum_{i=1}^n y_i = 21$. Then the posterior distribution is $B(27, 9)$. Let's see the plots (treba da se koregira grafikot)

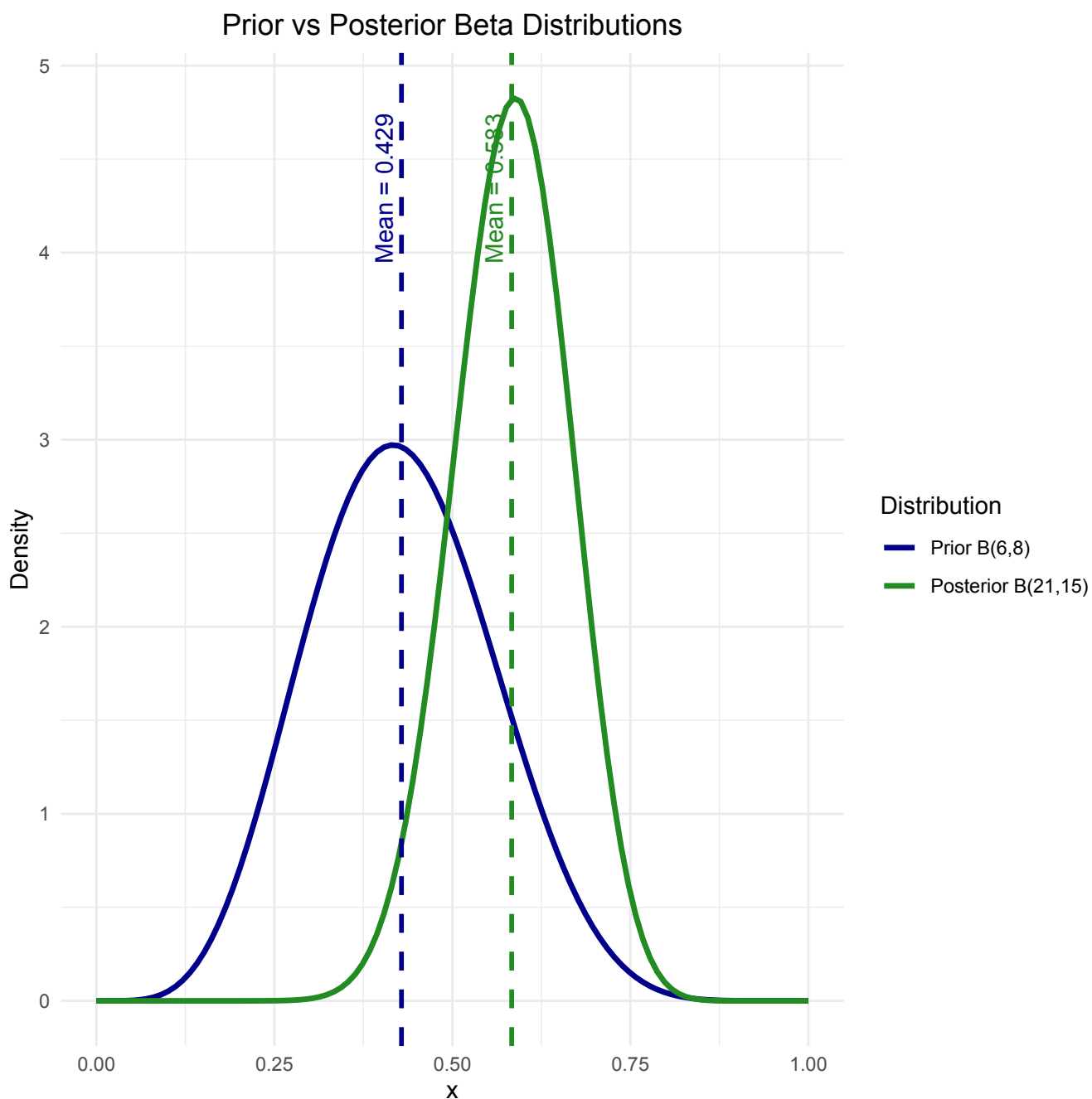


Figure 25: Prior vs posterior Beta distributions

For a Beta distribution with parameters α and β , the formulas for the mean and variance are given by:

$$\text{Mean } \mu = \frac{\alpha}{\alpha + \beta}$$

$$\text{Variance } \sigma^2 = \frac{\alpha \beta}{(\alpha + \beta)^2 (\alpha + \beta + 1)}$$

We can calculate

- Beta(6, 8): Mean ≈ 0.4286 , Variance ≈ 0.01633
- Beta(9, 15): Mean = 0.375, Variance ≈ 0.009375

- Beta(21, 15): Mean ≈ 0.5833 , Variance ≈ 0.00657

Think also what has influenced the change in the variance - what is the role of the prior and the data. How would you use the posterior to bet on the next coin flip ($E\theta > 0.5$ vs $E\theta < 0.5$)? $\square$

Lets see another example.

Example 14.4. An online marketing team wants to estimate the probability, θ , that a viewer clicks on a newly launched advertisement. Each time the ad is displayed to a user, the outcome is either a “click” (success) or “no click” (failure). Thus, the outcomes follow a Bernoulli distribution with success probability θ .

Based on previous campaigns, the team believes that the ad might not be very effective. They think that the click-through rate is around 40%. They choose a mildly informative prior for θ as a Beta(2, 3) distribution. This prior has a mean given by

$$E[\theta] = \frac{2}{2+3} = 0.4,$$

and thus the prior reflects the belief of the team about the click-through rate. The variance of this prior is , which reflects some uncertainty and thus we say that this prior is mildly informative (see the plot below)

The team has made some tests and the ad was shown to $n = 20$ users out of which $k = 10$ clicked the add (10 clicks are observed). Let us compute the posterior distribution. The likelihood for the data (assuming a Bernoulli model) is given by

$$p(y|\theta) \propto \theta^k(1-\theta)^{n-k} = \theta^{10}(1-\theta)^{10}.$$

With a Beta(2, 3) prior, whose density is proportional to

$$p(\theta) \propto \theta^{2-1}(1-\theta)^{3-1} = \theta^1(1-\theta)^2,$$

the posterior density is proportional to the product of the prior and the likelihood:

$$p(\theta \mid \text{data}) \propto \theta^{1+10}(1-\theta)^{2+10} = \theta^{11}(1-\theta)^{12}.$$

We remark that in this case we have omitted writing the normalizing constant, since we know the form of the density and we can still calculate the parameters of the posterior distribution, i.e.

$$\theta \mid \text{data} \sim \text{Beta}(2+10, 3+10) = \text{Beta}(12, 13).$$

For this posterior, the mean and variance are given by

$$E[\theta] = \frac{\alpha}{\alpha + \beta}, \quad \text{Var}(\theta) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}.$$

For the posterior Beta(12, 13):

$$E[\theta \mid \text{data}] = \frac{12}{12+13} = \frac{12}{25} = 0.48,$$

$$\text{Var}(\theta \mid \text{data}) = \frac{12 \cdot 13}{(25)^2 \cdot 26} = \frac{156}{16250} \approx 0.0096.$$

Lets see the plot of the prior and posterior distributions and discuss it

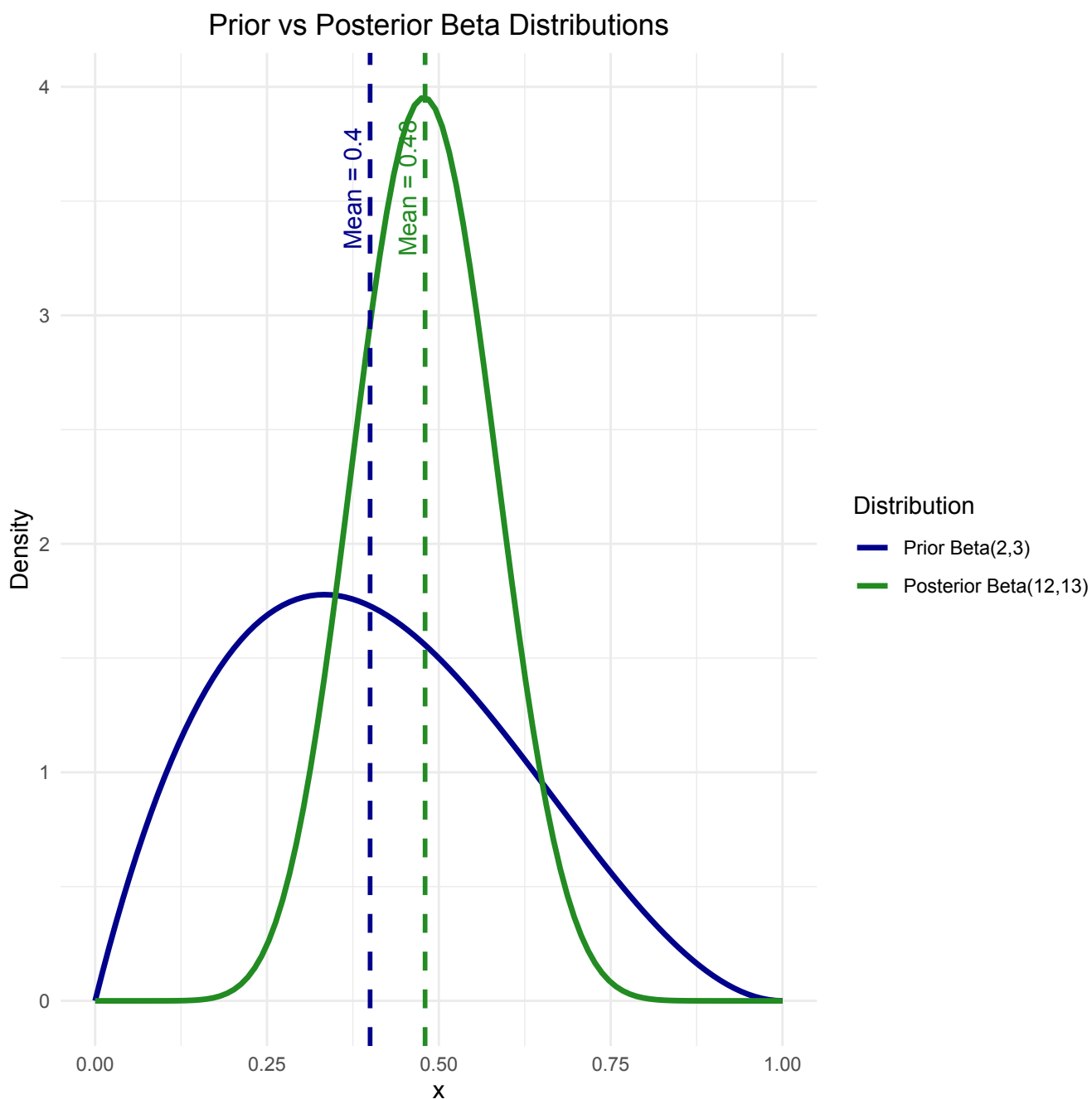


Figure 26: Prior vs posterior Beta distributions

The chosen prior, $\text{Beta}(2,3)$, reflects an initial belief that the click-through rate is around 40%. After observing 10 clicks out of 20 trials (which is a click rate of 50%), the posterior $\text{Beta}(12,13)$ represents a compromise between the prior belief and the observed data, yielding a posterior mean of 0.48.

Try some different priors and samples with different sizes to see how the posterior changes. For example, if we use a non-informative prior such as $\text{Beta}(1,1)$, the posterior would become $\text{Beta}(11,11)$ with a mean of 0.5. This reflects the observed data more strongly. The choice of prior is particularly influential when the sample size is small, but its influence diminishes as more data become available.

□

14.1 Choice of parameters of prior distribution

Let us consider an example of a coin toss with an unknown probability of a success θ and let the prior distribution for θ be some $Beta(\alpha, \beta)$.

1. We can think of α and β as the number of heads and tails in some previous experiment (or a hypothetical experiment that we believe could happen). Let us denote

$$\begin{aligned} N &= \alpha + \beta && \text{the size of that hypothetical sample} \\ m &= \frac{\alpha}{\alpha + \beta} && \text{proportion of heads (also mean)} \\ s^2 &= \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)} && \text{variance.} \end{aligned} \tag{9}$$

We may think that the coin is fair but we have to express our certainty through the distribution. If we believe strongly that the values of θ are close to 0.5, we should choose density which peaks at 0.5 and considers lower and higher values less probable. The density is peaked at 0.5 when $\alpha = \beta$. Smaller values of α and β gives us more evenly spread density thus reflects a bigger uncertainty. For example, if we have seen experiment with 3 heads and 3 tails, we may think that the coin is fair, but we may not be very certain since it is a small sample. On the other hand, if we have seen an experiment with 20 heads and 20 tails, we are more certain that the coin is fair and this is reflected in the prior density - the density is more concentrated around 0.5. See how the density of Beta distribution changes if we choose greater values of α or β .

2. We can also think in terms of average proportion of heads, m . Then

$$\alpha = m \cdot N \quad \text{and} \quad \beta = (1 - m) \cdot N \tag{10}$$

If we have some belief about m and N , that will translate into values for α and β . For example, for prior $Beta(5, 5)$ $N = 5 + 5 = 10$. This is effective sample size of the prior. The prior carries the same weight as observing 10 coin tosses (5 heads, and 5 tails), before observing real data - this might be some prior experiment or hypothetical experiment. Larger N would bring a stronger prior belief in the parameter (as it would happen if we have an information about a large experiment). This would also mean that we need a lot of data to overcome this stronger prior. Thus N quantifies how much new evidence is required to update our beliefs. A lower N would mean a weaker prior and in this case data would easily overpower the prior and maybe change our mind.

Thus, we can choose N based on how much hypothetical evidence would convince us to change our mind about the distribution of θ .

m , on the other hand is our choice for the proportion of success, the mean of the Beta distribution - or in this case our belief about the fairness of the coin and the probability to get a head. We can choose it based on previous information or our subjective belief.

3. We may choose the prior also based on our belief about the mean and the variance of the distribution. If we make a transformation from (9), we can express the parameters

$$\alpha = m \left(\frac{m(1 - m)}{s^2} - 1 \right) \quad \beta = (1 - m) \left(\frac{m(1 - m)}{s^2} - 1 \right) \tag{11}$$

We may have some historical data from which we can use the mean and the variance to calculate α and β . For example, if we have knowledge that $m = 0.6$ and $s = 0.02$ we can calculate that $\alpha = 6.6$ and $\beta = 4.4$ and thus obtain a prior $Beta(6.6, 4.4)$. This is equivalent to observing 11 pseudo-observations (6.6 "successes" and 4.4 "failures"). (Of course we may choose something else, but we have to justify it.)

Remark. We should be careful about this approach, since there are few limitations to have in mind. First of all, the highest possible variance in the Beta distribution family is achieved when $\alpha, \beta \rightarrow 0$ and this

limiting value is $1/4$. However, when both parameters are less than 1, we have the so called bimodal distribution which peaks at 0 and 1. Based on our data we have to decide whether this is a suitable model. If it is not and we need a unimodal distribution, we should choose both parameters to be greater or equal than 1. In this case, $Beta(1,1)$ has the highest value of variance $1/12 = 0.083$.

Additionally, for Beta distributions we have

$$s^2 = \frac{\alpha}{\alpha + \beta} \cdot \frac{\beta}{\alpha + \beta} \cdot \frac{1}{\alpha + \beta + 1} = m(1 - m) \frac{1}{\alpha + \beta + 1} \leq m(1 - m).$$

If our prior information does not satisfy this, we need to choose prior outside of the Beta distribution family - cannot model all mean-variance combinations. We may still choose the uniform prior and rely completely on our (new) experiment data or we can choose other types of priors and advanced methods to learn the posterior.

We should do this analysis for any type of likelihood model, prior information and prior distribution. Certainly we should also check the prior data - maybe there were outliers which increased the variance, maybe they are unreliable etc.

4. See what happens with the density of Beta, when $\alpha < 1$ and/or $\beta < 1$. For our Example this choice of the parameters is not suitable, but there is data for which such choice is appropriate. Read about bimodal Beta distribution.

Let's see how does the data (values of z and n) change the posterior? See how the prior changes the posterior distribution. What is the relationship between the prior and the posterior mean? The prior mean is $\frac{\alpha}{\alpha + \beta}$ and the posterior mean is given by $\frac{\alpha + z}{\alpha + \beta + n}$

$$\underbrace{\frac{\alpha + z}{\alpha + \beta + n}}_{\text{posterior mean}} = \underbrace{\frac{z}{n}}_{\text{data}} \cdot \underbrace{\frac{n}{n + \alpha + \beta}}_{\text{data weight}} + \underbrace{\frac{\alpha}{\alpha + \beta}}_{\text{prior mean}} \cdot \underbrace{\frac{\alpha + \beta}{n + \alpha + \beta}}_{\text{prior weight}} \quad (12)$$

This shows the posterior mean is a weighted average of the prior mean and the data mean, where the weights depend on the "strength" of the prior $\alpha + \beta$ relative to the sample size n . See what happens when we have large new sample $n \gg \alpha + \beta$.

Example 14.5. (Estimation of probability of free throws) A new player joins a basketball team and we want to calculate the probability that he will score when shooting a free throw (θ) (or we can estimate his end of season free throw average). The first three times he shoots, he misses. How can we estimate θ based on this sample?

- Classical approach

The given sample $y = (0, 0, 0)$ is data from Bernoulli distribution $B(1, \theta)$. If we use the classical approach, first we need to find the MLE estimator of θ which is $\hat{\theta} = \bar{Y}$ so the estimated value based on our sample is $\hat{\theta} = 0$. This leads to $P(Y = 1 | \hat{\theta} = 0) = 0$, that is the probability that the player could score is 0. This certainly seems extreme (in a professional league) but the results may be due to the small data set. The Bayesian approach in this case would work better.

- Bayesian approach

We will use a $Beta(\alpha, \beta)$ prior for the parameter θ (model which we choose before collecting the data). From previous statistics in the league, we know that free throw average for most of the players in the league range from 0.4 to 0.8. We consider our player to be an average player, so we may expect that his average score will be 0.6. We look for a Beta distribution with these characteristics.

Using the formula for mean of beta distribution we can calculate $\frac{\alpha}{\alpha+\beta} = 0.6$ i.e. $2\alpha = 3\beta$.
In R we can simulate beta distributions and see which describes our beliefs. Let's try

```
curve(dbeta(x, 3, 2))  
curve(dbeta(x, 15, 10))  
curve(dbeta(x, 30, 20))
```

Let us use the last one, i.e. take $\alpha = 30, \beta = 20$.

After we have observed the sample, the posterior parameters are $\alpha_1 = 30 + 0 = 30, \beta_1 = 20 + 3 = 23$.
Simulate `curve(dbeta(x, 30, 23))` to notice a slight shift of the density.

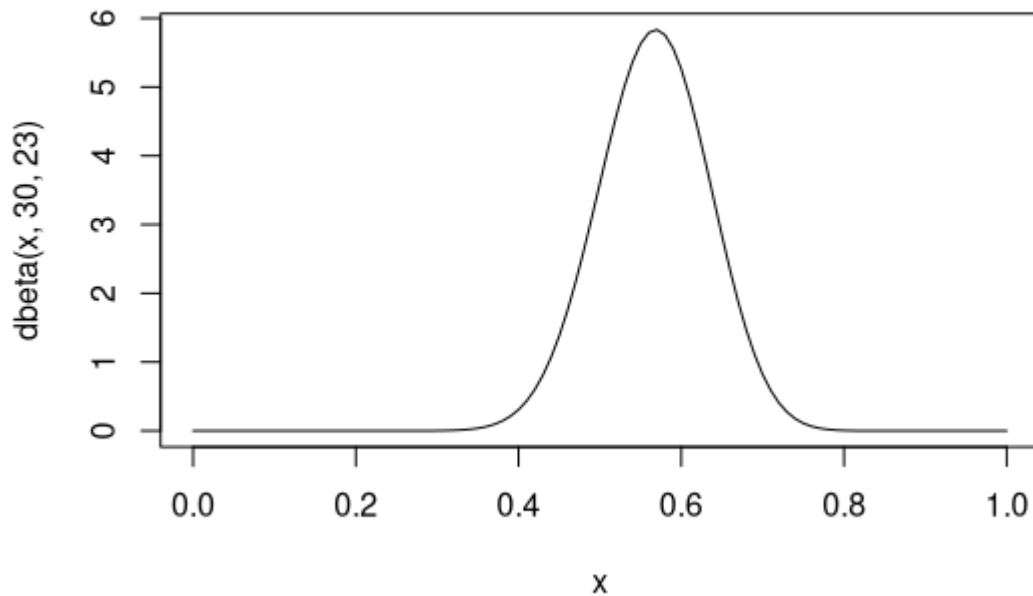


Figure 27: Density of Beta(30,23)

If we have some idea about the variance (maybe use some additional data from the league) we chose a prior which will reflect that. Maybe we expect (or know) that most of the data should be between 0.4 and 0.8, and we can set 0.05 and 0.95 quantiles to be 0.4 and 0.8 etc..

Remark. If we have some additional information for the quantiles of the beta distribution, we can actually calculate the parameters of the beta distribution. This is because the following holds:

Let the desired q_1 quantile be x_1 and the desired q_2 quantile be x_2 with $0 < q_1 < q_2 < 1$ and (therefore) $0 < x_1 < x_2 < 1$. Then there are unique α and β for which the $Beta(\alpha, \beta)$ distribution has these quantiles.

We can use the R function **beta.parms.from.quantiles** (you can download it here

<http://www.medicine.mcgill.ca/epidemiology/joseph/pbelisle/R/BetaParmsFromQuantiles.R>

For example calling the function with values for the 0.25 and 0.975 quantile

```
1 beta.parms.from.quantiles(c(.6,.9), plot=T)
```

produces the following plot

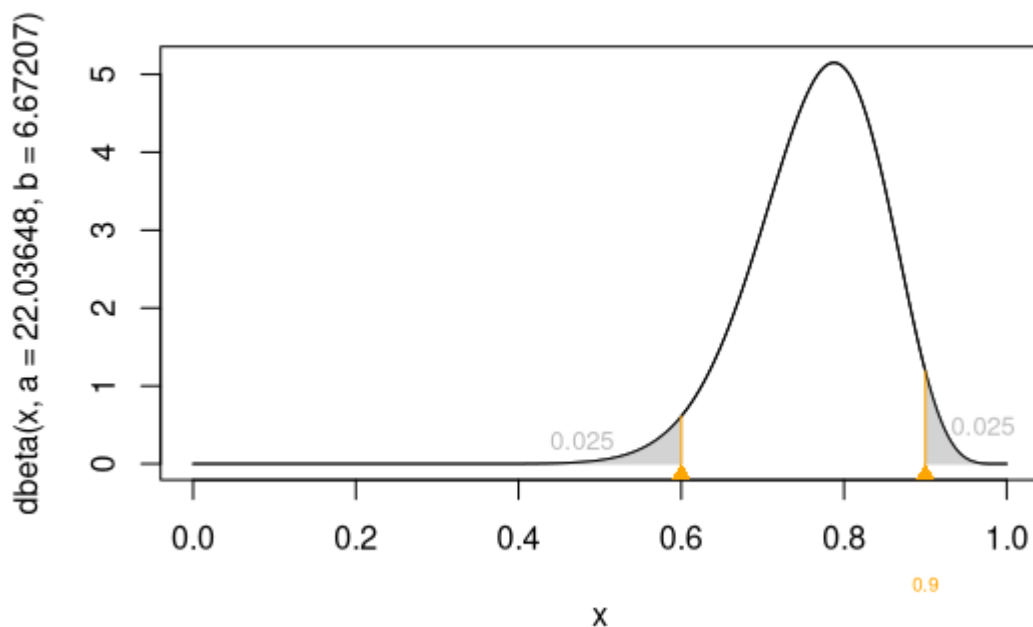


Figure 28: Beta density based on quantiles

Next. Simulate the posterior at mid season, when the player had 15 scores in 22 free throw attempts.

□

Exercise 14.6 (Bayesian Spam Filtering). A spam filter analyzes how often the word "free" appears in spam emails. Historical data suggests that 60% of spam emails contain "free" ($\theta \approx 0.6$).

In a new dataset, 75 out of 100 flagged spam emails contain "free."

Let us justify a Beta prior for θ using domain knowledge and derive the posterior distribution after observing the data. We will plot the prior, likelihood, and posterior and compute $P(\theta > 0.7 | \text{data})$.

We have prior information that θ is likely near 0.6 and our prior $\text{Beta}(\alpha, \beta)$ should reflect that. To match $\mathbb{E}[\theta] = 0.6$, we will set $\alpha/(\alpha + \beta) = 0.6$ i.e. choose a prior with this mean. There are many options for this, and our choice will depend on our confidence (reflected in the variance of the prior).

For moderate confidence, choose $\alpha + \beta = 10$ for e.g., $\text{Beta}(6, 4)$.

The likelihood can be derived from $\text{Bernouli}(1, \theta)$ using a sample of 100 with 75 successes. Thus the posterior is given by (try to derive this)

$$\text{Beta}(6 + 75, 4 + 25) = \text{Beta}(81, 29).$$

Let us plot the prior and the posterior

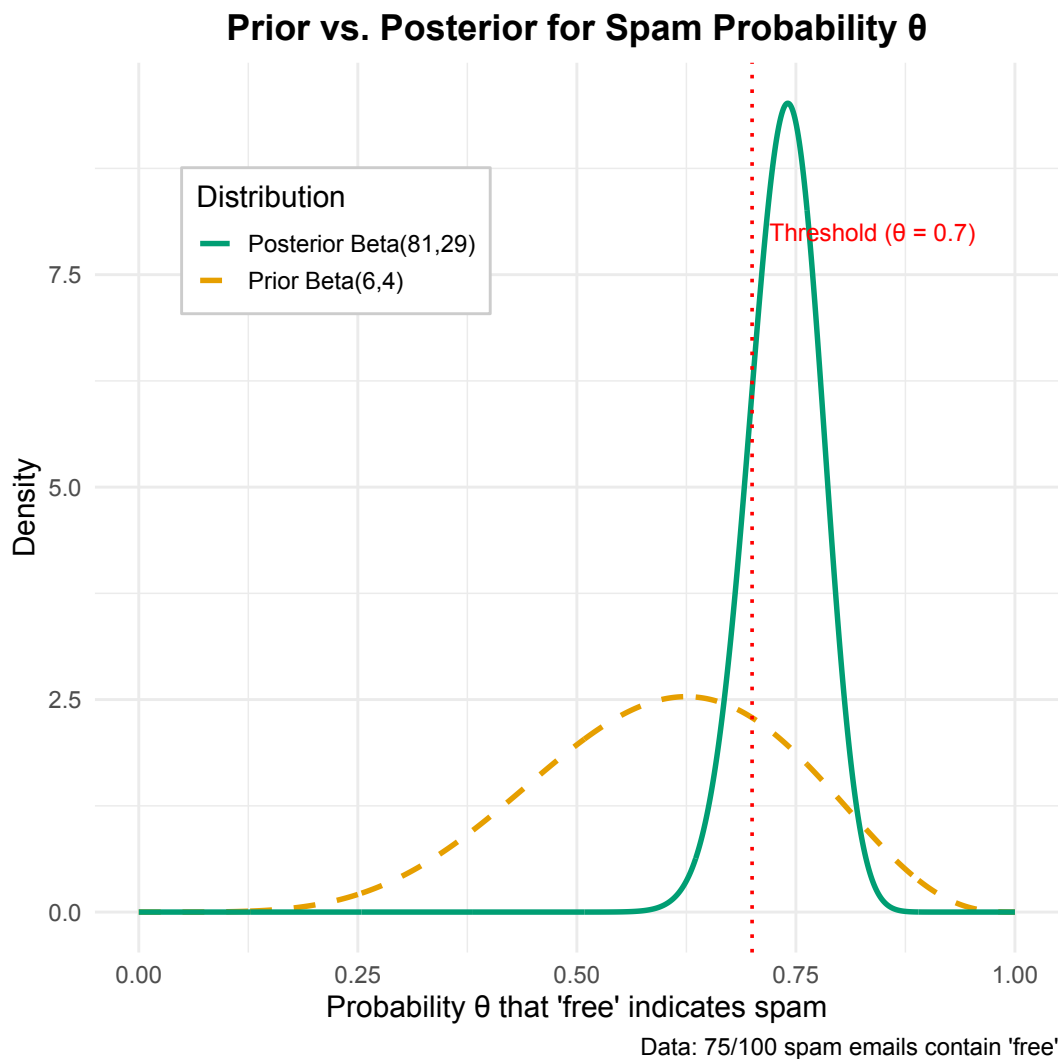


Figure 29: Prior vs posterior Beta distributions

Let us calculate

```
# P( > 0.7 | data)
1 - pbeta(0.7, 81, 29) # 0.92
```

This value is chosen for comparison, since we had a prior info of 60% spam and we have observed in the data 75% of spam. The posterior strongly favors $\theta > 0.7$ (92% probability), even though the prior centered at 0.6. The data (75/100) dominated the prior due to the large sample size. In practice, priors based on domain knowledge (e.g., $\text{Beta}(6,4)$) prevent extreme conclusions from sparse data.

Let us assume that we have a prior knowledge that 5% of non-spam emails contain the word "free". Let's see how we will incorporate this in the prior.

14.2 Prior predictive check, sensitivity analysis, data simulation

Prior predictive check involves generating simulated data using our model's likelihood function and our chosen prior distribution. This simulation gives us a picture of what data might look like if the world truly followed our prior beliefs. The goal is to verify that our prior, in combination with our model structure, produces reasonable predictions i.e. predictions that we believe in. If the simulated data are wildly implausible compared to what we expect to see in real life, it may indicate that our prior needs adjustment.

Next we will calculate the prior predictive probability for the data.

Definition 14.7. a) The **prior predictive probability** of data $\tilde{y}$ is defined as:

$$p(\tilde{y}) = \int p(\tilde{y} | \theta) p(\theta) d\theta, \quad (13)$$

where $p(\theta)$ is the prior distribution of the parameters θ , $p(\tilde{y} | \theta)$ is the likelihood of the data $\tilde{y}$.

b) Once we observe some data y and obtain the posterior $p(\theta|y)$ we can find the **posterior predictive probability** for observing $\tilde{y}$ is given by The posterior predictive distribution is given by:

$$p(\tilde{y} | y) = \int \underbrace{p(\tilde{y} | \theta)}_{\text{Likelihood of new data}} \underbrace{p(\theta | y)}_{\text{Posterior for } \theta} d\theta,$$

With this we can predict the distribution of possible data outcomes under the model and prior assumptions, before starting the experiment (i.e. before seeing any data).

Let us calculate the **prior predictive probability for our case of Bernoulli distributed data.**

To calculate the prior predictive probability when the likelihood follows a Bernoulli distribution, we assume a conjugate Beta prior for the Bernoulli parameter θ (probability of success).

Case 1. Single observation

Using the Bernoulli likelihood and the prior $Beta(\alpha, \beta)$ for a single observation $\tilde{y} \in \{0, 1\}$ we calculate:

$$\begin{aligned} p(\tilde{y}) &= \int_0^1 \theta^{\tilde{y}} (1 - \theta)^{1-\tilde{y}} \cdot \frac{1}{B(\alpha, \beta)} \theta^{\alpha-1} (1 - \theta)^{\beta-1} d\theta \\ &= \frac{1}{B(\alpha, \beta)} \underbrace{\int_0^1 \theta^{\tilde{y}+\alpha-1} (1 - \theta)^{(1-\tilde{y})+\beta-1} d\theta}_{\text{Beta function value } B(\tilde{y}+\alpha, (1-\tilde{y})+\beta)} \\ &= \frac{B(\tilde{y} + \alpha, (1 - \tilde{y}) + \beta)}{B(\alpha, \beta)}. \end{aligned}$$

Thus

- If $\tilde{y} = 1$:

$$p(\tilde{y} = 1) = \frac{B(\alpha + 1, \beta)}{B(\alpha, \beta)} = \frac{\alpha}{\alpha + \beta}.$$

- If $\tilde{y} = 0$:

$$p(\tilde{y} = 0) = \frac{B(\alpha, \beta + 1)}{B(\alpha, \beta)} = \frac{\beta}{\alpha + \beta}.$$

Hence, the prior predictive distribution for a single Bernoulli trial is

$$p(\tilde{y}) = \begin{cases} \frac{\alpha}{\alpha + \beta}, & \text{if } \tilde{y} = 1, \\ \frac{\beta}{\alpha + \beta}, & \text{if } \tilde{y} = 0. \end{cases}$$

This means that given the prior $Beta(\alpha, \beta)$ our data will have Bernoulli distribution $B\left(1, \frac{\alpha}{\alpha + \beta}\right)$ distribution.

Remark. We remark here, that we have actually calculated total probability for y , given all the possible scenarios (hypothesis) for θ . See Theorem 8.8 and Exercise 8.9.

Example 14.8. Let us consider the case with the basketball player. His success in free throw is θ . Let us consider the prior $Beta(2, 3)$. The prior predictive distribution for the data is Bernoulli $B(1, \frac{2}{5})$. If we simulate 100 values for the data, we should get approximately 40% of successful free throws. If this aligns with our believes about the basketball player, we can choose this prior. If expect that he is a better player, we may choose a different prior.

Next, suppose the player had 6 successful throws out of 10. Using this data, we obtain posterior distribution for θ $Beta(8, 7)$ and posterior predictive distribution for the data is $B(1, \frac{8}{15})$. If now we simulate 100 values for the data, we will get around 53.3%. The data has shifted the prior to a posterior which favors higher values for success and this is also reflected in the data.

We remark that the prior $Beta(20, 30)$ would lead to the same prior predictive distribution for the data Bernoulli $B(1, \frac{2}{5})$. However, the two priors would give us different posterior distributions, after we include the observed data. The weaker prior would be overpowered by the data, while the stronger prior would not allow a big shift away from it. Thus, we will also get different posterior predictive distribution for the data. If we take into account the data with 6 successes out of 10, the posterior in this case would be $Beta(26, 34)$ and the posterior predictive distribution for the data $B(1, \frac{26}{60})$. If we simulate 100 values now, the success would be around 43.3% - closer to the prior predictive probability for success. The prior in this case is stronger, we have small data set and and it will not influence the prior very much, which is also reflected in the data. $\square$

Let us now calculate the probability for k successes out of m independent trials.

Case 2. m independent trials (Binomial Likelihood)

If we want to find the probability of k successes in m trials, than the likelihood becomes a binomial:

$$p(k | \theta) = \binom{m}{k} \theta^k (1 - \theta)^{m-k}.$$

Thus, the prior predictive probability for $\tilde{k}$ successes out of m , $\tilde{k} \in \{0, 1, \dots, n\}$ is

$$\begin{aligned} p(\tilde{k}) &= \int_0^1 p(\tilde{k} | \theta) p(\theta) d\theta = \int_0^1 \binom{m}{\tilde{k}} \theta^{\tilde{k}} (1 - \theta)^{m-\tilde{k}} \frac{\theta^{\alpha-1} (1 - \theta)^{\beta-1}}{B(\alpha, \beta)} \\ &= \frac{\binom{m}{\tilde{k}} B(\alpha + \tilde{k}, \beta + m - \tilde{k})}{B(\alpha, \beta)} \underbrace{\int_0^1 \frac{\theta^{\alpha+\tilde{k}-1} (1 - \theta)^{\beta+m-\tilde{k}-1}}{B(\alpha + \tilde{k}, \beta + m - \tilde{k})} d\theta}_{=1, \text{ as integral of a density } \sim Beta(\alpha + \tilde{k}, \beta + m - \tilde{k})} \\ &= \frac{\binom{m}{\tilde{k}} B(\alpha + \tilde{k}, \beta + m - \tilde{k})}{B(\alpha, \beta)} \end{aligned} \tag{14}$$

This is the so called **Beta-Binomial distribution** $BetaBin(m, \alpha, \beta)$. The beta-binomial distribution is the binomial distribution in which the probability of success at each of n trials is not fixed but randomly drawn from a beta distribution.

Example 14.9. Let us consider the prior $Beta(2, 2)$, a symmetric prior. Then

- Prior predictive probability for $\tilde{y} = 1$ (success) is

$$p(\tilde{y} = 1) = \frac{2}{2+2} = 0.5.$$

- Prior predictive probability for $\tilde{k} = 3$ successes in $n = 5$ trials will be

$$p(\tilde{k} = 3) = \binom{5}{3} \frac{B(3+2, 5-3+2)}{B(2, 2)} = 10 \cdot \frac{B(5, 4)}{B(2, 2)}.$$

This can be calculate with a software or analytically, using the relation

$$B(a, b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)}$$

and $\Gamma(n) = (n-1)!$. We get

$$p(\tilde{k} = 3) = \frac{3}{14}$$

□

Example 14.10. The basketball player will throw 30 free throws in a row. We assume that the prior for the probability of success θ is $Beta(2, 3)$. Find the prior predictive probability distribution, its expected mean and the probability that he will score more than 20. Do this analytically, using R and by simulation.

- First we will use directly the formula for Beta-binomial distribution (14) and its properties

```

1 # Parameters
2 n <- 30      # Total free throws
3 alpha <- 2   # Beta prior parameters
4 beta <- 3
5
6 # Expected mean of Beta-Binomial
7 expected_mean <- n * alpha / (alpha + beta)
8 cat("Expected mean:", expected_mean, "\n")
9 # Output: Expected mean: 12
10
11 # Probability of making >20 (21-30 successes)
12 k_values <- 21:30
13 prob_more_than_20 <- sum(sapply(k_values, function(k) {
14   choose(n, k) * beta(k + alpha, n - k + beta) / beta(alpha, beta)
15 })))
16 cat("P(k > 20):", prob_more_than_20, "\n")
17 # Output: P(k > 20):

```

Listing 1: Manual Beta-Binomial Calculation

- Next we will use the **extraDistr** package. We will also plot the PMF

```

1 library(extraDistr)
2
3 # Parameters
4 n <- 30      # Total free throws
5 alpha <- 2   # Beta prior parameters
6 beta <- 3
7
8 # Expected mean (same formula)
9 expected_mean <- 30 * 2 / (2 + 3)
10 cat("Expected mean:", expected_mean, "\n")
11 # Output: Expected mean: 12
12
13 # Probability of >20 using cumulative distribution
14 prob_more_than_20 <- 1 - pbbinom(20, size = n,
15                                alpha = alpha, beta = beta)
16 cat("P(k > 20):", prob_more_than_20, "\n")
17 # Output: P(k > 20):
18
19 # Generate full PMF for plotting
20 k <- 0:n
21 pmf <- dbbinom(k, size = n, alpha = alpha, beta = beta)

```

Listing 2: Calculation with extraDistr Package

-Next we will do Monte Carlo simulation

```

1 set.seed(123) # Reproducible results
2 # Parameters
3 n <- 30      # Total free throws
4 alpha <- 2   # Beta prior parameters
5 beta <- 3
6
7 num_sim <- 1000 # Number of simulations
8
9 # Simulate theta from Beta prior
10 theta_samples <- rbeta(num_sim, alpha, beta)
11
12 # Simulate binomial outcomes
13 k_samples <- rbinom(num_sim, size = n, prob = theta_samples)
14
15 # Calculate empirical statistics
16 mc_mean <- mean(k_samples)
17 cat("Monte Carlo Expected Mean:", mc_mean, "\n")
18 # Output: Monte Carlo Expected Mean:
19
20 mc_prob <- mean(k_samples > 20)
21 cat("Monte Carlo P(k > 20):", mc_prob, "\n")
22 # Output: Monte Carlo P(k > 20):

```

Listing 3: Monte Carlo Simulation Approach

Using the analytical methods we get an expected number of 12 successful throws and $P(k > 20) = 0.115$ (averaged across all possible values of θ). Using the Monte Carlo simulation we get an expectation of 12.259 and $P(k > 20) = 0.127$.

□

What we have obtained in the previous discussion is the so called **compound distribution**. Let us define it.

Definition 14.11. Let us assume that data Y has distribution $F(\theta)$ with an unknown parameter θ . Let θ also be a random variable with a distribution G . The result of compounding of F with G is a distribution H with a density given by

$$p_H(y) = \int p_F(y|\theta)p_G(\theta)d\theta$$

This new, unconditional distribution H is obtained by integrating out the parameter θ or we say it results from marginalizing over G

Discuss the expectation of Y based on the prior/ posterior predictive distribution.

The compound distribution is a special case of marginal distribution (see for example the Wikipedia page). The term compound distribution is also used when talking about a random sum of random variables, convolutions, compound Poisson processes etc. For more details and examples, see <https://mathmodelsblog.wordpress.com/2010/01/17/an-introduction-to-compound-distributions/> or <https://probabilityandstats.wordpress.com/2025/03/21/compound-distributions/>

Examples:

- The Beta-Binomial distribution Beta-Binomial distribution is an example where a Binomial distribution's probability parameter is itself random and follows a Beta distribution.
- The Negative Binomial distribution Negative Binomial distribution can be seen as a compound distribution if it is constructed as a Poisson distribution with a Gamma-distributed rate parameter (often called a Poisson-Gamma mixture).

Next we give formula for the expectation and the variance of the compound distribution H of Y .

Theorem 14.12. a) Law of Total Expectation The overall expectation of Y is given by

$$E_H(Y) = E_G[E_F(Y|\theta)].$$

b) Law of Total Variance The overall variance of Y can be decomposed as follows:

$$\text{Var}_H(Y) = E_G[\text{Var}_F(Y|\theta)] + \text{Var}_G[E(Y_F|\theta)].$$

Proof. We will prove the first part only. Let F and G be probability distributions and let us denote the probability density functions as

$$f(y|\theta) = p_F(y|\theta) \quad \text{and} \quad g(\theta) = p_G(\theta),$$

respectively, and let $h(y)$ be the probability density of H . Then we have

$$\begin{aligned} E_H(Y) &\stackrel{\text{def.}}{=} \int_F y h(y) dy \\ &= \int_F y \left(\int_G f(y|\theta) g(\theta) d\theta \right) dy \\ &= \int_G \left(\int_F y f(y|\theta) dy \right) g(\theta) d\theta \\ &= \int_G E_F(Y | \theta) g(\theta) d\theta \\ &\stackrel{\text{def.}}{=} E_G(E_F(Y|\theta)) \end{aligned}$$

■

Let us illustrate this with an example.

Example 14.13. Let $Y \sim B(1, \theta)$ and $\theta \sim \text{Beta}(\alpha, \beta)$. The distribution of Y depends on θ and we should actually write $Y|\theta \sim B(1, \theta)$ - these are different random variables with a distribution that depends on θ . For their distribution we have

$$E(Y | \theta) = \theta.$$

This is again a random variable and thus we get

$$E(E(Y|\theta)) = E(\theta) = \frac{\alpha}{\alpha + \beta}$$

On the other hand, from the compounding, we have obtained that the unconditional distribution of Y is $B(1, \frac{\alpha}{\alpha+\beta})$. Thus

$$E(Y) = \frac{\alpha}{\alpha + \beta} = E(E(Y|\theta)).$$

This is actually a weighted expectation - we take into account "all possible" values of θ and the corresponding expectations of $Y|\theta$ and find a weighted average taking into account the distribution of θ $\square$

Example 14.14. Let us suppose that we have data Y modeled as $Y | \theta \sim B(n, \theta)$, where $\theta \sim \text{Beta}(\alpha, \beta)$.

We have seen that, by compounding we got the prior predictive distribution (unconditional distribution) of Y , which is the Beta-Binomial distribution with probability mass function

$$p(y) = \binom{n}{y} \frac{B(y + \alpha, n - y + \beta)}{B(\alpha, \beta)}.$$

Theoretically, the mean of this distribution is $\frac{n\alpha}{\alpha+\beta}$

Also

$$E(E(\underbrace{Y|\theta}_{\sim B(n, \theta)})) = E(n\theta) = nE(\underbrace{\theta}_{\sim \text{Beta}(\alpha, \beta)}) = n \frac{\alpha}{\alpha + \beta}$$

With this we have confirmed the law of total expectation.

Lets go back to Example 14.10 The basketball player will throw 30 free throws in a row. We have assumed a prior θ is $\text{Beta}(2, 3)$. We are interested in number of success in $n = 30$ attempts. The prior predictive probability for this is $K \sim \text{BetaBin}(30, 2, 3)$. The expected number of successful attempts is given by

$$E(K) = E(E(K|\theta)) = E(n\theta) = nE(\theta) = \frac{n\alpha}{\beta} = \frac{60}{5} = 12,$$

where we have used that $K|\theta \sim B(n, \theta)$. Thus the prior expected number of successful free throws out of 30 is 12 (based on the prior distribution).

After seeing some data we can calculate also the posterior expected number of data. $\square$

14.2.1 Maximum a posteriori estimation (MAP)

MAP estimate $\hat{\theta}_{MAP}$ is the value of θ that maximizes the posterior $p(\theta|y)$, i.e. it is the mode of the posterior distribution. We have seen that MLE method finds θ which maximizes the likelihood function $p(y|\theta)$. MAP can be seen as a Bayesian version of MLE, which also takes into account the data and the prior.

$$\begin{aligned} \hat{\theta}_{MAP} &= \arg \max_{\theta} p(\theta|y) = \arg \max_{\theta} \frac{p(y|\theta)p(\theta)}{p(y)} \\ &= \arg \max_{\theta} p(y|\theta)p(\theta) = \arg \max_{\theta} (\log(p(y|\theta)) + \log(p(\theta))) \end{aligned} \quad (15)$$

When the prior is uniform $\log(p(\theta))$ is a constant and does not affect the argument maximum. Thus, in this case $\hat{\theta}_{MAP}$ is identical to the ML estimate of θ .

The MAP estimator is consistent for reasonable choices of the prior (i.e. prior distributions that do not assign zero probability to the true parameter θ).

Let us calculate $\hat{\theta}_{MAP}$ for the case of Bernoulli data (Bernoulli likelihood) when a prior $Beta(\alpha, \beta)$ is chosen. In (15) we need to substitute the likelihood

$$p(y | \theta) = \theta^{\sum_{i=1}^n y_i} (1 - \theta)^{n - \sum_{i=1}^n y_i}$$

and the prior

$$p(\theta) = \frac{\theta^{\alpha-1} (1 - \theta)^{\beta-1}}{B(\alpha, \beta)}$$

We want to find the value of θ that maximizes the expression

$$\sum_{i=1}^n y_i \log(\theta) + (n - \sum_{i=1}^n y_i) \log(1 - \theta) + (\alpha - 1) \log(\theta) + (\beta - 1) \log(1 - \theta) - \log B(\alpha, \beta) \quad (16)$$

After differentiating and setting to zero at $\hat{\theta}_{MAP}$ we have

$$\begin{aligned} \frac{\sum_{i=1}^n y_i}{\hat{\theta}_{MAP}} - \frac{n - \sum_{i=1}^n y_i}{1 - \hat{\theta}_{MAP}} + \frac{\alpha - 1}{\hat{\theta}_{MAP}} - \frac{\beta - 1}{1 - \hat{\theta}_{MAP}} &= 0 \\ \hat{\theta}_{MAP} &= \frac{\sum_{i=1}^n y_i + \alpha - 1}{n + \beta - 1 + \alpha - 1} = \frac{n}{n + \alpha + \beta - 2} \frac{\sum_{i=1}^n y_i}{n} + \frac{\alpha - 1}{n + \alpha + \beta - 2} \\ &= \underbrace{\frac{n}{n + \alpha + \beta - 2}}_{\text{Data weight}} \cdot \hat{\theta}_{MLE} + \underbrace{\frac{\alpha - 1}{n + \alpha + \beta - 2}}_{\text{Prior weight}} \end{aligned}$$

In this case the ML estimate is given by $\hat{\theta}_{MLE} = \frac{\sum_{i=1}^n y_i}{n}$.

We have mentioned that a choice of $Beta(\alpha, \beta)$ can be understood as if we have seen a prior hypothetical sample with additional $\alpha - 1$ "1"s (i.e. heads) and $\beta - 1$ "0"s. If for example we choose $\alpha = 6, \beta = 8$ as in a previous discussion, this is as if we have started with 5 heads and 7 tails already (i.e with this sample of size 12 (1,1,1,1,1,0,0,0,0,0,0,0)).

Thus we can see that $\hat{\theta}_{MAP}$ is the ML estimate of a data set with additional $\alpha - 1$ "1"s (i.e. heads) and $\beta - 1$ "0"s (i.e. tails) (hypothetical sample). This can be seen from first part of the formula (see what is in the numerator and what in the denominator).

We can also see the influence of the data and the prior in the last equation.

For large samples from the Bernoulli distribution, as $n \rightarrow \infty$ we can see that

$$\lim_{n \rightarrow \infty} \hat{\theta}_{MAP} = \lim_{n \rightarrow \infty} \hat{\theta}_{ML} = \theta_{true} \quad (17)$$

since the ML estimator is consistent. Thus we conclude that the data will dominate and the point estimate of θ would be the same with frequentist and Bayesian approach.

Some discussion and simulations in Python can be found on <https://towardsdatascience.com/full-explanation-of-mle-map-and-bayesian-inference-1db9a7fb1d2b/>.

Example 14.15. Let us go back to the example with the basketball player, who had 0 successes in 3 attempts ($\hat{\theta}_{MLE}=0$).

For a weak prior $Beta(3, 2)$

$$\begin{aligned} \text{Prior Mean} &= \frac{3}{3 + 2} = 0.6 \\ \hat{\theta}_{MAP} &= \frac{3}{3 + 3 + 2 - 2} \cdot 0 + \frac{3 - 1}{6} = \frac{2}{6} \approx 0.333 \end{aligned}$$

For the stronger prior $Beta(30, 20)$

$$\text{Prior Mean} = \frac{30}{30 + 20} = 0.6$$

$$\hat{\theta}_{MAP} = \frac{3}{3 + 30 + 20 - 2} \cdot 0 + \frac{30 - 1}{3 + 30 + 20 - 2} = \frac{29}{51} \approx 0.569$$

Both priors have the same mean (0.6) but different strengths. The strong prior (N=50) dominates the data, keeping estimate near 0.6. The weak prior (N=5) shows more sensitivity to data, pulling estimate down to 0.333. All estimates are between the MLE (0) and prior mean (0.6). $\square$

We have worked so far with a so called conjugate prior for the likelihood model. Let us define it.

14.2.2 Bayesian confidence region aka credible intervals.

An interval $(l(y), r(y))$, based on the observed data y , has 90% Bayesian coverage for θ if

$$P(l(y) < \theta < r(y) | Y = y) = 0.90 \quad (18)$$

This interval describes our information about the position of the true value of θ , after observing the data. This is different from the frequentist approach for construction of confidence interval. The definition of the later is given by

$$P(l(Y) < \theta < r(Y) | \theta) = 0.90 \quad (19)$$

Note that in this last equation we are talking about the random confidence interval - its limits are random values $l(Y), r(Y)$. When we calculate a confidence interval based on a specific sample, we have a realization of this random interval. Thus $P(l(y) < \theta < r(y) | \theta)$ is 0 or 1 - the true value of θ is either in the interval or not. Thus the confidence interval itself does not tell us much about the uncertainty of the estimated value of θ .

In the previous example with the basketball player, the quantile based interval of the chosen posterior can be calculated with

```
1 qbeta(c(0.05, .95), 30, 23)
2 > 0.454 0.676
```

The interval (0.454, 0.676) means there is a 90% probability that θ lies between 0.454 and 0.676, given the observed data and prior beliefs. Mathematically: $P(0.454 < \theta < 0.676 | \text{Data}) = 0.90$.

A weaker prior (e.g., $Beta(3, 2)$) would widen the interval due to higher uncertainty.

The interval directly quantifies uncertainty about θ , unlike frequentist confidence intervals, which describe long-run behavior. It combines prior knowledge (league averages) and observed data (0/3 shots made). It tells us: "Given what we know, what is plausible for θ ?"

The Bayesian interval is a statement about the parameter θ itself- it is derived from the posterior distribution. The frequentist interval is a statement about the method (not about the calculated interval, which is often misunderstood).

In practice - we can use wider intervals (e.g., 90%) if we need more conservative estimates. We can recalculate the interval after every game to track the player's progress.

This quantification of uncertainty is especially important in ML.

14.2.3 Posterior analysis

- credible intervals, highest density interval, - Posterior probabilities $P(\theta > 0|y)$, $P(\theta_1 > \theta_2)$,
- Region of practical equivalence (ROPE),
- Check $P(\theta \in [-\delta, \delta] | y)$ against decision thresholds,
- Bayes factors, posterior odds and decision rules (based on loss function), - simulate data from posterior distribution,
- graphical representation,
- Model comparison.

14.2.4 ROPE

14.2.5 Posterior predictive check

14.3 Conjugate Priors

A **conjugate prior** is a prior distribution that, when combined with a likelihood function (via Bayes' theorem), produces a posterior distribution that is in the same family as the prior. In other words, if you choose a conjugate prior for your likelihood, the mathematical form of the prior is preserved in the posterior, simplifying the process of updating your beliefs with new data.

The following table represents some common combinations.

Table 22: Common Likelihood Models and Their Conjugate Priors

Data dist. (likelihood model)	Conjugate Prior
Bernoulli/Binomial	Beta
Multinomial	Dirichlet
Poisson	Gamma
Exponential (rate parameter)	Gamma
Normal (known variance, for the mean)	Normal
Normal (known mean, for the variance)	Inverse Gamma (or scaled Inverse- χ^2)
Normal (unknown mean and variance)	Normal-Inverse-Gamma (or Normal-Inverse- χ^2)
Multivariate Normal (for the covariance matrix)	Wishart (for the precision matrix) Inverse-Wishart (for the covariance matrix)

Next we will consider data from Binomial distribution

15 Data from Binomial distribution

Let $Y \sim \text{Bin}(m, \theta)$. The probability law of the binomial distribution is given by

$$p_\theta(Y = \tilde{y}) = p(\tilde{y}|\theta) = \binom{m}{\tilde{y}} \theta^{\tilde{y}} (1 - \theta)^{m - \tilde{y}}, \quad \tilde{y} \in \{0, 1, 2, \dots, m\}$$

Let $y = (y_1, y_2, \dots, y_n)$ be some sample from the binomial distribution. Then the likelihood function is given by

$$p(y|\theta) = \prod_{i=1}^n p_\theta(Y = y_i) = \prod_{i=1}^n \binom{m}{y_i} \theta^{y_i} (1 - \theta)^{m - y_i} = \left(\prod_{i=1}^n \binom{m}{y_i} \right) \theta^{\sum_{i=1}^n y_i} (1 - \theta)^{nm - \sum_{i=1}^n y_i}$$

We will show that the Beta distribution is conjugate prior for the binomial. Let $\theta \sim \text{Beta}(\alpha, \beta)$, i.e. the prior distribution of θ is

$$p(\theta) = \frac{\theta^{\alpha-1}(1-\theta)^{\beta-1}}{B(\alpha, \beta)}$$

where $B(\alpha, \beta)$ is the beta function. The posterior distribution of θ is given by

$$p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)} = \frac{\prod_{i=1}^n \binom{m}{y_i}}{B(\alpha, \beta)p(y)} \theta^{\overbrace{\alpha + \sum_{i=1}^n y_i}^{\alpha_1} - 1} (1-\theta)^{\overbrace{\beta + nm - \sum_{i=1}^n y_i}^{\beta_1} - 1}$$

This is density of Beta distribution $\text{Beta}(\alpha_1, \beta_1)$ where $\alpha_1 = \alpha + \sum_{i=1}^n y_i$, $\beta_1 = \beta + nm - \sum_{i=1}^n y_i$. Thus, the constant in front is $\frac{1}{B(\alpha_1, \beta_1)}$.

Taking $m = 1$ we will get the special case of Bernoulli distribution.

Using the properties of Beta distribution we see that the prior mean of θ is $\frac{\alpha}{\alpha+\beta}$ and similarly, the posterior mean of θ is $\mathbb{E}(\theta|y) = \frac{\alpha_1}{\alpha_1+\beta_1} = \frac{\alpha + \sum_{i=1}^n y_i}{\alpha + \beta + nm}$.

We will make a connection between the the two means and the MLE estimator for θ . The MLE estimator for the probability parameter θ in the Binomial distribution $\text{Bin}(m, \theta)$, based on the given sample is $\hat{\theta} = \frac{\sum_{i=1}^n y_i}{mn}$. Thus, the posterior mean of θ can be represented as a weighted sum of the prior mean and $\hat{\theta}$

$$\begin{aligned} \mathbb{E}(\theta|y) &= \frac{\alpha + \sum_{i=1}^n y_i}{\alpha + \beta + nm} = \frac{\alpha + \beta}{\alpha + \beta + nm} \frac{\alpha}{\alpha + \beta} + \frac{nm}{\alpha + \beta + nm} \frac{\sum_{i=1}^n y_i}{m} \\ &= \frac{\alpha + \beta}{\alpha + \beta + nm} \mathbb{E}(\theta) + \frac{nm}{\alpha + \beta + nm} \hat{\theta}_{MLE} \end{aligned}$$

Thus, when $n \rightarrow \infty$ the posterior expected value for θ tends to the MLE estimator of theta. As an exercise, simulate the posterior predictive distribution of Y for different sample sizes, including larger ones, as well as data from $\text{Bin}(m, \hat{\theta})$. Compare the distributions

So far we have discussed the parameter θ . What about the data? Let us next discuss what is the mean of the data itself and what is probability to observe some data from the prior or the posterior.

Example 15.1. We plan to observe the number of sunny days in the next 10 days. The data follows binomial distribution $\text{Bin}(10, \theta)$, $y \in 0, 1, 2, \dots, 10$. We will take prior of $\theta \sim B(3, 11)$ (in later examples we will discuss the choice of parameters). We would like to see how the parameter θ affects the data and analyze whether the model describes our "expectations" for the data. We can simulate data for different parameters θ and see how sensitive the model is to change of parameter values. Sometimes we have better understanding of our belief in the parameter when we see simulated data. Let us simulate values of Y for $\theta = 0.1$, $\theta = 0.2$ and $\theta = 0.3$. The following code gives side by side histograms, as presented bellow.

```

1 Y1<-rbinom(1000,10,0.1)
2 Y2<-rbinom(1000,10,0.2)
3 Y3<-rbinom(1000,10,0.3)
4 par(mfrow=c(1,3))
5 hist(Y1,xlab="Y1~Bin(10,0.1)",main = "")
6 hist(Y2,xlab="Y2~Bin(10,0.2)",main = "")
7 hist(Y3,xlab="Y3~Bin(10,0.3)", main = "")
8 par(oma=c(0,0,2,0))
9 title("Number of sunny days (samples of 1000), for different theta" , outer=TRUE)

```

Number of sunny days (samples of 1000), for different theta

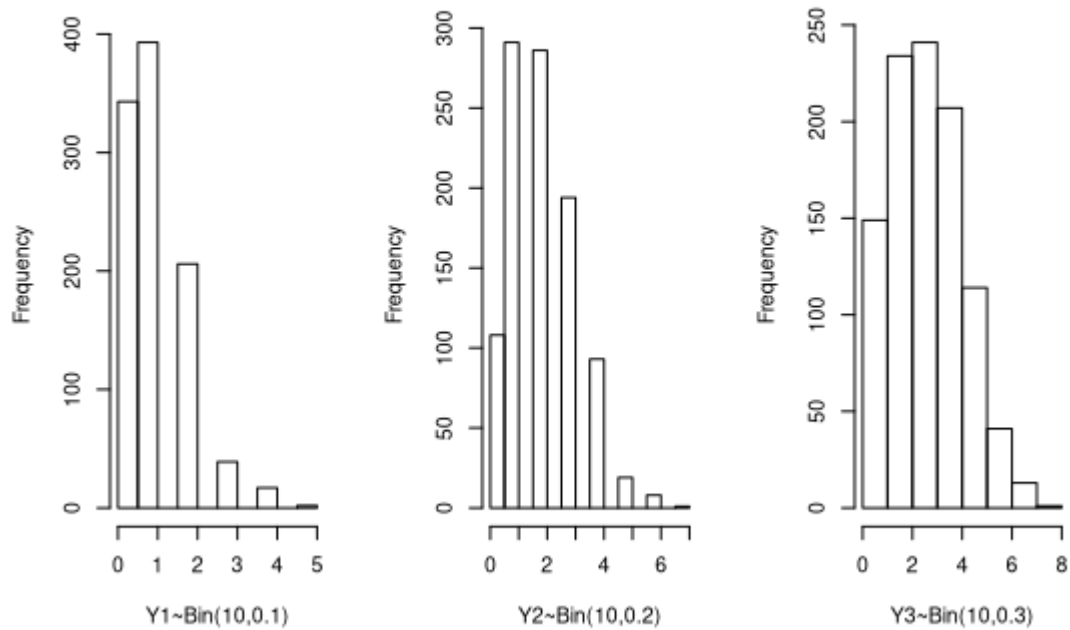


Figure 30: Side by side histograms of the data simulated for different values of theta.

However, what we need to do is simulate values from the prior predictive probability of the data. The following code of Monte Carlo simulations gives us 10000 values Y (of numbers of sunny days) based on the distribution of θ (across all scenarios of θ).

```
1 par(mfrow=c(1,1))
2 theta<-rbeta(100,3,11)
3 Y=c()
4 for(i in theta)
5 { r<-rbinom(100,10,i)
6   Y=c(Y,r)
7 }
8 hist(Y,main="",freq=T)
```

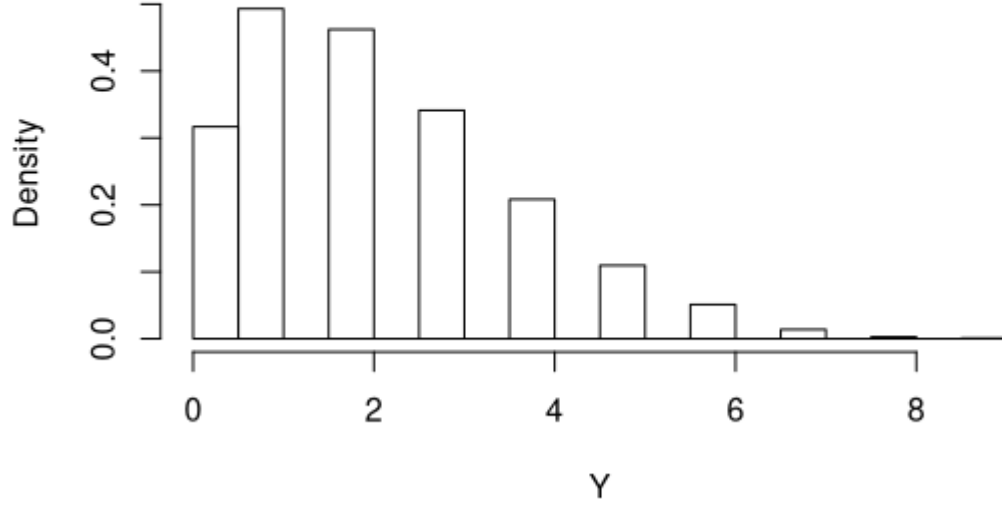


Figure 31: Density of number of sunny days simulated from the prior predictive distribution.

Next we observe 6 sunny days (out of 10) - thus we have a sample of size 1, i.e. $n = 1$ and $\sum y_i = y_1 = 6$. From the previous discussion we find that $\alpha_1 = 3 + 6 = 9$ and $\beta_1 = 11 + 10 - 6 = 15$. The prior predictive probability is

$$p(\tilde{y}) = \frac{\binom{10}{\tilde{y}} B(3 + \tilde{y}, 21 - \tilde{y})}{B(3, 11)}, \quad \text{beta binomial}$$

and the posterior predictive density we have

$$p(\tilde{y} | y) = \frac{\binom{10}{\tilde{y}} B(9 + \tilde{y}, 25 - \tilde{y})}{B(9, 15)}, \quad \text{beta binomial}$$

After seeing the data we may use the posterior predictive probability for forecasting.

Simulated data from the post. predictive probability

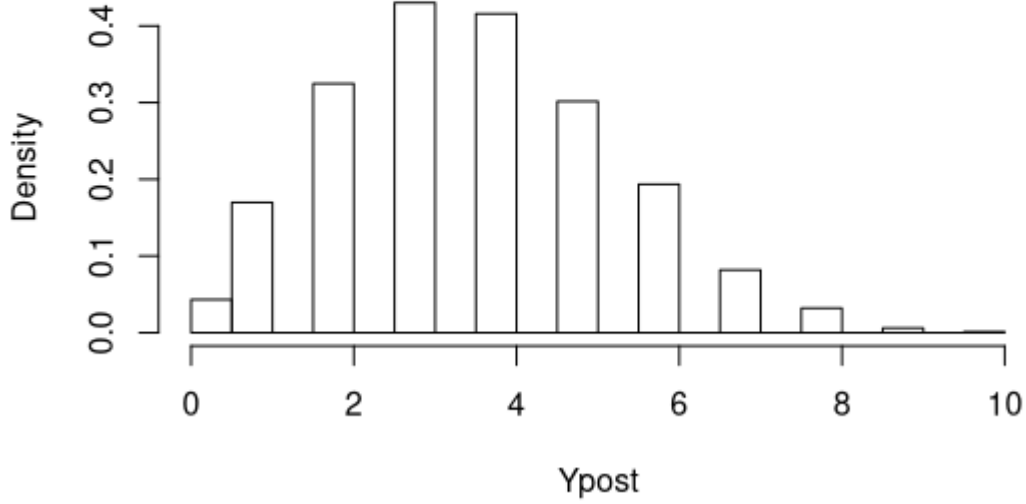


Figure 32: Density of number of sunny days simulated from the posterior predictive distribution.

The posterior mean of the data is $\frac{10.9}{9+15} = 3.75$, i.e. we expect 3.75 sunny days. For comparison the mean of the simulated data Y_{post} is $\bar{Y}_{post} = 3.67$. We can also use the simulated sample to calculate for example the probability that there will be less than 3 sunny days. The command

```
p<-sum(Ypost<3)/10000
```

gives $p=0.27$. Before observing the sample, this probability was higher, i.e. 0.59

Similar as in the previous example, we can calculate the $\hat{\theta}_{MAP}$.

$$\hat{\theta}_{MAP} = \frac{mn}{mn + \alpha + \beta - 2} \hat{\theta}_{MLE} + \frac{\alpha - 1}{mn + \alpha + \beta - 2} \quad (20)$$

where $\hat{\theta}_{MLE} = \frac{\sum_{i=1}^n y_i}{mn}$. You should be able to analyse how each of the parameters in the formula affects the estimated value and connect it with properties of the model (small sample, strong prior, large variance of the prior/ sampled data etc.)

Sometimes we have to decide on one value of θ from the posterior distribution, and simulate data for that chosen value of θ . Apart from MAP, there are other posterior estimates of θ . Here we will discuss only MAP. In our example, $\hat{\theta}_{MAP} = \frac{1}{4}$. Let us suppose that instead of the given sample, we have observed a sample of size 30 with $\sum_{i=1}^n y_i = 156$. For such a sample $\hat{\theta}'_{MAP} = 158/322 = 0.49$. The second term in the previous equation is very small (0.006), so this is approximately the true value of θ .

Let us sample data from binomial distribution with the two MAP estimators and compare them with the posterior predictive distribution of Y . The results are represented on the following figure.

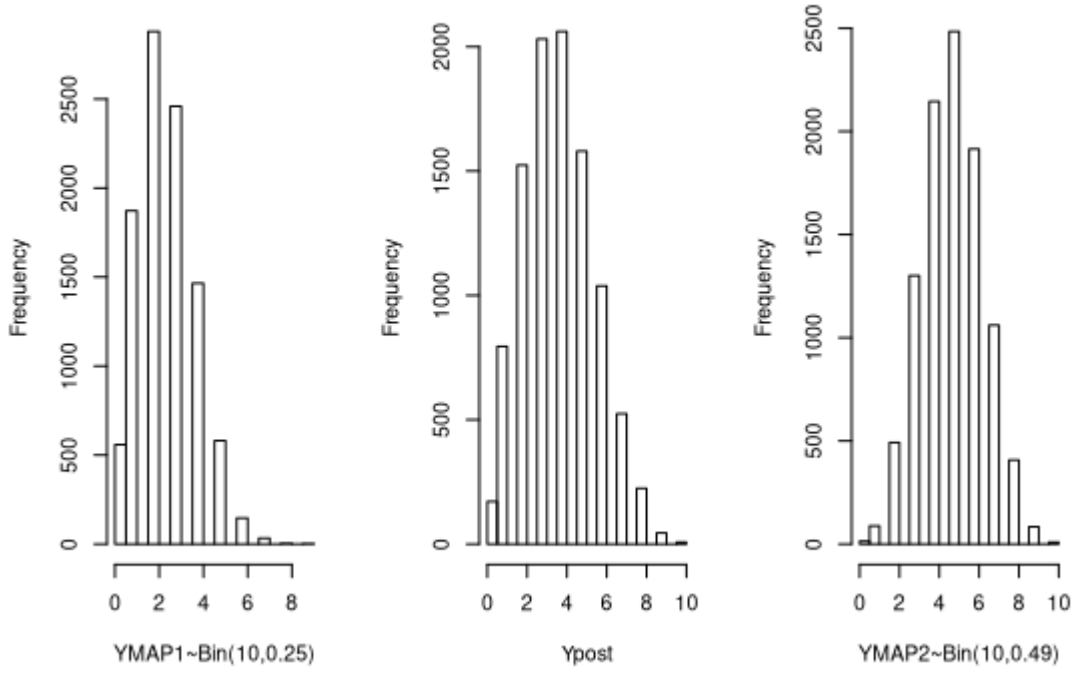


Figure 33: Histograms for the number of sunny days- for $\hat{\theta}_{MAP} = \frac{1}{4}$ (left), from the posterior predictive distribution (center), for $\hat{\theta}'_{MAP} = 0.49$.

As we can see, the left and the right histogram do not match the histogram of the posterior predictive distribution, which should be expected - both are simulations for one fixed value of θ , while the one for Y_{post} includes all possible scenarios for θ . Still, $\hat{\theta}'_{MAP} = 0.49$ is a better choice for simulation since it is based on a bigger sample and is very close to the true value of the parameter.

Additional exercise. Calculate credible intervals, HDI, probability that there will be less than 3 sunny days, prior and posterior predictive variance. $\square$

Check what effective sample size is, simulate with different priors (uniform, large and small variance, different parameters, see the influence to the data). Simulate with different sample sizes. Use different posterior values of theta to simulate data (MAP, posterior mean, median etc.) HDI, ROPE etc.)

16 Data from Geometrical distribution

Next, let us consider data from geometrical distribution $Y \sim Geom(\theta)$ i.e.

$$p(y|\theta) = P(Y = y|\theta) = \theta(1 - \theta)^{y-1}, \quad y \in \{0, 1, 2, \dots\}$$

If $y = (y_1, y_2, \dots, y_n)$ is a sample from this distribution, then the likelihood is

$$p(y|\theta) = \prod_{i=1}^n p_\theta(Y = y_i) = \theta^n (1 - \theta)^{\sum_{i=1}^n y_i - n}$$

We will show that the beta distribution is conjugate prior for the geometric distribution. Let $\theta \sim \text{Beta}(\alpha, \beta)$ i.e.

$$p(\theta) = \frac{\theta^{\alpha-1} (1 - \theta)^{\beta-1}}{B(\alpha, \beta)}$$

Then, the posterior for θ is given by

$$p(\theta|y) = \frac{\theta^n (1 - \theta)^{\sum_{i=1}^n y_i - n} \theta^{\alpha-1} (1 - \theta)^{\beta-1}}{B(\alpha, \beta) p(y)} = \frac{\theta^{\alpha+n-1} (1 - \theta)^{\beta+\sum_{i=1}^n y_i - n-1}}{B(\alpha, \beta) p(y)} \sim \text{Beta}(\alpha_1, \beta_1) \quad (21)$$

where $\alpha_1 = \alpha + n$, $\beta_1 = \beta + \sum_{i=1}^n y_i - n$. The constant term $B(\alpha, \beta) p(y)$ is the normalizing term of the posterior distribution and thus $B(\alpha, \beta) p(y) = B(\alpha_1, \beta_1)$.

Since $\theta \sim \text{Beta}(\alpha_1, \beta_1)$, the posterior mean for θ is $\frac{\alpha_1}{\alpha_1 + \beta_1}$

To analyze the model, we can calculate the prior predictive probability and simulate data.

The prior predictive probability for a new observation $\tilde{y}$ is computed by marginalizing out θ :

$$p(\tilde{y}) = \int_0^1 p(\tilde{y}|\theta) p(\theta) d\theta.$$

We substitute the expressions for the likelihood $p(\tilde{y}|\theta)$ and the prior $p(\theta)$:

$$\begin{aligned} p(\tilde{y}) &= \int_0^1 \theta (1 - \theta)^{\tilde{y}-1} \frac{\theta^{\alpha-1} (1 - \theta)^{\beta-1}}{B(\alpha, \beta)} d\theta \\ &= \frac{1}{B(\alpha, \beta)} \int_0^1 \theta^\alpha (1 - \theta)^{\tilde{y}+\beta-2} d\theta. \end{aligned}$$

The integral in this equation can be calculated using the Beta function since

$$\int_0^1 \theta^\alpha (1 - \theta)^{\tilde{y}+\beta-2} d\theta = B(\alpha + 1, \tilde{y} + \beta - 1).$$

Thus, the prior predictive probability is given by

$$p(\tilde{y}) = \frac{B(\alpha + 1, \tilde{y} + \beta - 1)}{B(\alpha, \beta)} \quad \tilde{y} \in \{0, 1, 2, \dots\}. \quad (22)$$

This is the so called **Beta-geometric distribution**, $\text{BetaGeom}(\alpha, \beta)$. Let us calculate its mean and variance.

Mean of Beta-geometric distribution. By the law of total expectation, for the expectation of Y we have

$$E(Y) = E(E(Y | \theta)) = E\left(\frac{1}{\theta}\right).$$

Let us calculate $E\left(\frac{1}{\theta}\right)$

$$\begin{aligned} E\left(\frac{1}{\theta}\right) &= \int_0^1 \frac{1}{\theta} p(\theta) d\theta = \frac{1}{B(\alpha, \beta)} \int_0^1 \frac{1}{\theta} \theta^{\alpha-1} (1 - \theta)^{\beta-1} d\theta \\ &= \frac{1}{B(\alpha, \beta)} \int_0^1 \theta^{\alpha-2} (1 - \theta)^{\beta-1} d\theta = \frac{B(\alpha-1, \beta)}{B(\alpha, \beta)} \end{aligned}$$

Next, we express the Beta function in terms of Gamma functions:

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}$$

and

$$B(\alpha - 1, \beta) = \frac{\Gamma(\alpha - 1)\Gamma(\beta)}{\Gamma(\alpha + \beta - 1)}.$$

Thus,

$$E\left(\frac{1}{\theta}\right) = \frac{\Gamma(\alpha - 1)\Gamma(\alpha + \beta)}{\Gamma(\alpha)\Gamma(\alpha + \beta - 1)}.$$

Now, using the relation $\Gamma(z + 1) = z\Gamma(z)$, we have:

$$\Gamma(\alpha) = (\alpha - 1)\Gamma(\alpha - 1) \quad \text{and} \quad \Gamma(\alpha + \beta) = (\alpha + \beta - 1)\Gamma(\alpha + \beta - 1).$$

Substitute these into our expression:

$$E\left(\frac{1}{\theta}\right) = \frac{\Gamma(\alpha - 1)(\alpha + \beta - 1)\Gamma(\alpha + \beta - 1)}{(\alpha - 1)\Gamma(\alpha - 1)\Gamma(\alpha + \beta - 1)} = \frac{\alpha + \beta - 1}{\alpha - 1}.$$

Hence $E(Y) = \frac{\alpha + \beta - 1}{\alpha - 1}$.

We have found the mean analytically, and the same can be done for the variance. However we can also simulate data from the distribution and estimate the mean. Install the package *VGAM* in *R* and see the use of **rbetageom**, **dbetageom**, **pbetageom**.

We will make simulations in one of our next examples.

The method for calculation of the **posterior predictive distribution** $p(\tilde{y}|y)$ is analogous with the one for prior predictive distribution, since the prior and the likelihood will be of the same form. Thus for a posterior $Beta(\alpha_1, \beta_1)$ and a geometric likelihood, we will again obtain Beta-geometric distribution $BetaGeom(\alpha_1, \beta_1)$

$$p(\tilde{y}|y) = \frac{B(\alpha_1 + 1, \beta_1 + \tilde{y} - 1)}{B(\alpha_1, \beta_1)}, \quad \tilde{y} \in \{1, 2, \dots\}$$

with a posterior predictive expectation

$$E(Y|y) = \frac{B(\alpha_1 - 1, \beta_1)}{B(\alpha_1, \beta_1)}$$

which can be easily calculated in *R*.

Example 16.1. Suppose a company displays an advertisement to website visitors. For each visitor, we record the number of times the ad is shown to the visitor (number of impressions) needed until the visitor finally clicks the ad. Let us assume that the number of impressions Y follows a geometric distribution with success probability θ ; that is,

$$P(Y = k | \theta) = (1 - \theta)^{k-1}\theta, \quad k = 1, 2, \dots,$$

where k is the number of impressions until the first click.

The company has seen some historical data with a probability for click of around 50%, however, they are not really sure about it. Let us choose the prior $\theta \sim \text{Beta}(3, 3)$. Then, using (22) for the prior predictive probability mass function for Y we get

$$P(Y = \tilde{y}) = \frac{B(4, \tilde{y} + 2)}{B(3, 3)},$$

Using the law of total expectation we can calculate

$$E(Y) = \frac{3 + 3 - 1}{3 - 1} = \frac{5}{2} = 2.5.$$

For the prior predictive variance, the law of total variance gives

$$\text{Var}(X) = 7.5 + 3.75 = 11.25.$$

For exercise - Simulate the prior predictive distribution using Monte Carlo sampling. Analyze the distribution and its statistical properties (mean, variance, quantiles, some characteristic probabilities).

Now suppose we collect data from this process. For example, assume we observe the number of impressions until the first click for $n = 5$ independent visitors:

$$y_1, y_2, y_3, y_4, y_5,$$

with observed values: 2, 3, 1, 2, and 4. Since the data has distribution $\text{Geom}(\theta)$ the probability of each observation (conditioned on θ) is

$$P(Y = y \mid \theta) = (1 - \theta)^{y-1} \theta.$$

For a vector of independent observations $y = (y_1, \dots, y_n)$, the likelihood is

$$p(y \mid \theta) = \prod_{i=1}^n \theta (1 - \theta)^{y_i-1} = \theta^n (1 - \theta)^{\sum_{i=1}^n (y_i-1)}.$$

Using the conjugate Beta prior $p \sim \text{Beta}(\alpha, \beta)$, the posterior is

$$p(\theta \mid y) \propto \theta^{\alpha+n-1} (1 - \theta)^{\beta + \sum_{i=1}^n y_i - n - 1}.$$

Thus, the posterior distribution is

$$p(\theta \mid y) \sim \text{Beta}\left(\alpha + n, \beta + \sum_{i=1}^n y_i - n\right).$$

With our chosen prior $\text{Beta}(3, 3)$ and data $(y_1, \dots, y_5) = (2, 3, 1, 2, 4)$, we have

$$n = 5 \quad \text{and} \quad \sum_{i=1}^5 y_i - 5 = 7.$$

Therefore,

$$p(y \mid \theta) \sim \text{Beta}(3 + 5, 3 + 7) = \text{Beta}(8, 10).$$

Posterior predictive distribution

The posterior predictive distribution for a new observation $\tilde{y}$ is obtained by integrating over the posterior for θ :

$$p(\tilde{y} \mid y) = \int_0^1 p(\tilde{y} \mid \theta) p(\theta \mid y) d\theta.$$

Since

$$p(\tilde{y} \mid \theta) = (1 - \theta)^{k-1} \theta \quad \text{and} \quad \theta \sim \text{Beta}(8, 10),$$

we get

$$p(\tilde{y} \mid y) = \frac{B(8 + 1, 10 + k - 1)}{B(8, 10)} = \frac{B(9, k + 9)}{B(8, 10)}.$$

Posterior predictive mean and variance

Using the law of total expectation similar to the prior predictive case, we can calculate the posterior predictive distribution and mean. Note that $E[Y|\theta, y] = 1/\theta$ is a random variable where θ now is considered with the posterior distribution. Then

$$E[Y|y] = E(E(Y|\theta, y)) = \left(\frac{1}{\theta}\right)$$

For $\theta \sim \text{Beta}(8, 10)$, we have

$$E\left(\frac{1}{\theta}\right) = \frac{\alpha_1 + \beta_1 - 1}{\alpha_1 - 1} = \frac{8 + 10 - 1}{8 - 1} = \frac{17}{7} \approx 2.429.$$

Similarly, provided $\alpha > 2$,

$$E\left(\frac{1}{\theta^2}\right) = \frac{(\alpha_1 + \beta_1 - 1)(\alpha_1 + \beta_1 - 2)}{(\alpha_1 - 1)(\alpha_1 - 2)} = \frac{(17)(16)}{(7)(6)} = \frac{272}{42} \approx 6.476.$$

Thus,

$$E(\text{Var}(Y | \theta)) = E\left(\frac{1-p}{p^2}\right) = E\left(\frac{1}{p^2}\right) - E\left(\frac{1}{p}\right) \approx 6.476 - 2.429 = 4.047.$$

Also,

$$\text{Var}(E[Y | \theta]) = \text{Var}\left(\frac{1}{p}\right) = E\left(\frac{1}{p^2}\right) - \left(E\left(\frac{1}{p}\right)\right)^2 \approx 6.476 - (2.429)^2 \approx 6.476 - 5.901 = 0.575.$$

Therefore, the posterior predictive variance is

$$\text{Var}(Y | y) = 4.047 + 0.575 \approx 4.622.$$

For additional exercise - simulate the prior and posterior for the parameter and the prior and posterior predictive distributions for the data. Analyze the distributions and their statistical properties. □

17 Data from Exponential distribution

Let $Y \sim \mathcal{E}(\theta)$, be exponentially distributed data, with a density

$$p_\theta(y) = \theta e^{-\theta y}, y \in (0, +\infty)$$

For the sample $y = (y_1, y_2, \dots, y_n)$, the likelihood function is given by

$$p(y|\theta) = \prod_{i=1}^n \theta e^{-\theta y_i} = \theta^n e^{-\theta \sum_{i=1}^n y_i}$$

The conjugate prior for exponential distribution is the Gamma distribution. If $\theta \sim \text{Gamma}(\alpha, \beta)$, the density is given by

$$p(\theta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{\alpha-1} e^{-\beta\theta}, \quad \theta > 0, \quad \text{with} \quad \mathbb{E}(\theta) = \frac{\alpha}{\beta}, \quad \text{Var}(\theta) = \frac{\alpha}{\beta^2} \quad (23)$$

Hence, the posterior of θ will be obtained as follows

$$p(\theta|y) = \frac{\beta^\alpha}{\Gamma(\alpha)p(y)} \theta^{\alpha+n-1} e^{-\theta(\beta+\sum_{i=1}^n y_i)} \sim \text{Gamma}(\underbrace{\alpha+n}_{\alpha_1}, \underbrace{\beta+\sum_{i=1}^n y_i}_{\beta_1})$$

For the posterior predictive density we have

$$\begin{aligned} p(\tilde{y}|y) &= \int_0^\infty p(\tilde{y}|\theta, y) p(\theta|y) d\theta \\ &= \frac{\beta_1^{\alpha_1}}{\Gamma(\alpha_1)} \frac{\Gamma(\alpha_1+1)}{(\beta_1+\tilde{y})^{\alpha_1+1}} \underbrace{\int_0^\infty \frac{(\beta_1+\tilde{y})^{\alpha_1+1}}{\Gamma(\alpha_1+1)} \theta^{\alpha_1+1-1} e^{-\theta(\beta_1+\tilde{y})} d\theta}_{=1(*)} = \frac{\alpha_1 \beta_1^{\alpha_1}}{(\beta_1+\tilde{y})^{\alpha_1+1}} \end{aligned} \quad (24)$$

(*) as an integral of the density of $\text{Gamma}(\beta_1+\tilde{y}, \alpha_1+1)$.

Next - find predictive expectation of data, make simulations for the prior and posterior distribution of the parameter, prior and posterior predictive distributions of the data and analyze the distributions statistically.

18 Data from Poisson distribution

Let $Y \sim \mathcal{P}(\theta)$ be a random variable from Poisson distribution with a probability mass function

$$p_\theta(y) = P(Y = y|\theta) = \frac{\theta^y}{y!} e^{-\theta}, \quad \theta > 0, \quad y \in 0, 1, 2, \dots,$$

For $y = (y_1, y_2, \dots, y_n)$, the likelihood function is given by

$$p(y|\theta) = \prod_{i=1}^n \frac{\theta^{y_i}}{y_i!} e^{-\theta} = \frac{\theta^{\sum_{i=1}^n y_i} e^{-n\theta}}{\prod_{i=1}^n y_i!}$$

Conjugate prior for the Poisson distribution is Gamma distribution, with a density $\text{Gamma}(\alpha, \beta)$

$$p(\theta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \theta^{\alpha-1} e^{-\beta\theta}, \quad \theta > 0, \quad \text{with} \quad \mathbb{E}(\theta) = \frac{\alpha}{\beta}, \quad \text{Var}(\theta) = \frac{\alpha}{\beta^2} \quad (25)$$

Here $\Gamma(\alpha)$ is the gamma function, generalization of the factorial. For integer values it holds $k \in \mathbb{N}$, $\Gamma(k) = (k-1)!$.

For the posterior we have

$$\begin{aligned} p(\theta|y) &= \frac{\theta^{\sum_{i=1}^n y_i} e^{-n\theta} \beta^\alpha \theta^{\alpha-1} e^{-\beta\theta}}{\left(\prod_{i=1}^n y_i!\right) \Gamma(\alpha) p(y)} = \frac{\beta^\alpha}{\left(\prod_{i=1}^n y_i!\right) \Gamma(\alpha) p(y)} \theta^{\alpha+\sum_{i=1}^n y_i-1} e^{-\theta(\beta+n)} \\ &= \frac{\beta_1^{\alpha_1}}{\Gamma(\alpha_1)} \theta^{\alpha_1-1} e^{-\theta\beta_1} \sim \text{Gamma}(\alpha_1, \beta_1), \quad \text{where } \alpha_1 = \alpha + \sum_{i=1}^n y_i, \beta_1 = \beta + n \end{aligned} \quad (26)$$

The posterior predictive probability is given by

$$\begin{aligned}
p(\tilde{y}|y) &= \int_0^\infty p(\tilde{y}|\theta, y)p(\theta|y)d\theta = \int_0^\infty \frac{\theta^{\tilde{y}}}{\tilde{y}!} e^{-\theta} \frac{\beta_1^{\alpha_1}}{\Gamma(\alpha_1)} \theta^{\alpha_1-1} e^{\theta\beta_1} d\theta \\
&= \frac{\beta_1^{\alpha_1}}{\Gamma(\alpha_1)\tilde{y}!} \frac{\Gamma(\alpha_1 + \tilde{y})}{(\beta_1 + 1)^{\alpha_1 + \tilde{y}}} \underbrace{\int_0^1 \frac{(\beta_1 + 1)^{\alpha_1 + \tilde{y}}}{\Gamma(\alpha_1 + \tilde{y})} \theta^{\alpha_1 + \tilde{y} - 1} e^{-\theta(\beta_1 + 1)} d\theta}_{=1(*)} \\
&= \frac{\Gamma(\alpha_1 + \tilde{y})}{\Gamma(\alpha_1)\Gamma(\tilde{y} + 1)} \left(\frac{\beta_1}{\beta_1 + 1}\right)^{\alpha_1} \frac{1}{(\beta_1 + 1)^{\tilde{y}}} \quad \text{negative binomial for } \alpha_1 \in \mathbb{N} \quad (27)
\end{aligned}$$

(*) - as an integral of the density of $\text{Gamma}(\beta_1 + 1, \alpha_1 + \tilde{y})$.

For integer values of α_1 , this is the negative binomial distribution $NBin(\alpha_1, p = \frac{\beta_1}{\beta_1 + 1})$, with a probability law $P(Y = \tilde{y}) = \binom{\tilde{y}-1}{\alpha_1-1} \left(\frac{\beta_1}{\beta_1+1}\right)^{\alpha_1} \left(\frac{1}{\beta_1+1}\right)^{\tilde{y}}$. The constant $\frac{\Gamma(\alpha_1 + \tilde{y})}{\Gamma(\alpha_1)\Gamma(\tilde{y} + 1)}$ in (27) is actually the binomial coefficient.

If we know the form of the posterior predictive distribution, we can directly simulate from it and use its properties. However, we can also use Monte Carlo sampling.

Example 18.1. (From the book "A first course of Bayesian Statistical Analysis", by Peter D. Hoff) An agency of social services gathered data on the educational attainment and number of children of 155 women who were 40 years of age at the time of the survey. We would like to compare the women with college degrees to those without, in terms of their number of children. Thus, we have data from two random variables Y_1 (number of children of women with lower than bachelor's) and Y_2 (number of children of women with bachelor's or higher). The empirical distribution is given in the following figure:

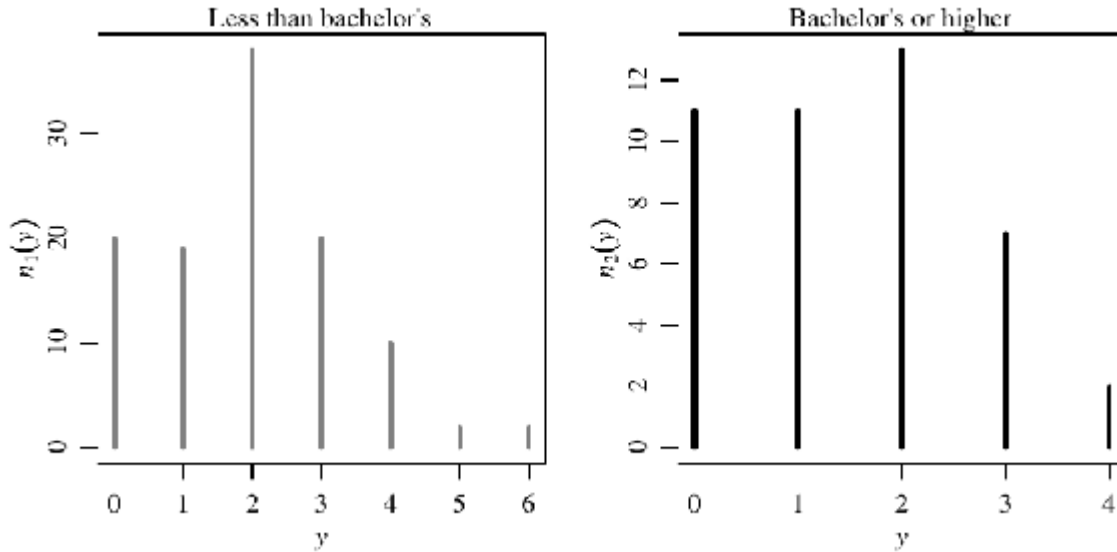


Figure 34: Empirical distribution - number of children for the two groups.

The collected sample is summarized bellow

$$\begin{aligned}
\text{(number of children of women with lower than bachelor's)} \quad y_1 &: n_1 = 111, \sum_{i=1}^{n_1} y_{i,1} = 217, \bar{y}_1 = 1.95 \\
\text{(number of children of women with bachelor's or higher)} \quad y_2 &: n_2 = 44, \sum_{i=1}^{n_2} y_{i,1} = 66, \bar{y}_2 = 1.5
\end{aligned}$$

Let us assume that the data comes from the following model

$$\begin{aligned} Y_{1,1}, \dots, Y_{n_1,1} | \theta_1 &\sim i.i.d. \text{Poisson}(\theta_1) \\ Y_{1,2}, \dots, Y_{n_2,2} | \theta_2 &\sim i.i.d. \text{Poisson}(\theta_2) \end{aligned} \quad (28)$$

The mean of the birth rates (expected birth rates) is θ_1 and θ_2 for each group, respectively. We will discuss later whether this is an appropriate model.

If we assume that $(\theta_1, \theta_2) \sim i.i.d. \text{Gamma}(2,1)$, then from the previous analysis the posterior distribution for the parameters θ_1, θ_2 are given with

$$\begin{aligned} \theta_1 | y_1 &\sim \text{Gamma}(2 + 217, 1 + 111) = \text{Gamma}(219, 112) \\ \theta_2 | y_2 &\sim \text{Gamma}(2 + 66, 1 + 44) = \text{Gamma}(68, 45) \end{aligned}$$

For exercise - simulate data from the prior and posterior distribution of the parameters, prior and posterior predictive distribution of the data, calculate the MAP estimates for both parameters and based on all the known data, discuss whether it is a good approximation of the true parameters.

We give here posterior means, modes and 95% quantile based confidence intervals:

```

1 a<-2 ; b<-1      # prior parameters
2 n1<-111 ; sy1 <-217      # data in group 1
3 n2<-44 ; sy2 <-66      # data in group 2
4
5
6 (a+sy1)/(b+n1)      # posterior mean
7 [1] 1.955357
8 (a+sy1-1)/(b+n1 ) # posterior mode
9 [1] 1.946429
10 qgamma(c(0.025, 0.975), a+sy1, b+n1)      # posterior 95% CI
11 [1] 1.704943 2.222679
12
13
14 (a+sy2)/(b+n2)
15 [1] 1.511111
16 (a+sy2-1)/(b+n2)
17 [1] 1.488889
18 qgamma(c(0.025,0.975), a+sy2, b+n2)
19 [1] 1.173437 1.890836

```

The expected birth rates for both groups are θ_1 and θ_2 (the data are Poisson distributed). Next we will represent the posterior densities for the population means of the two groups and the posterior predictive probabilities. The following code gives those figures.

```

1 # posterior densities
2 par(mfrow=c(1,1))
3 #posterior theta1
4 curve(dgamma(x, shape=219, rate=112), from=0, to=4, ylab="", xlab="theta1/theta2")
5 #posterior theta2
6 curve(dgamma(x, shape=68, rate=45), from=0, to=4, col="grey", add=TRUE)
7
8 curve(dgamma(x, shape=2, rate=1), from=0, to=4, lty=2, add=TRUE) #prior
9 legend(2.5, 3, legend=c("posterior theta1", "posterior theta2", "prior"),
10       col=c("gray", "black", "black"), lty=c(1,1,2), cex=0.8)
11
12
13 # posterior predictive probabilities for the number of children
14 par(mfrow=c(1,2))
15 theta1<-rgamma(100,219,112)
16 theta2<-rgamma(100,68,45)

```



```

17 Y1=c()
18 Y2=c()
19 for(i in theta1)
20 { r1<-rpois(100,i)
21 Y1=c(Y1,r1)
22 }
23 for(j in theta2)
24 { r2<-rpois(100,j)
25 Y2=c(Y2,r2)
26 }
27 hist(Y1,main="",freq=F,breaks=100, ylim=c(0,4),xlab="Y1 (less than bachelor's)")
28 hist(Y2,main="",freq=F,col="black",breaks=100, ylim=c(0,4),
29 xlab="Y2(bachelor's and higher)")
30
31 #posterior predictive probabilities using the formula
32 #for the posterior negative binomial distribution
33 par(mfrow=c(1,2))
34 Y1post<-rnbinom(10000,219,mu=1.95) #second parameter is the mean, it is an option in R
35 Y2post<-rnbinom(10000,68,mu=1.51)
36 hist(Y1post,main="",freq=F,breaks=100,xlim=c(0,10),
37 ylim=c(0,4),xlab="Y1 (less than bachelor's)")
38 hist(Y2post,main="",freq=F,breaks=100,col="black",xlim=c(0,10),
39 ylim=c(0,4),xlab="Y2(bachelor's and higher)")

```

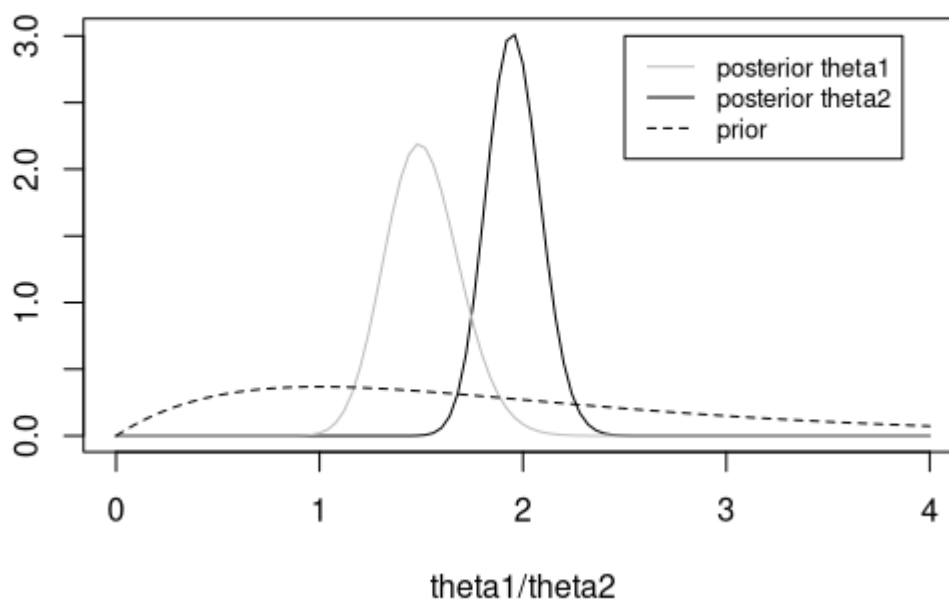


Figure 35: Posterior densities of mean birth rates

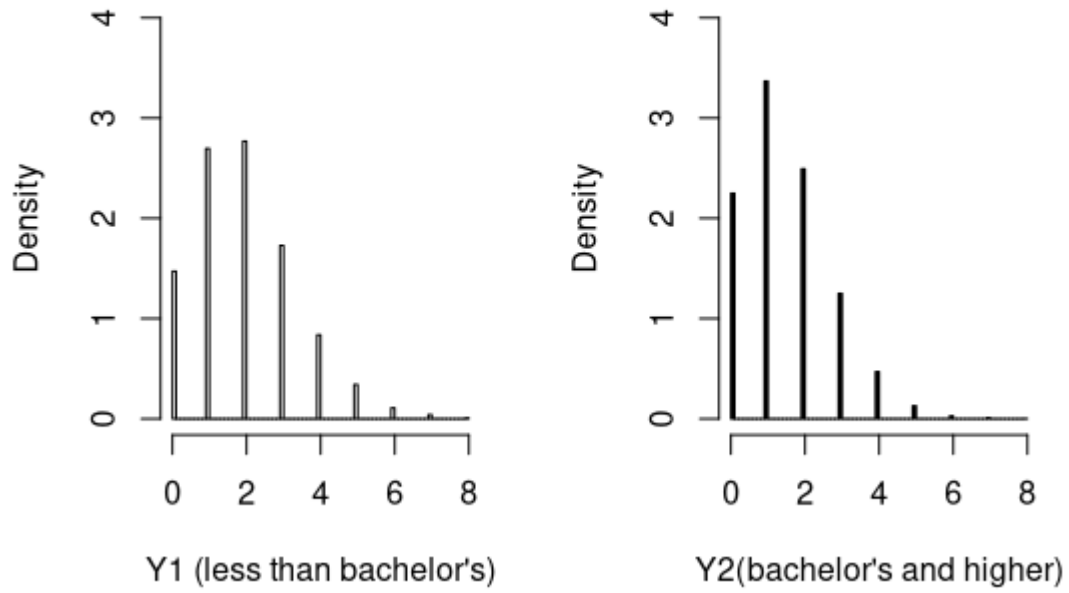


Figure 36: Posterior predictive probabilities for number of children for both groups, simulated indirectly

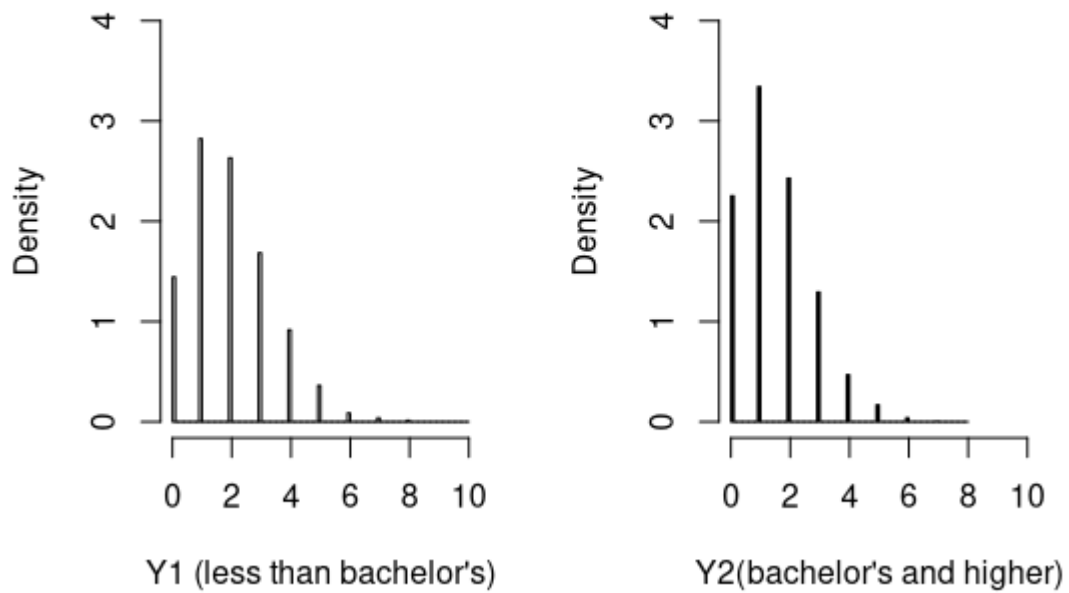


Figure 37: Posterior predictive probabilities for number of children for both groups, using the formulas for the posterior negative binomial distribution

The posterior densities of the mean rates give evidence that $\theta_1 > \theta_2$. We can calculate the probability $P(\theta_1 > \theta_2 | y_1, y_2)$. Also we want to know, for two randomly sampled individuals from both groups, what is the probability that one has more, equal or lower number of children. The code in the next lines

```

1 theta1<-rgamma(10000,219,112)
2 > theta2<-rgamma(10000,68,45)
3 > mean(theta1>theta2)
4 [1] 0.972
5 > mean(Y1post>Y2post)
6 [1] 0.4848
7 > mean(Y1==Y2)
8 [1] 0.2188
9 > mean(Y1post<Y2post)
10 [1] 0.3049

```

We can see that even though there is a strong evidence that $\theta_1 > \theta_2$, the difference between the data from the two groups is not that large.

Next, let us discuss if the Poisson is a good choice for this data. Let us see again the histograms for the empirical distribution Figure 18.1 and posterior predictive distribution of Y_1 (number of children in the group with lower education) Figure 18.1. In the empirical distribution, the number of women who have 2 children is twice the number of women with one child (38 vs. 19). In the posterior distribution the probability to have 2 children in this group is slightly higher than the probability for one child. The large number of women with 2 children may be just by chance (from the sampling variability). But this may be a characteristic in the population. There is no Poisson distribution with such a high peak at $Y = 2$, so maybe the model is wrong. This motivates us to see whether Poisson model is good choice for the data.

Let for a vector of length $n=111$, let $t(y)$ denote the ratio of number of women who have 2 children to the number of women with one child. For our observed data, this number is 2. We will simulate posterior predictive data sets and for each sample, calculate the value of t . The procedure is as following For each $i \in \{1, 2, \dots, n\}$

1. sample value for θ_i from the posterior distribution $\theta_i \sim p(\theta | data)$
2. sample a vector $\tilde{Y}^{(i)} = (\tilde{y}_1^{(i)}, \dots, \tilde{y}_n^{(i)}) \sim i.i.d. p(y | \theta_i)$ (posterior predictive data sample of size n)
3. compute $t^{(i)} = t(\tilde{Y}^{(i)})$ (with this we have samples from the posterior predictive distributions of $t(\tilde{Y})$)

The code

```

1 t<-NULL
2 for(s in 1:10000){
3   theta1<-rgamma(1,219,112)
4   y1.mc<-rpois(n1,theta1)
5   t<-c(t,sum(y1.mc==2)/sum(y1.mc==1))
6 }
7 hist(t,freq=F)

```

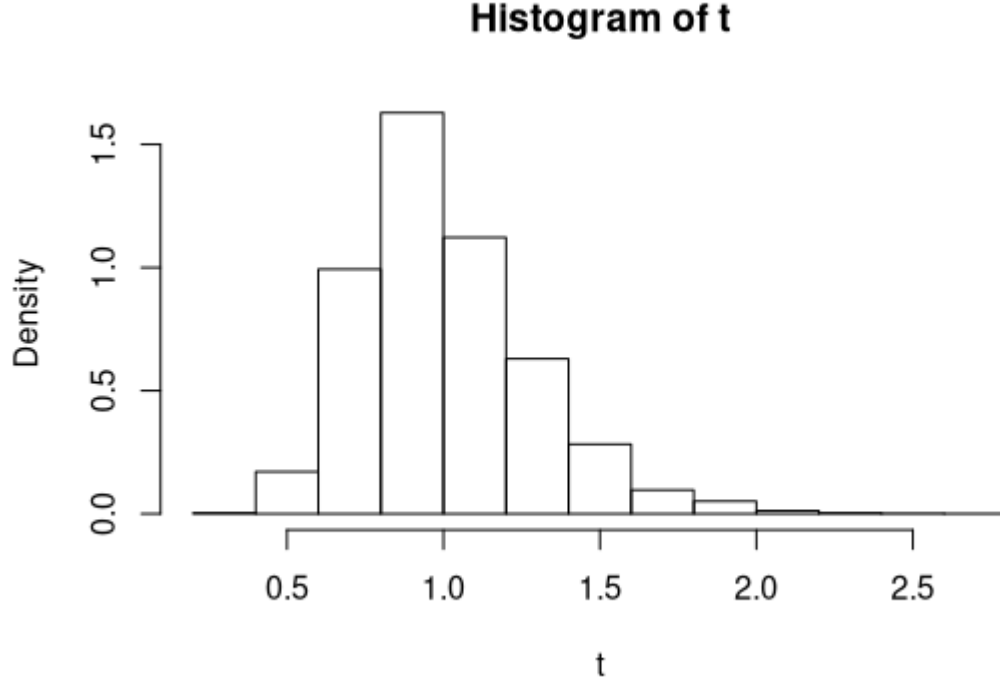


Figure 38: Posterior predictive distribution of the empirical odds of having 2 children vs one child, in a data set of size n

As we can see from the last histogram (based on sample of t of size 10000), we will hardly ever observe this ratio to be 2 (about 0.5 percent). Thus, the Poisson model may not be good if we want to inference the true probability distribution of Y . Still, if we calculate the mean and the variance of the observed sample, they are very close (calculate it for both data sets). Thus, the Poisson model may still be good if we want to infer on the mean, the variance and some other aspects from the true probability. For more details we refer to Chapter 4 from "A first Course in Bayesian Statistical Methods" by Peter Hoff. $\square$

19 Normal - normal model

Let the data be normally distributed $Y \sim \mathcal{N}(\mu, \sigma^2)$, μ is unknown and σ^2 is known. Let us take a prior distribution for μ , $\mu \sim \mathcal{N}(\mu_p, \sigma_p^2)$, where both parameters are either known or can be estimated. To simplify the calculations, let us take one observation y ($n=1$). Then for the posterior we have

$$p(\mu|y) = \frac{p(y|\mu)p(\mu)}{p(y)} = \frac{e^{-\frac{(y-\mu)^2}{2\sigma^2}} e^{-\frac{(\mu-\mu_p)^2}{2\sigma_p^2}}}{p(y)} \quad (29)$$

We will transform the exponential term

$$\begin{aligned}
\frac{(y - \mu_p)^2}{2\sigma^2} + \frac{(\mu - \mu_p)^2}{2\sigma_p^2} &= \frac{1}{2} \left[\frac{y^2 - 2y\mu + \mu^2}{\sigma^2} + \frac{\mu^2 - 2\mu\mu_p + \mu_p^2}{\sigma_p^2} \right] \\
&= \frac{1}{2} \left[\left(\frac{1}{\sigma^2} + \frac{1}{\sigma_p^2} \right) \mu^2 - 2 \left(\frac{y}{\sigma^2} + \frac{\mu_p}{\sigma_p^2} \right) \mu + \left(\frac{y^2}{\sigma^2} + \frac{\mu_p^2}{\sigma_p^2} \right) \right] \\
&= \frac{1}{2} \left[\frac{\sigma^2 + \sigma_p^2}{\sigma^2 \sigma_p^2} \mu^2 - 2 \left(\frac{y}{\sigma^2} + \frac{\mu_p}{\sigma_p^2} \right) \mu + \left(\frac{y^2}{\sigma^2} + \frac{\mu_p^2}{\sigma_p^2} \right) \right] \\
&= \frac{\sigma^2 + \sigma_p^2}{2\sigma^2 \sigma_p^2} \left[\mu^2 - 2 \frac{\sigma^2 \sigma_p^2}{\sigma^2 + \sigma_p^2} \left(\frac{y}{\sigma^2} + \frac{\mu_p}{\sigma_p^2} \right) \mu + \underbrace{\frac{\sigma^2 \sigma_p^2}{\sigma^2 + \sigma_p^2} \left(\frac{y^2}{\sigma^2} + \frac{\mu_p^2}{\sigma_p^2} \right)}_B \right] \\
&= \frac{\sigma^2 + \sigma_p^2}{2\sigma^2 \sigma_p^2} \left[\mu^2 - 2 \underbrace{\left(\frac{\sigma_p^2}{\sigma^2 + \sigma_p^2} y + \frac{\sigma^2}{\sigma^2 + \sigma_p^2} \mu_p \right)}_A + \underbrace{\left(\frac{\sigma_p^2}{\sigma^2 + \sigma_p^2} y + \frac{\sigma^2}{\sigma^2 + \sigma_p^2} \mu_p \right)^2}_{A^2} \right] + \underbrace{\frac{\sigma^2 \sigma_p^2}{\sigma^2 \sigma_p^2} [B - A^2]}_C \\
&= \frac{\sigma^2 + \sigma_p^2}{2\sigma^2 \sigma_p^2} \left[\mu - \underbrace{\left(\frac{\sigma_p^2}{\sigma^2 + \sigma_p^2} y + \frac{\sigma^2}{\sigma^2 + \sigma_p^2} \mu_p \right)}_{A = \mu_{post}} \right]^2 + K
\end{aligned} \tag{30}$$

Thus, the posterior distribution of μ is

$$p(\mu|y) = C e^{-\frac{\sigma^2 + \sigma_p^2}{2\sigma^2 \sigma_p^2} \left[\mu - \left(\frac{\sigma_p^2}{\sigma^2 + \sigma_p^2} y + \frac{\sigma^2}{\sigma^2 + \sigma_p^2} \mu_p \right) \right]^2} = C e^{-\frac{\left[\mu - \left(\frac{\sigma_p^2}{\sigma^2 + \sigma_p^2} y + \frac{\sigma^2}{\sigma^2 + \sigma_p^2} \mu_p \right) \right]^2}{\frac{2\sigma^2 \sigma_p^2}{\sigma^2 + \sigma_p^2}}} = C e^{-\frac{(\mu - \mu_{post})^2}{2\sigma_{post}^2}} \tag{31}$$

The posterior distribution of μ is normal $\mathcal{N}(\mu_{post}, \sigma_{post}^2)$, where $\mu_{post} = \left(\frac{\sigma_p^2}{\sigma^2 + \sigma_p^2} y + \frac{\sigma^2}{\sigma^2 + \sigma_p^2} \mu_p \right)$ and $\sigma_{post}^2 =$

$$\frac{\sigma^2 \sigma_p^2}{\sigma^2 + \sigma_p^2}.$$

20 Inference and decision making

Example 20.1. (from "An introduction to Bayesian thinking" by Clyde et al., modified)

A car manufacturer plans to introduce 5 new model and is interested how many people would buy the model A. The management thinks the proportion is either 10% or 20%. We have to make a decision on the true proportion, based on which we may get a bonus. We have some budget which can be spend on data collection, but it is limited. We have to decide whether it is necessary to collect a big sample or if we can make a decision based on a smaller sample.

Frequentist approach We define the null and the alternative hypotheses

- H_0 : 10% of people would buy model A
- H_a : 20% would buy model A

We will set $\alpha = 0.05$. Suppose we have observed the following data: $n=5$ were asked about the model they would buy, one of them chose model A, thus $k=1$. If the null hypothesis is true, i.e. $p=0.1$, we would expect to see around 10 percent of the customers to buy model A. Given a sample of size 5, that is less than 1. We would like to find the probability that in such a sample size, we would observe 1 or more extreme

value, given the null hypothesis is true. This is the so called p values, which in this case is calculated as following

$$P(k \geq 1|n = 5, p = 0.1) = 1 - P(k = 0|n = 5, p = 0.1) = 1 - 0.9^5 \approx 0.41$$

(The significance probability, or p-value, is the chance of observing data that are as or more supportive of the alternative hypothesis than the data that were collected, when the null hypothesis is true.) This value is high, even more it is greater than 0.05. Thus, based on such a small sample, the null hypothesis can not be rejected. If we had a large sample, we could perform the test for proportions. However, that is not applicable here. What is important in this frequentist approach is that the test does not confirm the null hypothesis. We just say that we do not have enough evidence to reject it. Thus, based on this sample, we may decide that $p=10\%$.

Bayesian approach. We have the hypothesis

- H_0 : 10% of people would buy model A
- H_a : 20% would buy model A

and we assign a prior probability to both of them. Let us assume that they are equally probable. Then $P(H_0) = P(H_a) = 0.5$. If we observed a sample as above ($k=1, n=5$) then the likelihoods are given with

$$\begin{aligned} P(k = 1|H_0) &= \binom{5}{1} 0.1^1 0.9^4 \approx 0.33 \\ P(k = 1|H_a) &= \binom{5}{1} 0.2^1 0.8^4 \approx 0.41 \end{aligned} \quad (32)$$

Then, the posterior probabilities for the hypothesis, given the data are

$$\begin{aligned} P(H_0|k = 1) &= \frac{P(k = 1|H_0)P(H_0)}{P(k = 1)} = \frac{0.33 \cdot 0.5}{0.33 \cdot 0.5 + 0.41 \cdot 0.5} \approx 0.45 \\ P(H_a|k = 1) &= 1 - P(H_0|k = 1) = 0.55 \end{aligned} \quad (33)$$

The posterior probabilities are close to each other, we had equal prior and small sample size. Thus we can not make a decision with a high confidence. However, if we have to make a decision, we have to go with the hypothesis that has higher posterior probability. In our case, that is $p=0.2$, which contradicts the frequentist approach.

The following table represents how the results would look like if we had chosen larger sample sizes. in each case, the p value in the frequentist approach is higher than 0.05, and we will not reject the null hypothesis with any of the 4 samples. The Bayesian methods are more consistent, i.e. in all four cases the posterior probability for H_a is higher.

Frequentist and Bayesian probabilities for larger sample sizes			
	Frequentist	Bayesian H_0	Bayesian H_a
Observed data	$P(k \text{ or more} n, H_0)$	$P(H_0 n, k)$	$P(H_1 n, k)$
$n=5, k=1$	0.41	0.45	0.55
$n=10, k=2$	0.26	0.39	0.61
$n=15, k=3$	0.18	0.34	0.66
$n=20, k=4$	0.13	0.29	0.71

We also have to mention the following - the results in the frequentist approach would be different if we define the hypothesis differently. Let us take $H_0: p = 20\%$ and $H_a: p < 20\%$. Make the calculations to see that we would get different results. This happens since the frequentist approach is highly sensitive to the null hypothesis, while in the Bayesian approach it is not important in which order we consider the hypothesis.

□

Read about odds, log odds and Bayesian Factor from the MIT lecture notes.

Bayesian hypothesis testing procedure: Suppose we want to test

$$\begin{aligned} H_0 : \theta \in \Theta_1 \quad vs \\ H_a : \theta \in \Theta_2, \end{aligned}$$

where Θ_1, Θ_2 are some given sets. Also, suppose θ has a prior distribution $p(\theta)$ and posterior distribution $p(\theta|y)$, given some sample y . We compute the following

$$\begin{aligned} \alpha_0 &= P(H_0|y) = P(\theta \in \Theta_1|y) \\ \alpha_1 &= P(H_1|y) = P(\theta \in \Theta_2|y) \end{aligned} \tag{34}$$

Calculate the Bayes factor $\frac{\alpha_0}{\alpha_1} = \frac{P(H_0|y)}{P(H_1|y)}$. If it is less than 1 we reject the null hypothesis, otherwise we accept it.

Example 20.2. (from "Mathematical Statistics with Applications" by Ramachrandran)

A student taking a standardized test is classified as gifted if he or she scores at least 100 out of a possible score of 150. Otherwise the student is classified as not gifted. Suppose the prior distribution of the scores of all students is a normal with mean 100 and standard deviation 15. It is believed that scores will vary each time the student takes the test and that these scores can be modeled as a normal distribution with mean μ and variance 100. Suppose the student takes the test and scores 115. Test the hypothesis that the student can be classified as a gifted student.

We want to test

$$\begin{aligned} H_0 : \theta < 100 \quad vs \\ H_a : \theta \geq 100, \end{aligned}$$

The posterior distribution of θ is $\mathcal{N}(110.4, 69.2)$ (from the formula from previous chapter).

Using that $\theta \sim \mathcal{N}(100, 225)$ we can calculate

$$\alpha_0 = P(\theta < 100|y = 115) = P\left(\frac{\theta - 110.4}{\sqrt{69.2}} < \frac{100 - 110.4}{\sqrt{69.2}}\right) = P\left(z \leq \frac{-10.4}{\sqrt{69.2}}\right) = 0.106$$

and

$$\alpha_1 = P(\theta \geq 100|y = 115) = 1 - \alpha_0 = 0.894 \tag{35}$$

Thus, the Bayes factor is $\frac{\alpha_0}{\alpha_1} = 0.119 < 1$ and we reject H_0 .

□

Example 20.3. (An example from decision theory)

Suppose you own a small stall at a flea market that is open only on weekends. If the weather is good, you make a profit of \$ 200, and if it is bad, you close your stall and you make no (zero) profit. However, you have the option of buying, from an insurance company, weather insurance that costs \$ 75. The company pays you \$ 210 if the weather is bad. Suppose you believe that the probability of good weather on a particular weekend is p . Compute the expected gain if you insure and if you do not. What is the best course of action? Arrive at a decision.

We will model θ with Bernoulli distribution, $\theta = 1$ if the weather is good (probability p), and 0 otherwise. Let us make a table

Weather		
Parameter /Decision	Good weather	Bad weather
	$(\backslash \theta_1)$	$(\backslash \theta_2)$
Insurance (d.1)	200-75=125 (\$)	210-75=135 (\$)
No Insurance (d.2)	200 (\$)	0 (\$)

For the conditional expected gain we have

$$\begin{aligned}\mathbb{E}(\text{gain}|\text{insurance}) &= 125p + 135(1 - p) = 135 - 10p \\ \mathbb{E}(\text{gain}|\text{noinsurance}) &= 200p\end{aligned}$$

Thus, we would prefer the insurance if $135 - 10p > 200p$, i.e. the probability of good weather $p < \frac{135}{210} = 0.643$.

Next we should look into some historical data for the weather (or sample ourselves) and estimate p . □

21 Important formulas

Beta Distribution

Let $X \sim \text{Beta}(\alpha, \beta)$, $\alpha, \beta > 0$, with support $x \in [0, 1]$. Its density is

$$f_X(x) = \frac{x^{\alpha-1} (1-x)^{\beta-1}}{B(\alpha, \beta)}, \quad B(\alpha, \beta) = \int_0^1 t^{\alpha-1} (1-t)^{\beta-1} dt.$$

Moments:

$$\mathbb{E}[X] = \frac{\alpha}{\alpha + \beta}, \quad \text{Var}(X) = \frac{\alpha \beta}{(\alpha + \beta)^2 (\alpha + \beta + 1)}.$$

Properties:

- Conjugate prior for a Binomial(n, p) likelihood.
- Symmetric about $x = \frac{1}{2}$ when $\alpha = \beta$.
- Special cases:
 - Beta(1, 1) $\equiv$ Uniform(0, 1).
 - Beta($\alpha, 1$) has density $\propto x^{\alpha-1}$.

Gamma Distribution

Let $Y \sim \text{Gamma}(\kappa, \theta)$ with shape $\kappa > 0$, scale $\theta > 0$, support $y \geq 0$. Its density (shape–scale form) is

$$f_Y(y) = \frac{1}{\Gamma(\kappa) \theta^\kappa} y^{\kappa-1} e^{-y/\theta}, \quad y \geq 0,$$

where

$$\Gamma(\kappa) = \int_0^\infty t^{\kappa-1} e^{-t} dt, \quad \Gamma(n) = (n-1)! \quad (n \in \mathbb{N}).$$

In the alternative shape–rate parametrization $\text{Gamma}(\alpha, \lambda)$, $\lambda = 1/\theta$, the density is

$$f_Y(y) = \frac{\lambda^\alpha}{\Gamma(\alpha)} y^{\alpha-1} e^{-\lambda y}.$$

Moments:

$$\mathbb{E}[Y] = \kappa \theta, \quad \text{Var}(Y) = \kappa \theta^2.$$

Properties:

- Additivity: If $Y_1 \sim \text{Gamma}(\kappa_1, \theta)$ and $Y_2 \sim \text{Gamma}(\kappa_2, \theta)$ are independent, then

$$Y_1 + Y_2 \sim \text{Gamma}(\kappa_1 + \kappa_2, \theta).$$

- Special cases:
 - $\text{Gamma}(1, \theta) \equiv \text{Exponential}(1/\theta)$.
 - Integer κ yields the Erlang distribution.
- Conjugate prior for the rate parameter of a Poisson or Exponential model.

22 Improvement of the material

More exercises on convolution, total probability, conditional expectation, simulations

https://courses.cs.washington.edu/courses/cse312/20su/files/student_drive/5.5.pdf

23 Bayesian decision theory, loss function, minimizing expected loss

24 Markov Chain Monte Carlo (MCMC) simulations with applications

25 Application in Machine Learning

26 Notes on R and additional reading

- The codes from Kruschke’s book (R functions) can be found here:

https://github.com/boboppie/kruschke-doing_bayesian_data_analysis/find/master

- The code in Kruschke’s book is probably written for windows. The command `windows()` is for Windows only. For Linux `x11` works well so substitute it in your code. Some details can be found on Kruschke’s blog:

<http://doingbayesiandataanalysis.blogspot.com/2012/01/graphics-plots-in-r-for-macos-linux-and.html>

- The book "Bayesian Computation in R" by Jim Albert has detailed codes and figures with a lot of exercises. The author uses the package LearnBayes. Some codes with exercises can be found here <https://rdrr.io/cran/LearnBayes/f/inst/doc/DiscreteBayes.pdf>
- Important discussion by Gelman, Bayes holes https://sites.stat.columbia.edu/gelman/research/unpublished/bayes_holes_2.pdf
- Easy explanation of Metropolis algorithm, applied on Binomial proportion <https://www.quantstart.com/articles/Markov-Chain-Monte-Carlo-for-Bayesian-Inference-The-Metropolis-Algorithm/>
- Important discussion by Gelman, choice of prior in the practical context http://www.stat.columbia.edu/~gelman/research/unpublished/prior_context_2.pdf
- Great lecture notes, detailed mathematics, examples, R, more advanced topics (more unknown parameters, simulations). <http://www.mas.ncl.ac.uk/~nmf16/teaching/mas3301/week5.pdf>
- Lecture on MonteCarlo with examples, but with MatLab http://halweb.uc3m.es/esp/Personal/personas/mwiper/docencia/English/PhD_Bayesian_Statistics/ch6_2009.pdf

References

- [1] Doing Bayesian Data Analysis, John K. Kruschke
- [2] A First Course in Bayesian Statistical Methods, Peter D. Hoff
- [3] Bayesian Computation with R, Jim Albert
- [4] (Exercises) Mathematical Statistics with Applications, Ramachandran, Tsokos

27 Bayesian Linear regression

27.1 Convolution of Random Variables

If X and E are independent with densities $f_X(x)$ and $f_E(e)$, then

$$Y = X + E, \quad f_Y(y) = (f_X * f_E)(y) = \int_{-\infty}^{\infty} f_X(x) f_E(y - x) dx.$$

27.2 Additive Noise Model

Observe

$$y = x + e, \quad e \sim N(0, \sigma^2), \quad x \sim f_X(x | \theta).$$

- **Conditional:** $y | x \sim N(x, \sigma^2)$,

$$p(y | x, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y-x)^2}{2\sigma^2}\right).$$

- **Prior:** $x \sim f_X(x \mid \theta)$ (e.g. Normal, Uniform, Exponential).
- **Marginal:**

$$p(y \mid \theta, \sigma) = \int p(y \mid x, \sigma) f_X(x \mid \theta) dx = (N(0, \sigma^2) * f_X)(y).$$

27.3 Likelihood for a Sample

For n iid observations $y_1, \dots, y_n$, the joint likelihood is

$$L(\theta, \sigma \mid y_{1:n}) = \prod_{i=1}^n \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - x)^2}{2\sigma^2}} f_X(x \mid \theta) dx.$$

27.4 Bayesian Linear Regression

Data: (x_i, y_i) , noise $e_i \sim N(0, \sigma^2)$.

27.4.1 One-Dimensional Regression ($p = 1$)

Model:

$$\begin{aligned} y_i &= \beta x_i + e_i, \quad e_i \sim N(0, \sigma^2), \\ \beta &\sim N(0, \tau^2), \quad p(y \mid X, \beta, \sigma) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - \beta x_i)^2}{2\sigma^2}}. \end{aligned}$$

Posterior (conjugate) is Gaussian and the MAP coincides with ridge when taken at mode.

27.4.2 Multivariate Regression ($p > 1$)

Let $X \in \mathbb{R}^{n \times p}$, $\beta \in \mathbb{R}^p$.

$$\begin{aligned} \mathbf{y} &= X\beta + \mathbf{e}, \quad \mathbf{e} \sim N(0, \sigma^2 I_n), \\ \beta &\sim N(0, \tau^2 I_p), \quad p(\beta) \propto e^{-\frac{1}{2\tau^2} \beta^\top \beta}, \\ p(\beta, \sigma \mid y, X) &\propto p(y \mid X, \beta, \sigma) p(\beta) p(\sigma). \end{aligned}$$

27.5 Connection to Ridge and Lasso

- *Ridge:* MAP under Gaussian prior $\beta \sim N(0, \frac{\sigma^2}{\lambda} I)$ gives

$$\hat{\beta} = \arg \min_{\beta} \|y - X\beta\|_2^2 + \lambda \|\beta\|_2^2.$$

- *Lasso:* MAP under Laplace prior $p(\beta_j) \propto e^{-\frac{\lambda}{\sigma^2} |\beta_j|}$ gives

$$\hat{\beta} = \arg \min_{\beta} \|y - X\beta\|_2^2 + \lambda \|\beta\|_1.$$

27.6 Summary

1. Convolution arises from summing independent variables.
2. Marginal $p(y)$ is convolution of noise and prior on x .
3. Bayesian linear regression uses priors on β ; conjugacy yields closed-form posteriors.
4. Ridge and Lasso correspond to Gaussian and Laplace priors, respectively.